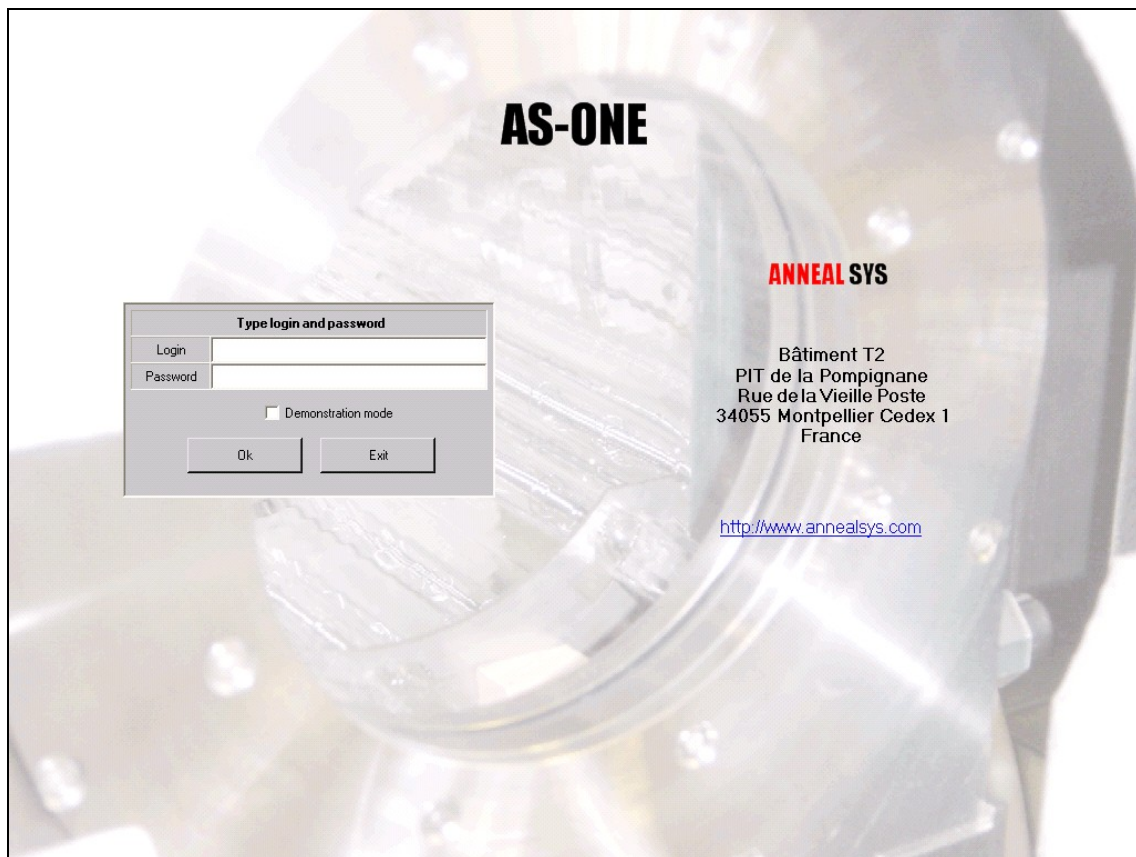


# AS-One

**Rapid Thermal Processor**

## **Software user's manual**



**Software version: 1.5.2**

**Manual version: 1.5.2.0**

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## 1. CONSTRUCTOR'S FOREWORD

|                                                                                   |                                                                                                                                                                                                                               |
|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <b>⚠ WARNING</b>                                                                                                                                                                                                              |
|                                                                                   | <b>SAFETY INSTRUCTIONS</b><br>The user must carefully read the following instructions.<br>The user is responsible for installing the system in safe conditions and in an environment that complies with the local regulation. |

The Annealsys system can be used for various applications. Many different types of gases can be used in the system; consequently, the user must take all the necessary precautions to prevent hazardous mixtures occurring inside the process chamber.

The constructor declines all responsibility for any incidents caused by insufficient precautions or handling errors, and their consequences.

The constructor specifies that the Annealsys system has no protection against any possible toxic emanations.

The responsibility for installing the machine in an environment which complies with the relating legislation is left entirely to the initiative and charge of the user, who is considered to be aware of the effects of the gases that he uses, as well as those of the decomposition products and gases generated by the processes in operation.

The user shall be responsible for connecting the exhaust line and the pump exhaust of the system to a gas scrubbing installation which is compatible with the process gases and gas flows and that complies with local regulation.

The user is responsible for ensuring that supply pipes and cables as well as exhaust are routed in such a way to eliminate trip hazard. Also, floor surface is to be suitable and must reduce as far as possible any slip hazard for the operator or others in the vicinity of the machine. Adequate lighting and any fire suppression system is also the responsibility of the user

The constructor also specifies that the quartz, ceramic and graphite parts and all parts in contact with vacuum or process gases must only be handled with gloves to avoid any pollution.

All maintenance and servicing work should be carried out by skilled personnel and, where specified, in relation with Annealsys Customer Service Department.

## 2. SAFETY INSTRUCTIONS

The signal words for the safety instructions and labels are DANGER, WARNING and CAUTION.

### **DANGER**

DANGER is the signal word used to indicate an imminently hazardous situation that, if not avoided, will result in death or severe injury.

### **WARNING**

WARNING is the signal word used to indicate a potentially hazardous situation which, if not avoided, could result in death or severe injury.

### **CAUTION**

CAUTION is the signal word used to indicate a potentially hazardous situation which, if not avoided, could result in moderate or minor injury. It may also be an alert against unsafe practices.

### 2.1 Pictogram definitions

|                                                                                     |                                                                                                                                                             |
|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    | <b>Hot surface</b><br><b>Burn hazard while touching hot parts.</b><br><b>This label warns the user from eventual burning hazards.</b>                       |
|  | <b>Electricity, Electrical hazard</b><br><b>Contact may cause electric shock or burn</b><br><b>This label warns the user of eventual electrical hazards</b> |
|  | <b>Toxic material, poison</b><br><b>This label warns the user from the danger and lethal risks involved when using toxic or corrosive chemicals</b>         |
|  | <b>Pinch hazard</b><br><b>Possible pinch hazard with mechanical moving parts.</b><br><b>This label warns the user of eventual mechanical hazards</b>        |
|  | <b>General warning</b><br><b>This label warns the user from eventual hazards with possible injury or damage to the system.</b>                              |

### 2.2 Personnel qualifications

All installation or maintenance operations should only be carried out by persons who have suitable technical training and the necessary experience.

## 2.3 Safety information

### 2.3.1 Process gases

|                                                                                   |                                                                                                                                                                                                                                                                                                          |
|-----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <p style="text-align: center;"><b>⚠ WARNING</b></p> <p>Toxic gases<br/>Some gases are toxic and inhaling them should be avoided.<br/>Take steps to ascertain whether or not the gases being used are known toxic substances. Refer to the Material Safety Data Sheets covering the gases in question</p> |
|-----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### 2.3.2 Burn hazard

|                                                                                   |                                                                                                                                                                                                                                                                                                                     |
|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <p style="text-align: center;"><b>⚠ WARNING</b></p> <p>Burn hazard<br/>Substrate and quartz window may be at high temperature.<br/>After manual heating of after a process both substrate and quartz window may still be at high temperature.<br/>Wait for sufficient cool down of the substrate for unloading.</p> |
|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### 2.3.3 Electrical shock hazard

|                                                                                     |                                                                                                                                                                                                                                                                                                            |
|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <p style="text-align: center;"><b>⚠ WARNING</b></p> <p>Electrical shock hazard<br/>The power supply line and the furnace circuit breakers must be switched off and locked before starting maintenance operations.<br/>These operations must be carried out by trained and authorized technicians only.</p> |
|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### 2.3.4 Eyes damage

|                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                          |
|-------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <p style="text-align: center;"><b>⚠ WARNING</b></p> <p>Possible eye damage<br/>The lamp furnace emits strong visible and infrared light during operations. Some light can be seen from the rear exhaust duct.<br/>Never look furnace light from the rear of the system during operation. This may lead to eye damage. A warning label affixed on the backside lid warns the user of potential hazard</p> |
|-------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

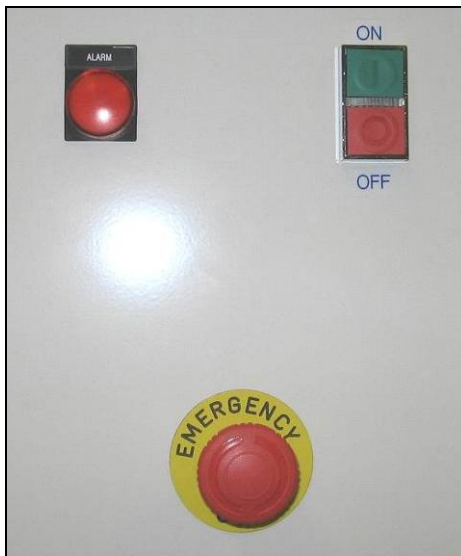


### 3. SYSTEM START-UP

Before you start-up the system, you must check that all facilities are available:

- Electricity
- Water with a pressure of 2 to 4 bars
- Compressed air with a pressure of 6 bars
- Purge gas with a pressure of 1 bar
- Process gases with a pressure of 1 to 2 bars

Then close the rear panel circuit breaker and switch-on the system using the front panel green push button.



When the operator closes the general circuit breaker located at the rear panel, the PLC runs a diagnostic routine that takes several seconds. After the initialization cycle is completed the alarm indicator turns to red and the system can be switched on using the green push button located at the front panel. The alarm can be reset after powering up the system by clicking on the alarm indicator on the software screen. Otherwise it will be automatically reset when the first process is performed.

After powering ON the system, it is necessary to download the configuration and tables from the PC to the PLC. This is done by downloading a recipe. Any recipe is fine.

Each time the system is switched off (by the red push button); the alarm indicator turns to red to inform the user of a system power off.

When the system is switched on, the ON indicator lights on.

Note: If the system cannot be switched on check if the Emergency button is in pushed position. In this case, it can be released by turning it in clockwise direction.

If the system is installed at an altitude that is not closed to the sea level then the atmospheric pressure must be set in the configuration mode. The pressure of the chamber must reach the atmospheric pressure to unlock the chamber. If the atmospheric pressure is lower than the threshold value the chamber cannot be opened. The atmospheric pressure can be set in the configuration mode.

#### 4. SOFTWARE CONTROL ARCHITECTURE

The control system of the AS-One is made of:

- a PC which is the interface between the user and the equipment
- a PLC that controls process operation and safeties

The operator interface is a PC running under Windows XP Professional.

The PC and the PLC are connected by an Ethernet communication line.

A RS485 communication line between the PLC and the temperature controller allows entering the temperature control parameters to the temperature controller directly from the computer.

This control architecture offers high reliability and easy operations. The AS-One can run the process under safe conditions even if there is a problem with the PC.

The computer must be connected to the RTP furnace and the furnace must be powered to allow communication with the PC.

Start the software using the icon that has been automatically installed during the software setup.



## 5. SOFTWARE ACCESS

When the software starts the computer displays the home page and requests the access information. The user must enter his login and his password to have access to the software functions.



The software has 4 access levels:

- Operator mode
- Engineering mode
- Administrator mode
- Maintenance mode

Each access level requires a login and a password. Logins and passwords are case insensitive.

Each user can be assigned a private folder in which recipes and historicals files are stored. Only the administrator is granted to assigning user private folders.

The user must exit from the software to change the access level.

The software can also be used in demonstration mode without connection to the furnace. This feature is useful to prepare a recipe and for process historical analysis on another computer.

**Additional run times for off-line computers are free of charge.**

### 5.1 Operator mode

Access to the operator mode requires a login and the associated password. The operator's login and password are selected in the configuration mode. If the operator information has not been provided, the access is not possible. The operator mode gives access to:

- Process mode
- Manual mode : visualization only
- Process historicals and alarm list

Default operator login and password at the system delivery:

- Login: "ope"
- Password: "oooo"

### 5.2 Engineering mode

The access to engineering mode requires a login and the associated password.

This mode gives access to all the software options except maintenance mode and special administrator mode options.

Default engineer login and password at the system delivery:

- Login: "eng"
- Password: "eeee"

### 5.3 Maintenance Mode

This mode is reserved to Annealsys engineers for maintenance and special diagnostic operations.

### 5.4 Administrator mode

The access to the administrator mode requires a login and the associated password.

|                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <b>⚠ CAUTION</b>                                                                                                                                                                                                                                                                                                                                                                                                                       |
|                                                                                     | <b>ADMINISTRATOR MODE</b><br><b>This mode gives access to all the software options including the maintenance mode. It allows direct control of the PLC outputs without any safety interlocks.</b><br><b>The password and login must be changed when the system is installed and the access to this mode restricted to people who will not make any maintenance operation without the agreement of Annealsys or its representative.</b> |

In addition of the maintenance mode the administrator mode gives access to the administration of the login and password list.

Default administrator login and password at the system delivery:

- Login: "adm"
- Password: "aaaa"

**It is strongly recommended to change the administrator password at system installation.**

## 5.5 Demonstration mode

If this mode is selected the software runs without any communication with the PLC. This allows using the software for training or on another computer for recipe edition and process historicals visualization. Login and password are still required.

Check the "Demonstration mode" box on the software home page to run the software in demonstration mode. The demonstration mode gives access to the different functions.

Some functions that require communication with the system (process, manual mode and temperature controller) are not active in the manual mode but can be displayed.

Any file modified or created using the demonstration mode must be transferred to the AS-One computer to be used by the system.

**Important notice: The "Demonstration mode" box must be unchecked to use the furnace.**

## 6. SOFTWARE NAVIGATION

When the software starts the computer displays the downloading form:

| Temperatures |             | Gas (scm) | Setpoint | Readout | Gauge name | mBar      | Recipe parameters |             | Machine parameters |  |     |
|--------------|-------------|-----------|----------|---------|------------|-----------|-------------------|-------------|--------------------|--|-----|
| Setp. (%)    | 0.0         | N2        | 0.0      | 0.0     | Full range | 0.500E+00 | Cooling           | Not cooling | Idle               |  | Com |
| Power %      | 1.0         | O2        | 0.0      | 0.0     | Capa 1     | 1.33E+01  | Step time         | 0.0         | System ON          |  |     |
| TC1 °C       | Not present |           |          |         |            |           | Recipe time       | 0.0         | POWER              |  |     |
| Pyro1 °C     | 104.0       |           |          |         |            |           | Step number       | 0           |                    |  |     |
| TC2 °C       | Not present |           |          |         |            |           | Recipe name       | RTA_1000    | Eng                |  |     |
|              |             |           |          |         |            |           | Open door         | Safe        | Engineer           |  |     |

| < Last downloaded recipe >         |                         | < Recipe to download >             |                         |
|------------------------------------|-------------------------|------------------------------------|-------------------------|
| Name                               | RTA_1000                | Name                               | RTA_1000                |
| Date of last download              | 11/05/2007              | Creation date                      | 03/05/2007 - 09:30:21   |
| Hour of last download              | 15:18:17                | Modification date                  | 09/05/2007 - 17:22:22   |
| User name                          | Adm                     | User name                          | Adm                     |
| PID table                          | SUPyroHT400-1000        | PID table                          | SUPyroHT400-1000        |
| Thermocouple calibration table     | Default_TC_Table_4-20mA | Thermocouple calibration table     | Default_TC_Table_4-20mA |
| Pyrometer 1 calibration table (HT) | CALSU                   | Pyrometer 1 calibration table (HT) | CALSU                   |

RTA test 1000C 60s

RTA test 1000C 60s

Door locked

Start purge

**Turbo pump**

State : Stopped

Start

|         |         |             |               |             |           |
|---------|---------|-------------|---------------|-------------|-----------|
| Recipes | Process | Historicals | Configuration | Manual Mode | Shut down |
|---------|---------|-------------|---------------|-------------|-----------|

Software version : 1.5.0  
PCD Version : 1.5.0

The screen is divided in 4 main parts:

- The upper part of the screen is the system status panel. It displays the main system parameters
- The center part of the screen is the information panel. It displays the selected form with related action buttons and information
- The lower part of the screen is the primary navigation panel. It contains buttons to select the desired information panel.
- The right part of the screen is the secondary navigation panel. It contains buttons to select the desired sub-information panel. It may also contain commands.

The software gives full access to the different modes at any time and provides very friendly utilization.

## 6.1 Main system status panel

The upper part of the screen displays the main system parameters.

| Temperatures |             | Gas (sccm) | Setpoint | Readout | Gauge name | mBar     | Recipe parameters |             | Machine parameters | Com |
|--------------|-------------|------------|----------|---------|------------|----------|-------------------|-------------|--------------------|-----|
| Setp. [%]    | 0.0         | N2         | 0.0      | 0.0     | Full range | 6.58E+00 | Cooling           | Not cooling | Idle               |     |
| Power %      | 1.0         | O2         | 0.0      | 0.0     | Capa 1     | 1.33E+01 | Step time         | 0.0         | System ON          | Com |
| TC1 °C       | Not present |            |          |         |            |          | Recipe time       | 0.0         | POWER              |     |
| Pyro1 °C     | 104.0       |            |          |         |            |          | Step number       | 0           |                    |     |
| TC2 °C       | Not present |            |          |         |            |          | Recipe name       | RTA_1000    | Eng                |     |
|              |             |            |          |         |            |          | Open door         | Safe        | Engineer           |     |

Parameters in the left column:

| Temperatures |             |
|--------------|-------------|
| Setp. [%]    | 0.0         |
| Power %      | 1.0         |
| TC1 °C       | Not present |
| Pyro1 °C     | 104.0       |
| TC2 °C       | Not present |

- Setpoint %: Heating setpoint (temperature or power)
- Power %: Readout of the power to the lamps
- TC1 °C: Temperature from thermocouple 1 in °C (\*)
- Pyro1 °C: Temperature from pyrometer 1 in °C
- TC2 °C: Temperature from thermocouple 2 in °C (\*)
- Pyro2 °C: Temperature from pyrometer 2 in °C (Option)

(\*) Note that "Not present" can mean that the TC is broken.

Parameters in the second column:

| Gas (sccm) | Setpoint | Readout |
|------------|----------|---------|
| N2         | 0.0      | 0.0     |
| O2         | 0.0      | 0.0     |

- Gas 1 name, gas 1 setpoint in sccm, gas 1 readout in sccm
- Gas 2 name, gas 2 setpoint in sccm, gas 2 readout in sccm
- Gas 3 name, gas 3 setpoint in sccm, gas 3 readout in sccm
- Gas 4 name, gas 4 setpoint in sccm, gas 4 readout in sccm
- Gas 5 name, gas 5 setpoint in sccm, gas 5 readout in sccm

(\*) Number of lines displayed depends on lines installed on the system and set in the configuration mode.

Parameters in the third column:

The third column displays the values from the different pressure sensors (up to 4). As a standard feature the system is equipped with a constraint gauge. Capacitance manometer is optional.

**Important notice:** When the system is equipped with a turbo pump, the gauge 1 is a full range gauge. As this gauge is filament type it must be isolated from process gases. The valve GIV (Gas isolation valve) isolates the full range gauge when gases are introduced inside the process chamber. A rough pumping step of more than 30 seconds is necessary for the isolation valve to open.

When the valve is closed the pressure from the full range gauge is not the pressure of the chamber. In order for the operator to be informed that the valve is closed and that the pressure indication is not the pressure of the reactor, the background of the pressure value turns to red. It turns to black when the valve opens. The isolation valve must be open to start a turbo pumping step.

Pressure values (isolation valve closed)

| Gauge name | mBar     |
|------------|----------|
| Full range | 6.58E+00 |
| Capa 1     | 1.33E+01 |

Pressure values (isolation valve open)

| Gauge name | mBar     |
|------------|----------|
| Full range | 3.10E-01 |
| Capa 1     | 1.00E+00 |

Both gauge names and unit can be configured. (See configuration at chapter 11)

Parameters in the fourth column:

| Recipe parameters |                                |   |
|-------------------|--------------------------------|---|
| Cooling           | Not cooling                    |   |
| Step time         | 0.0                            | ← |
| Recipe time       | 0.0                            | ← |
| Step number       | 0                              |   |
| Recipe name       | TEST_cool_down                 |   |
| Open door         | Chamber is not at ATM pressure |   |

Count down  
Progress bars

- Cooling information:
  - Not cooling
  - Remaining cooling time after using the heating in process or manual mode. The time is set to 180 s during manual heating and process and count down progress bar is displayed
  - Cooling and waiting for other condition than timer count down (temperature condition)
- Step time: remaining time to the end of the step (process mode only)
- Recipe time: Only active in process mode. This is the remaining estimated time to the end of the process. It does not include time to reach temperature or pressure end of step conditions, or operator actions on hold or next step buttons.
- Step number: number of the current process step (process mode only)
- Recipe name (last recipe downloaded to the furnace PLC)
- Open door: Give information on the condition status that allow or not to unlock the chamber

Parameters in the fifth column:

| Machine parameters |     |
|--------------------|-----|
| Idle               | Com |
| System ON          |     |
| POWER              |     |
| Eng                |     |
| Engineer           |     |

- Machine state: "Idle" or "Process"
- Main switch status: "System ON" or "System OFF"
- Temperature control mode: "POWER", "TC1", "PYRO1" or "PYRO2" (Option)
- Login of the current user ("Eng" in the example)
- Login mode: "Operator", "Engineer", "Administrator" or "Maintenance"

At the upper right: Com is communication status indicator with a flashing frequency of 2 Hz:

- Green when the Ethernet communication between the PC and the machine is OK.
- Red when there is no communication between the PC and the machine
- Blue when the PC software is used in demonstration mode

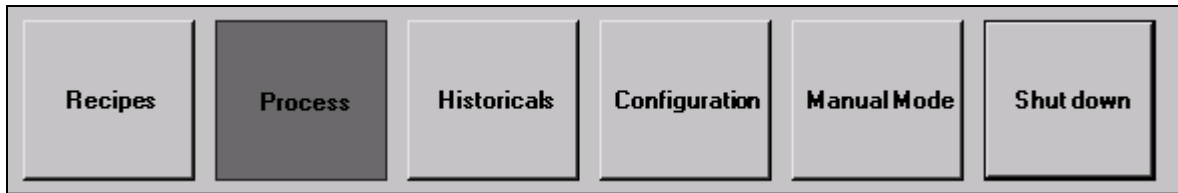
The green square in the lower right angle is the alarm signal. It is green when there is no alarm and turns to red when an alarm occurs.

**When the furnace is powered on (after PLC initialization), or when system is powered off (button on front panel), the software displays an alarm. The alarm indicator turns red. The alarm can be reset after powering up the system by clicking on the alarm indicator. Otherwise it will be automatically reset when the first process is performed.**

## 6.2 Information panel

This is the main central part of the screen. It changes depending on primary and secondary navigation selection.

## 6.3 Primary navigation panel



Software functions are chosen by selecting them with mouse and pressing the left button.

The main navigation panel offers the following functions:

- **Recipes** to create, modify, display or delete a recipe
- **Process** to download a recipe and to run the process
- **Historicals** to display process historicals and alarms list
- **Configuration** to enter the calibration tables and the configuration parameters
- **Manual Mode** for manual control of the system and temperature control adjustment
- **Shut-down** to exit from the control software and return to Windows

The different functions are described hereafter.

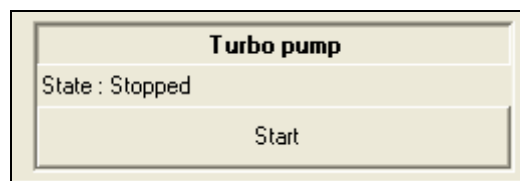
## 6.4 Secondary navigation panel

At the right part of the page are some additional buttons. These buttons are secondary navigation buttons. They are used to select sub-functions.

The secondary navigation is described hereafter for each mode.

## 6.5 Turbo pump option

If the system is equipped with the turbo pump option it is possible to start and stop the turbo pump from the downloading page. This gives access to the turbo pump control by the operator that has no access to the manual mode control.



## 6.6 Software version

The PLC and software versions are displayed on the bottom right side of the screen.



## 6.7 Print screen

Right-clicking on the context menu of the status bar pops up the context menu from which we can select the "Print screen" function.

If the SHIFT button is held down, the function saves the current screen shot to a file in BITMAP format.



## 7. RECIPES EDITION

For the AS-One each recipe is made of:

- A name
- An associated thermocouple calibration table
- An associated pyrometer 1 calibration table
- An associated pyrometer 2 calibration table (If option is installed)
- An associated PID parameter table
- A comment (up to 100 characters)
- Up to 100 steps

The above recipe parameters (except steps) are the recipe header and they will remain the same for the whole duration of the recipe.

When the recipe mode is selected the screen displays the recipe header:

| Temperatures |             | Gas (scon) | Setpoint | Readout | Gauge name | mBar     | Recipe parameters |             | Machine parameters |  |     |
|--------------|-------------|------------|----------|---------|------------|----------|-------------------|-------------|--------------------|--|-----|
| Setp. [°C]   | 0.0         | N2         | 0.0      | 0.0     | Full range | 4.92E+00 | Cooling           | Not cooling | Idle               |  | Com |
| Power %      | 1.0         | O2         | 0.0      | 0.0     | Capa 1     | 1.33E+01 | Step time         | 0.0         | System ON          |  |     |
| TC1 °C       | Not present |            |          |         |            |          | Recipe time       | 0.0         | POWER              |  |     |
| Pyro1 °C     | 199.9       |            |          |         |            |          | Step number       | 0           |                    |  |     |
| TC2 °C       | Not present |            |          |         |            |          | Recipe name       |             | Adm                |  |     |
|              |             |            |          |         |            |          | Open door         | Safe        | Administrator      |  |     |

< Recipe Header Information >

**System** AS-ONE

Creation date 03/05/2007 - 09:30:21

Modification date 03/05/2007 - 10:14:16

**User Name** Adm

**Recipe name** RTA\_1100

Thermocouple calibration table Default\_TC\_Table\_0-10V

Pyrometer 1 calibration table (HT) CALS1

PID control parameter table SITC

< Recipe comments >

RTA test 1100C 60s

< Commands >

New Browse... Delete Print Save as Save

Header

Steps

Profile

Recipes

Process

Historicals

Configuration

Manual Mode

Shut down

Software version : 1.5.0  
PCD Version : 1.5.0

At the right side of the page there are 3 buttons for the secondary navigation.

At the lower part of the header there are 6 commands buttons that are used for the recipe management.

### 7.1 Recipe mode secondary navigation buttons

The recipe mode secondary navigation buttons are:

- Header: To display the recipe header
- Steps: To display the step parameters
- Profile: To display the heating graphic profile of the recipe

## 7.2 Recipe header command buttons

There are several command buttons in the bottom of the recipe header form.



### 7.2.1 New

This button is used to create a new recipe. Clicking on "New" will reset all the recipe parameters and the user can create a new recipe selecting each parameter.

### 7.2.2 Browse

To open any folder on the computer to load a recipe

This command allows each user having his own recipe folder.

### 7.2.3 Delete

To delete the recipe that is currently displayed on the screen

### 7.2.4 Print

To print the recipe header only

Go to step mode to print the recipe header with all the step parameters.

### 7.2.5 Save as

To save the displayed recipe with a new name

The "Save as" command can also be used to save a new recipe in a folder that is different from the default folder.

### 7.2.6 Save

"Save" will save a new recipe in the default recipe folder: "c:\Program Files\Annealsys\Recipes".

## 7.3 Recipe header parameters

When operator enters the recipes mode he can:

- Select one of the recipes in the default recipe list (using the "Recipe name" dropdown box arrow)
- Select a recipe in any folder of the computer using the "Browse" at the bottom of the page.
- Create a new recipe selecting the new button at the bottom of the page

When an existing recipe is selected, it can then be modified, it is also possible to save it with a new name ("Save as" command) and to modify the new recipe created by this way.

When the "New recipe" command is selected all the parameters are reset.

### 7.3.1 Recipe name

The recipe name can have up to 20 characters. This is the name under which the recipe is saved as "name.rec".

The extension of the recipe files is ".rec".

### 7.3.2 Associated thermocouple calibration table

To each recipe is associated a thermocouple calibration table. This feature is useful in special applications.

**The system is provided with K type thermocouples and K type thermocouple converters as standard features. The K type thermocouple calibration table is set at Annealsys factory using a thermocouple calibrator. It should usually not be modified by the operator.**

For special applications and requirements about thermocouple calibration table contact Annealsys.

"tcc" is the extension for the thermocouple calibration table files.

|                                                                                   |                                                                                                                                                                                                                                       |
|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <b>⚠ CAUTION</b>                                                                                                                                                                                                                      |
|                                                                                   | <p><b>Thermocouple calibration table.</b><br/> <b>A wrong thermocouple calibration table can lead to unexpected thermal behavior of the system.</b><br/> <b>Always take great care to enter the calibration table parameters.</b></p> |

### 7.3.3 Associated pyrometer 1 calibration table

The AS-One is always delivered with one pyrometer which is pyrometer 1.

To each recipe is associated a pyrometer 1 calibration table. The pyrometer temperature measurement depends on substrate type as well as some process conditions. This function allows having the right calibration table associated to the recipe for the process and for the right type of substrate. The pyrometer 1 calibration tables are created and saved using the configuration mode.

"cal" is the extension for the pyrometer calibration table files.

### 7.3.4 Associated pyrometer 2 calibration table

The pyrometer 2 is an optional feature. The pyrometer 2 calibration table will be proposed to the user only if the pyrometer 2 option is installed and selected in the hardware configuration mode.

To each recipe is associated a pyrometer 2 calibration table. The pyrometer temperature measurement depends on substrate type as well as some process conditions. This function allows having the right calibration table associated to the recipe for the process and for the right type of substrate. The pyrometer 2 calibration tables are created and saved using the configuration mode.

"cal" is the extension for the pyrometer calibration table files.

### 7.3.5 Associated PID parameter table

To each recipe is associated a PID parameters table. The temperature control parameters depend on the process conditions and the substrate type (susceptor or not for example).

This function allows having the right temperature control parameters associated to the recipe for the process and for the type of substrate. The PID parameters tables are created and saved using the configuration mode.

"pid" is the extension for the PID table files.

### 7.3.6 Comment (up to 100 characters)

The operator can enter a 100-character comment for the description of the recipe.

## 7.4 To create a new recipe

Procedure to create a new recipe:

- Click on the "New" button. The system resets the header parameters and displays a default name made with the current date and time like: "new\_20051129140917"
- Enter a new name if you want a different name than the default one
- Select a thermocouple calibration table
- Select a pyrometer 1 calibration table
- Select a pyrometer 2 calibration table (if the option is installed)
- Select a PID table (according to process conditions and sample)
- Enter a comment (not mandatory)
- Click on "Save" to save the recipe in the default folder or on "Save as" to save it in a different folder
- Click on "Steps" in the secondary navigation panel and enter the step parameters
- Save again when all parameters have been set

## 7.5 To modify a recipe

Procedure to modify a recipe:

- Select the recipe in the default folder using the dropdown list or use the "Browse" to load a recipe from a different folder.
- Change one of the header parameter if needed
- Click on "Steps" in the secondary navigation buttons and enter the step parameters
- Change the step parameters if needed
- Save the recipe

## 8. STEPS EDITION

Click on the "Steps" button in the recipe secondary navigation panel.

The screen displays the step parameters (for readability convenience, status and primary navigation panels have been removed from the following view):

| Recipe name                                                                                                                                                                                                                                                                                                              |            | RTA_1000   |            |               |            |            |  |  |  |  |  |  |  |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|------------|------------|---------------|------------|------------|--|--|--|--|--|--|--|
| Step number                                                                                                                                                                                                                                                                                                              | < Step 1 > | < Step 2 > | < Step 3 > | < Step 4 >    | < Step 5 > | < Step 6 > |  |  |  |  |  |  |  |
| Step comment                                                                                                                                                                                                                                                                                                             | Pumping    | Purge      | Pumping    | Fill with gas | Pre-heat 1 | Pre-heat 2 |  |  |  |  |  |  |  |
| End of step condition                                                                                                                                                                                                                                                                                                    | Duration   | Duration   | Duration   | Duration      | Duration   | Duration   |  |  |  |  |  |  |  |
| Value                                                                                                                                                                                                                                                                                                                    | 30 s       | 30 s       | 30 s       | 100 s         | 10 s       | 20 s       |  |  |  |  |  |  |  |
| Temperature control                                                                                                                                                                                                                                                                                                      | TC (°C)    | TC (°C)    | TC (°C)    | TC (°C)       | Power (%)  | Power (%)  |  |  |  |  |  |  |  |
| Setpoint - Alarm                                                                                                                                                                                                                                                                                                         | 0          | 0          | 0          | 0             | 100        | 100        |  |  |  |  |  |  |  |
| Vacuum                                                                                                                                                                                                                                                                                                                   | Rough pump | No pumping | Rough pump | No pumping    | No pumping | No pumping |  |  |  |  |  |  |  |
| Purge valve                                                                                                                                                                                                                                                                                                              | off        | ON         | off        | off           | off        | off        |  |  |  |  |  |  |  |
| Process gas valve 1                                                                                                                                                                                                                                                                                                      | off        | off        | off        | off           | off        | off        |  |  |  |  |  |  |  |
| N2 - Alarm                                                                                                                                                                                                                                                                                                               | 0          | 0          | 0          | 1000          | 100        | 100        |  |  |  |  |  |  |  |
| O2 - Alarm                                                                                                                                                                                                                                                                                                               | 0          | 0          | 0          | 0             | 0          | 0          |  |  |  |  |  |  |  |
| <div style="display: flex; justify-content: space-between;"> <span>&lt; Commands &gt;</span> <div> <span>&lt;&lt; First</span> <span>&lt; Previous</span> <span>Next &gt;</span> <span>&gt;&gt; Last</span> <span>Copy</span> <span>Insert</span> <span>Delete</span> <span>Print</span> <span>Save</span> </div> </div> |            |            |            |               |            |            |  |  |  |  |  |  |  |

Header  
  
Steps  
  
Profile

**Note:** Only one step is enabled (the one with white boxes). The 2 previous steps and 3 following steps (with grey boxes) are disabled and only displayed for a better readability of the recipe.

Each recipe can have up to 100 steps. For each step the operator can select:

- Comment (up to 20 characters)
- End of step condition
- End of step condition value
- Temperature control mode
- Temperature setpoint
- Temperature band alarm
- To make vacuum or not
- To purge the chamber or not
- Gas flow setpoints, ramping option and gas flow band alarm

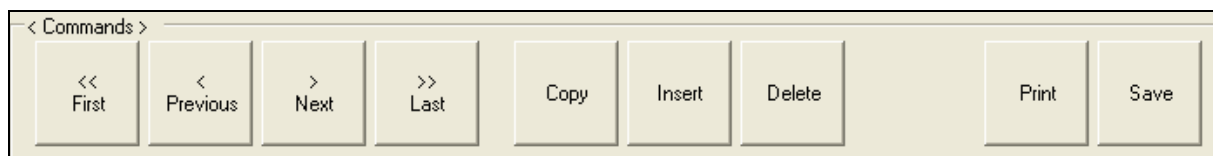
And as options:

- Pumping with turbo pump
- Pressure setpoint (if pressure control with throttle valve is installed)

The step edition form displays the current step, the 2 previous steps and the 3 next steps. This allows better view of the sequence of the recipe steps.

### 8.1 Step edition form buttons

Several command buttons in the lower part of the form allow navigation inside the recipe and some additional functions that are described hereafter.



#### 8.1.1 First step

To display the first step as current step in the step editor

#### 8.1.2 Previous Step

Shortcut <Ctrl+B>. To display the previous step as current step in the step editor

#### 8.1.3 Next step

Shortcut <Ctrl+N>. To display the next step as current step in the step editor.

#### 8.1.4 Last step

To display the last step as current step in the step editor

#### 8.1.5 Copy step

To copy all the parameter values from previous step to the current step. All the parameters of the step will be then set to the same values than the parameters of the previous step except for the comment. This function allows faster recipe edition by copying and just modifying the requested parameters.

#### 8.1.6 Insert step

To insert a step before the current step **CAUTION: The step 100 will be deleted.**

#### 8.1.7 Delete step

To delete the current step A new step 100 will be created with all parameters set to zero.

#### 8.1.8 Print

To print the recipe header with all the steps and their parameters. In the recipe header mode the print command only prints the recipe header.

#### 8.1.9 Save

To save the recipe file

## 8.2 Step parameters

The user can set the parameters of the current step which is the only one enabled in the step editor.

| Recipe name RTA_1100  |            |            |            |
|-----------------------|------------|------------|------------|
| Step number           | < Step 1 > | < Step 2 > | < Step 3 > |
| Step comment          | Pumping    | Purge      | Pumping    |
| End of step condition | Duration   | Duration   | Duration   |
| Value                 | 30 s       | 30 s       | 30 s       |
| Temperature control   | Power (%)  | Power (%)  | Power (%)  |
| Setpoint - Alarm      | 0 —        | 0 —        | 0 —        |
| Vacuum                | Rough pump | No pumping | Rough pump |
| Purge valve           | off        | ON         | off        |
| Process gas valve 1   | off        | off        | off        |
| N2 - Alarm            | 0 — —      | 0 — —      | 0 — —      |
| O2 - Alarm            | 0 — —      | 0 — —      | 0 — —      |
| Pressure              | Angle (°)  | Angle (°)  | Angle (°)  |
| Setpoint - Alarm      | 0 —        | 0 —        | 0 —        |

The step parameters are:

- Step comment
- End of step condition (duration, pressure or temperature) and value
- Temperature control mode and setpoint
- Temperature band alarm
- Vacuum
- Purge
- Gas setpoints, ramping option and band alarm
- Pressure control (option)

### 8.2.1 Step comment

A 20 characters comment can be associated to each step.

This comment is displayed on the screen during the process. It allows an easier follow up of the process sequence when the process is performed.

### 8.2.2 End of step condition

The user can choose different conditions to end the step and to switch to the next step. The user can select one of the following options from the list (activated with the arrow):

- Duration: Step duration
- Pressure: Pressure to be reached
- Temp. TC1: Temperature to be reached by TC1
- Temp. Pyro1: Temperature to be reached by Pyro1
- Temp. Pyro2: Temperature to be reached by Pyro2 (option)

The default value is "Duration".

A dropdown menu with a downward arrow on the right. The menu is open, showing a list of options: 'Duration' (highlighted in blue), 'Pressure', 'Temp. TC1', and 'Temp. Pyro1'.

### 8.2.2.1 Step duration

The step will end when the duration set for the step is elapsed.

- Step duration range: 0 to 9999.9 s

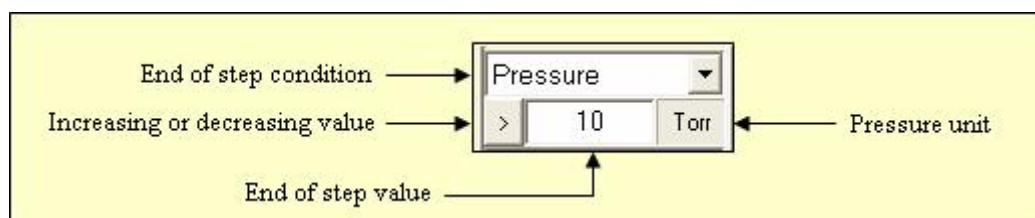
The step duration can be increased by increment of 0.1 second. The default value for the step duration is "0". If the operator enters a value out of range it is automatically reset and a message with the right range of values is displayed.

**The minimum step duration is 0.4 second. Even if shorter step duration is set in the recipe the system will run a 0.4 second step. When the recipe switches to one step to the next one during the process, there are a lot of operations to be performed by the system (temperature and gas setpoint to be set, vacuum conditions to be checked and control of valves, etc.). This is the limitation for step duration.**

For step duration of more than 0.4 second, the duration can be set by increment of 0.1 second.

### 8.2.2.2 Pressure to be reached

When this option is selected, the step ends when the pressure reaches the end of step condition value. The system must not be in pressure control mode; if it is then the process stops immediately and the error message "End step pressure condition under pressure control is not allowed" is displayed to the operator.



The pressure to be reached to end the step can be set into the end of step condition value box.

- Pressure range: 1.00E0 to 1.00E5 Pa

If the operator enters a value out of range it is automatically reset and a message with the right range of values is displayed.

The user can select pressure increasing or pressure decreasing end of step condition by clicking on the ">" sign to switch it to "<". The parameters for the pressure end of step condition are:

- P0: Pressure when steps starts
- PS: Pressure threshold value to be reached at the end of the step
- PC: Current pressure during the step

The step will end according to the conditions describes in the following table:

| Condition           | Case         | Step ends               | Comment                        |
|---------------------|--------------|-------------------------|--------------------------------|
| Pressure increasing | $P0 \geq PS$ | Immediately             | Alarm message and stop process |
|                     | $P0 < PS$    | As soon as $PC \geq PS$ |                                |
| Pressure decreasing | $P0 \leq PS$ | Immediately             | Alarm message and stop process |
|                     | $P0 > PS$    | As soon as $PC \leq PS$ |                                |

Example:

| Condition           | P0       | PS       | Step ends                     | Comment                        |
|---------------------|----------|----------|-------------------------------|--------------------------------|
| Pressure increasing | 500 mbar | 400 mbar | Immediately                   | Alarm message and stop process |
| Pressure increasing | 500 mbar | 900 mbar | As soon as $PC \geq 900$ mbar |                                |
| Pressure decreasing | 500 mbar | 900 mbar | Immediately                   | Alarm message and stop process |
| Pressure decreasing | 500 mbar | 20 mbar  | As soon as $PC \leq 20$ mbar  |                                |

**The operator must select a pressure that is compatible with the vacuum pump equipment and the vacuum gauges installed on the system.**

The pressure unit is indicated at the right of the end of step value. This is the standard pressure unit which is set in the configuration mode.

If the system has no turbo pump option the pressure for the end of step condition is measured by the constraint vacuum gauge.

If the system is equipped with a turbo pump then the pressure for the end of step condition is measured by:

- Full range gauge if the isolation valve (GIV) is open
- Capacitance manometer (CAPA1) if the full range isolation valve (GIV) is closed

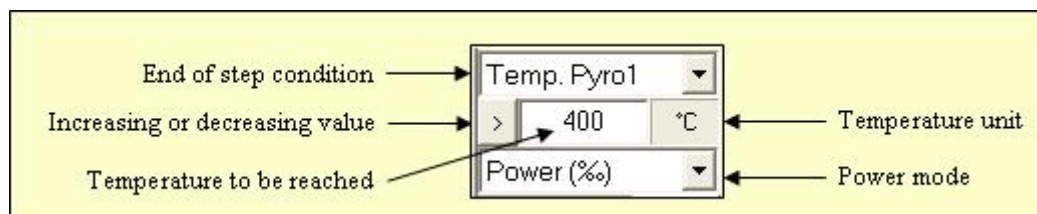
|                                                                                   |                                                                                                                                                                                                                                                                                                                  |
|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <b>⚠ CAUTION</b>                                                                                                                                                                                                                                                                                                 |
|                                                                                   | <p><b>End of step condition.</b></p> <p><b>Possible long heating or overheating of the substrate</b></p> <p><b>If the end of step condition is set to pressure the temperature or power setpoint is set to a fixed value for the duration of the step until the system reaches the end of step pressure.</b></p> |

#### 8.2.2.3 Temperature to be reached

When the operator selects the temperature to be reached as end of step condition, the step ends as soon as the temperature value is reached.

When the temperature end of step condition is selected the heating control mode is automatically set to power mode (no temperature control).

The operator can select any of the temperature control sensor to measure the temperature for the end of step condition: TC1, Pyro1 or Pyro2 (option).



The user can select temperature increasing or temperature decreasing end of step condition by clicking on the ">" sign to switch it to "<".

The temperature to be reached to end the step can be set into the end of step condition value box.

- Temperature range: 0 to 1400°C

If the operator enters a value out of range it is automatically reset and a message with the right range of values is displayed.

|                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <b>⚠ CAUTION</b>                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                                                                                     | <p><b>The system may not reach the end of step condition</b></p> <p><b>The temperature ranges are subject to change depending on the substrates nature and on the temperature calibration tables. They must be checked by the user to be sure that the system will be able to measure it and to reach it.</b></p> <p><b>For example the end of step temperature must not be set to 250°C if the selected sensor is the high temperature pyrometer.</b></p> |



The operator must select a temperature that is compatible with the temperature sensor that is selected for the end of step condition. The sensor temperature ranges are usually:

- Thermocouple (K type): 0 to 1000°C
- Low temperature pyrometer: 150°C to 1100°C
- High temperature pyrometer: 400°C to 1400°C

The parameters for the temperature end of step condition are:

- T0: Actual temperature when step starts
- TS: Temperature target value to end the step
- TC: Current temperature

The step will end according to the conditions described in the following table:

| Condition    | Case         | Step ends               | Comment                               |
|--------------|--------------|-------------------------|---------------------------------------|
| Ramping up   | $T0 \geq TS$ | Immediately             | Alarm message and stop process        |
|              | $T0 < TS$    | As soon as $TC \geq TS$ |                                       |
| Ramping down | $T0 \leq TS$ | Immediately             | Alarm message and stop process        |
|              | $T0 > TS$    | As soon as $TC \leq TS$ | Maximum duration of the step = 10 mn* |

*\* The step will end automatically after 10 minutes if the sensor cannot reach the target temperature in order to keep the system running. This may happen when using the pyrometer for low temperature end of step condition or if the heating power has not been set to a sufficient value.*

Examples:

| Condition    | T0    | TS    | Step ends                              | Comment                         |
|--------------|-------|-------|----------------------------------------|---------------------------------|
| Ramping up   | 300°C | 280°C | Immediately                            | Alarm message and stop process  |
| Ramping up   | 250°C | 300°C | As soon as $TC \geq 300^\circ\text{C}$ |                                 |
| Ramping down | 250°C | 300°C | Immediately                            | Alarm message and stop process  |
| Ramping down | 250°C | 100°C | As soon as $TC \leq 300^\circ\text{C}$ | End after maximum of 10 minutes |

### 8.2.3 Temperature control mode and setpoint

The temperature and the power can be ramp or dwell. The setpoint calculation will be different if the control mode changes from step n-1 to step n and will also depend on the end of step condition.

The heating setpoint is updated every 100 ms by the PLC.

The heating setpoint is calculated according to the following rules:

| Step n-1 control mode | Step n control mode | Condition for end of step n | Initial heating setpoint for step n     | Final heating setpoint for step n      |
|-----------------------|---------------------|-----------------------------|-----------------------------------------|----------------------------------------|
| Temperature control   | Temperature control | Duration                    | Recipe temperature setpoint of step n-1 | Recipe temperature setpoint for step n |
| Temperature control   | Temperature control | Pressure                    | Recipe temperature setpoint for step n  | Recipe temperature setpoint for step n |
| Power mode            | Temperature control | Duration                    | Actual temperature at end of step n-1   | Recipe temperature setpoint for step n |
| Power mode            | Temperature control | Pressure                    | Recipe temperature setpoint for step n  | Recipe temperature setpoint for step n |
| Temperature control   | Power mode          | Duration                    | Actual power at end of step n-1         | Recipe power setpoint for step n       |
| Temperature control   | Power mode          | Pressure                    | Recipe power setpoint for step n        | Recipe power setpoint for step n       |
| Power mode            | Power mode          | Duration                    | Recipe power setpoint for step n-1      | Recipe temperature setpoint for step n |
| Power mode            | Power mode          | Pressure                    | Recipe power setpoint for step n        | Recipe power setpoint for step n       |
| Power mode            | Power mode          | Temperature                 | Recipe power setpoint for step n-1      | Recipe power setpoint for step n       |

The user first selects the temperature control mode:

- Power ‰(open loop)
- TC1 in °C
- Pyro1 in °C
- Pyro2 in °C (option)

|                     |             |             |             |
|---------------------|-------------|-------------|-------------|
| Temperature control | Pyro 1 (°C) | Pyro 1 (°C) | Pyro 1 (°C) |
| Setpoint - Alarm    |             |             | Power (‰)   |
| Vacuum              | off         | off         | TC (°C)     |
|                     |             |             | Pyro 1 (°C) |

When the operator selects a different mode the setpoint is set to zero and the setpoint range is switched to power per thousand (power ‰) or °C (thermocouple and pyrometer).

The pyrometer 1 is proposed only if this feature is installed in the system.

The default value is "Pyro 1".

Then the user has to select the setpoint and to choose to activate or not the band alarm:

|                          |         |                                   |
|--------------------------|---------|-----------------------------------|
| Temperature control mode | TC (°C) |                                   |
| Setpoint                 | 600     | A                                 |
|                          |         | Temperature band alarm activation |

#### 8.2.3.1 Power setpoint

If the power mode is selected the temperature setpoint is set to zero and the system asks for a power setpoint (0.0 to 1000 ‰ of the full power).

The power setpoint is set in ‰ per thousand and not % (percent) in order to set a value that is closer to a temperature value and to limit consequences of wrong parameter setting.

|                                                                                     |                                                                                                                                                                                                                                                                                                                                                 |
|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <b>⚠ CAUTION</b>                                                                                                                                                                                                                                                                                                                                |
|                                                                                     | <p><b>Power mode.</b></p> <p><b>Possible overheating of the substrate and the furnace</b></p> <p><b>The user must check the maximum power that can be handled by the substrate and the system. At 500 ‰ power the temperature is in the range of 1200°C. Long step at high power may lead to damages to the substrate or to the system.</b></p> |

If the operator enters a value out of range it is automatically reset and a message with the right range of values is displayed.

### 8.2.3.2 Temperature setpoint

If one temperature sensor is set for the temperature control then the user sets the temperature setpoint.

- Temperature range: 0°C to 1400°C

The temperature increment is 1°C.

|                                                                                   |                                                                                                                                                                                                                          |
|-----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <b>⚠ CAUTION</b>                                                                                                                                                                                                         |
|                                                                                   | <b>High temperature process.</b>                                                                                                                                                                                         |
|                                                                                   | <b>Possible overheating of the furnace</b>                                                                                                                                                                               |
|                                                                                   | <b>Note that the recipe offers to possibility to set the temperature at high value for a long duration. The user should check the maximum duration for the selected temperature as indicated in the table hereafter.</b> |
|                                                                                   | <b>Longer duration can lead to equipment damage</b>                                                                                                                                                                      |

### Maximum temperature versus time:

|             | AS-One 100       | AS-One 150       |
|-------------|------------------|------------------|
| Temperature | Maximum duration | Maximum duration |
| 1300°C      | 60 s             | Out of range     |
| 1250°C      | 180 s            | 60 s             |
| 1200°C      | 5 mn             | 5 mn             |
| 1150°C      | 6 mn             | 6 mn             |
| 1100°C      | 7 mn             | 7 mn             |
| 1050°C      | 10 mn            | 10 mn            |
| 1000°C      | 12 mn            | 12 mn            |
| 900°C       | 60 mn            | 60 mn            |
| < 800°C     | 120 mn           | 120 mn           |

The operator must select a temperature that is compatible with the temperature sensor that is selected for the temperature control. The sensor temperature ranges are usually:

- Thermocouple: 0 to 1000°C
- Low temperature pyrometer: 150°C to 1100°C
- High temperature pyrometer: 400°C to 1400°C

### 8.2.3.3 Temperature band alarm

The temperature band alarm can be activated by clicking on the small box at the right of the setpoint box.

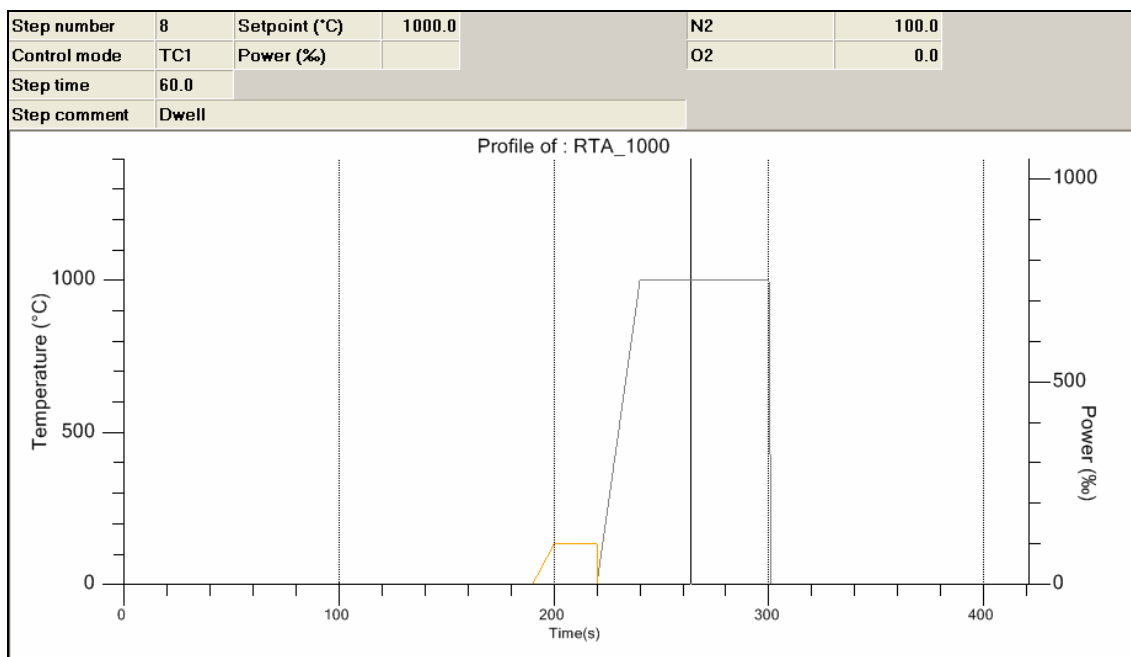
The box displays "-" if the alarm is not activated and "A" if the alarm is activated.

- Temperature band alarm range: Active ("A") or not ("-")

It can be activated or not for each step. The band alarm parameters are set in configuration mode. See the configuration mode for further information on the temperature band alarm.

### 8.2.4 Recipe profile

The secondary navigation button "Profile" displays the recipe heating profile. The parts of the curve in yellow are performed in power mode and the parts of the curve in grey are performed in temperature control mode. The profile cannot display actual profile when the temperature control mode changes.



### 8.2.5 Vacuum

The vacuum function allows selecting the vacuum mode among the following options.

#### 8.2.5.1 System without turbo pump

If the system is not equipped with a turbo pump:

- Vacuum: "Yes" or "No"

The default value is "No". The vacuum valve is open during the step if the parameter is set to "Yes" and is closed if the parameter is set to "No".

#### 8.2.5.2 System with turbo pump

If the system is equipped with a turbo pump there are 3 possible options:

- No pumping: All vacuum valves are closed
- Rough pumping: The rough pump vacuum valve (VV) is open
- Turbo pumping: The turbo pump valve (TPV) is open

Thanks to software interlocks, the system will open only one of the valves (VV or TPV) and will never open the rough pump vacuum valve and the turbo pump valve simultaneously.

When the system is equipped with a turbo pump there is a full range gauge to measure the low vacuum. The full range gauge is a filament gauge and is isolated from the process chamber by an isolation valve (GIV) in order to avoid filament damage by the process gases.

The gauge isolation valve (GIV) will open if one of the following conditions are fulfilled:

- The rough pump vacuum valve (VV) is open **and** all the gas valves (from the process gas lines) are closed for more than 30 seconds. The GIV valve opens if these 2 conditions are fulfilled simultaneously for 30 seconds
- The turbo pump valve is open

The gauge isolation valve automatically closes if one of the process gas valves opens.

Before a turbo pumping step the user must set a primary pumping step with the following conditions:

- The primary pumping step must be long enough in order to reach a pressure  $< 200$  Pascal.
- No process gases should be introduced during this step
- The step duration must be at least 35 seconds to allow opening the GIV valve

If these conditions are not fulfilled the turbo pump valve will not open and the process will stop.

### 8.2.6 Purge

- Purge: "Yes" or "No"

The default value is "No". The purge valve is open during the step if the parameter is set to "Yes" and is closed if the parameter is set to "No".

### 8.2.7 Process gas valve (option)

The process gas valve is installed with the UHV kit option. This valve isolates the reactor gas inlet from the process gas line. The user can decide to open or not this valve at each step of the recipe. This valve must of course be open to flow process gases inside the process chamber.

- Process gas valve: "Yes" or "No"

The default value is "No". The process gas valve is open during the step if the parameter is set to "Yes" and is closed if the parameter is set to "No".

**Particular case:** see chapter 8.2.8.2 Gas and turbo pump

### 8.2.8 Gas flow setpoints

Setpoint

|                  |  |   |   |  |  |     |   |   |
|------------------|--|---|---|--|--|-----|---|---|
| N2 - Alarm       |  | - | - |  |  | 200 | - | - |
| N2:H2-3% - Alarm |  | - | - |  |  | 100 | R | - |
| Ar - Alarm       |  | - | - |  |  | 500 | - | A |

Ramping setpoint

Gas band alarm activation

For each gas the user can set the following parameters:

- Setpoint from 0 to MFC range in sccm
- Gas band alarm
- Ramping (or not)

The system can receive up to 3 MFC. The gas lines parameters are displayed only if the name of the corresponding gas is different from null. (See configuration mode).

#### 8.2.8.1 Gas setpoint

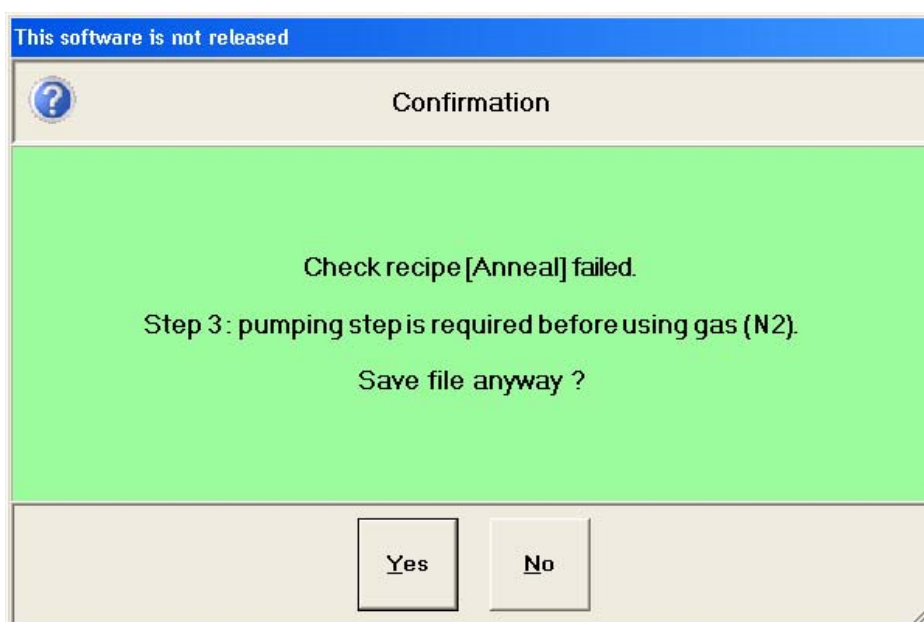
- Gas flow setpoint range: 0 to MFC range in sccm

If the operator enters a setpoint value out of range it is automatically reset and a message with the right range of values is displayed.

|                                                                                   |                                                                                                                                                                                                                                                                                                              |
|-----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <b>⚠ WARNING</b>                                                                                                                                                                                                                                                                                             |
|                                                                                   | <b>GAS HAZARD</b><br>Gas and air mixing interlock.<br>Some gases should not be mixed with air and a pumping step is required before introducing these gases to the process chamber in order to avoid dangerous mixture. The user must check gas compatibility and avoid any dangerous mixture in the recipe. |

The system can be provided with some special software interlocks to avoid dangerous gas mixtures.

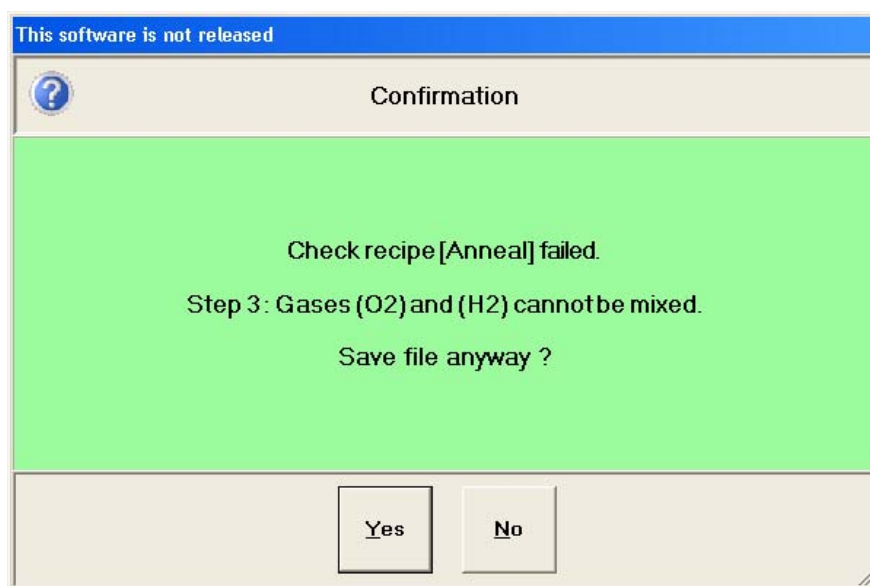
If the option air mixture interlock is activated for a gas, the following message will be displayed when saving the recipe if the operator has not asked for any pumping step before setting a gas flow for this gas.



Although you are allowed to save the file, downloading and running such a recipe is not possible.

|                                                                                     |                                                                                                                                                                                |
|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <b>⚠ WARNING</b>                                                                                                                                                               |
|                                                                                     | <b>GAS HAZARD</b><br>Gas mixing interlock.<br>Some gases should not be mixed together.<br>The user must check gas compatibility and avoid any dangerous mixture in the recipe. |

If the option "gas mixture with another gas" interlock is activated for the current gas and if the gas not to be mixed has a setpoint different from zero, the following message will be displayed



**Although you are allowed to save the file, downloading and running such a recipe is not possible.**

#### 8.2.8.2 Gas and turbo pump

The process gases cannot be introduced in the process chamber when the system is under turbo pumping. More over, a blank step of at least 2 seconds must follow a turbo pumping before you can introduce gas into the chamber. This blank step is required to let the system correctly close and manage its vacuum valves before you can open PGV and flow some gas.

#### 8.2.8.3 Gas flow band alarm

The gas flow band alarm can be activated or not for each step and for each gas. Click on the box at the right side of the setpoint box to activate the alarm (box turns to "A" when alarm is activated).



Band alarm parameter value is set in configuration mode. See the configuration mode for further information on the gas flow band alarm.

#### 8.2.8.4 Gas flow ramping

By default the gas flow setpoint is applied for the duration of the step. In order to smoothly increase the gas flow or the pressure inside the process chamber, the user can choose to have a ramp on the gas flow.

The gas flow ramping can be activated or not for each step and for each gas. Click on the second box at the right side of the setpoint box to activate the ramp (box turns to "R" when ramp is activated).



When the ramp is activated for the step, the gas flow setpoint at the beginning of the step is the setpoint value of the previous step and the gas flow setpoint at the end of the step is the flow set in the setpoint box.

### 8.2.9 Pressure control mode (optional)

An optional pressure control system with a throttle valve can be installed as an optional feature. If it is installed the software allows controlling the pressure or the valve angle.

|                  |           |           |           |
|------------------|-----------|-----------|-----------|
| Pressure         | Angle (°) | Angle (°) | Angle (°) |
| Setpoint - Alarm |           |           | 0         |

The operator can select:

- Pressure control mode: Pressure or valve angle
- Setpoint: Pressure or valve angle
- Band alarm

#### 8.2.9.1 Pressure control mode

- Pressure control mode: Pressure or valve

When the operator selects a different mode the pressure setpoint is set to zero and the setpoint range is switched to pressure range (set in the configuration mode) or valve angle (0 to 90°).

The default value is "Angle".

#### 8.2.9.2 Pressure control setpoint

The operator sets the pressure or the valve angle depending on the pressure control mode that has been selected.

- Pressure range: 0 to Full scale of the capacitance manometer
- Valve angle range: 0 to 90°

The full scale of the capacitance manometer is set in the configuration mode.

This option is displayed only if it is installed and selected in the configuration mode.

The system uses the pressure unit that has been selected in the configuration mode.

#### 8.2.9.3 Pressure alarm

The pressure band alarm can be activated or not for each step. Click on the box at the right side of the setpoint box to activate the alarm (box turns to "A" when alarm is activated).

Band alarm parameter value is set in configuration mode. See the configuration mode for further information on the gas flow band alarm.

**This option is not enabled in this software version.**



## 9. PROCESS

### 9.1 Recipe download

The process parameters (recipe, calibration tables, PID table and configuration) are saved on the computer's hard disk. During the process the system is controlled by the PLC and the data must be downloaded to the PLC before starting the process.

When the process mode is selected the system displays the download form.

The screen displays the last recipe that has been downloaded on the left side. The recipe is always downloaded with the thermocouple calibration table, the pyrometer calibration table, the PID table and the configuration.

| < Last downloaded recipe >     |                  | < Recipe to download >         |                       |
|--------------------------------|------------------|--------------------------------|-----------------------|
| Name                           | CALSITC          | Name                           | CALSITC               |
| Date of last download          | 30/08/2007       | Creation date                  | 15/03/2007 - 21:25:40 |
| Hour of last download          | 14:42:40         | Modification date              | 30/08/2007 - 14:31:09 |
| User name                      | Administrator    | User name                      | Administrator         |
| PID table                      | SITCPID          | PID table                      | SIPYLTPIID            |
| Thermocouple calibration table | Default_TC_Table | Thermocouple calibration table | Default_TC_Table      |
| Pyrometer 1 calibration table  | CALSIPYLT-01     | Pyrometer 1 calibration table  | CALSIPYLT-01          |

Buttons: Download, Start process, Browse..., Profile

Bottom buttons: Door locked, Start purge, Turbo pump (State: Ready, Start)

Recipe temperature profile can be viewed from Download window using the profile button on the right side

When the option "Force pick and download" is **enabled** (in the configuration), the operator must pick and download the recipe before starting a process. The selected recipe clears from the listbox whenever:

- the operator navigates to another page
- the process is started
- the selected recipe changes

When the option "Force pick and download" is **disabled**, consecutive "Starts" are possible.

If one parameter from a recipe or one of the associated tables or of the configuration has been modified, it is necessary to download again the recipe in order for the system to take into account the modification in the new process.

The recipe to download can be selected in the default list (with the arrow) or in any folder of the computer using the "Browse" button. When a recipe is selected the screen displays the recipe header to help the user in the recipe selection.

The downloading will start by clicking on the "Download" button.

## 9.2 Substrate loading

|                                                                                   |                                                                                                                                                                                                                                                                                |                                                                                     |
|-----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
|  | <p style="text-align: center;"><b>⚠ WARNING</b></p> <p><b>Burn hazard</b><br/>Quartz window may be at high temperature if some process has been performed before loading the substrate.<br/>Wear high temperature gloves to load the substrate inside the process chamber.</p> |  |
|-----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|

Before starting the process the user must load the sample and close the chamber.

The thermocouple must be installed if it is going to be used for the temperature control during the process.

The chamber is locked by clicking on the button "Chamber unlocked" or will be locked automatically when the process starts.

### 9.3 Start process

When the recipe has been downloaded or if the same process is being performed, the operator can start the process with the "Start process" button.

Then the computer will display the following form:

| Historical data                     |                                         |
|-------------------------------------|-----------------------------------------|
| Path                                | c:\program files\annealsys\historicals\ |
| Historical Name                     | his_20070504132714                      |
| Operator name                       | Eng                                     |
| Logging mode                        | Engineer                                |
| Recipe name                         | RTA_1000                                |
| Thermocouple calibration table name | Default_TC_Table_4-20mA                 |
| Pyro 1 table name                   | CALSU                                   |
| PID parameter table name            | SUPyroHT                                |
| Historical comment                  |                                         |

< Historicals >

- his\_20070427110950
- his\_20070427170008
- his\_20070430180156
- test1\_20070427110711
- test2\_20070427112033
- test3\_20070427112033
- test4\_20070427112033
- test5\_20070427112033
- test6\_20070427112033
- test7\_20070427112033
- test8\_20070427112033
- test9\_20070427112033
- test\_cool\_down
- test\_cool\_down\_2
- test\_cool\_down\_3
- test\_cool\_down\_4
- test\_cool\_down\_5
- test\_cool\_down\_6
- test\_cool\_down\_7
- test\_cool\_down\_8
- test\_gas\_01
- test\_gas\_02
- test\_gas\_03
- test\_gas\_04

Browse... Start Cancel

During the process all the parameters are logged and saved. The historical file name and the directory must be selected before starting the process.

All the process historicals are saved on the hard disk of the computer. The system proposes a default process historical file name and will save the process historical in the default directory C:\Program Files\Annealsys\historicals.

The user can enter any process historical name and can select a different directory on the computer using the "Browse..." function.

**The "Start" button will start performing the process.**

The "Cancel" button will return to the downloading mode.

#### 9.4 Wafer traceability (option)

Traceability is implemented to help plant managers track their wafers and lots.

The traceability is an additional module to the software. It is optional and not provided as a standard feature.

When the traceability is enabled, a dedicated area is displayed in "Start process" window. A new record is added to the traceability file when a new process is actually started.

The screenshot shows the 'Start process RTA\_1000' window. It has a blue title bar and standard Windows window controls. The main area is divided into two sections: 'Traceability data' and 'Historical data'.

**Traceability data section:**

|                   |  |
|-------------------|--|
| Wafer number      |  |
| Lot number        |  |
| Project reference |  |

**Historical data section:**

|                                     |                                         |
|-------------------------------------|-----------------------------------------|
| Path                                | c:\program files\annealsys\historicals\ |
| Historical Name                     | his_20070511174616                      |
| Operator name                       | Eng                                     |
| Logging mode                        | Engineer                                |
| Recipe name                         | RTA_1000                                |
| Thermocouple calibration table name | Default_TC_Table_4-20mA                 |
| Pyro 1 table name                   | CALSU                                   |
| PID parameter table name            | SUPyroHT400-1000                        |
| Historical comment                  |                                         |

At the bottom of the window are three buttons: 'Browse...', 'Start', and 'Cancel'.

On the right side, there is a list box titled '< Historicals >' containing a list of historical records, such as 'his\_20070427110950', 'his\_20070427170008', etc.

One to three traceability parameters can be added to the "Start process" window (depending on the configuration). In the above example: Wafer number, lot number and project reference have been added.

The traceability file is located in the directory **C:\Program Files\Annealsys** and has the following name:

- Traceability\_YYYYMM.csv (where YYYY is the year and MM is the month).

The file has "Read only" attribute.

In case of writing record failure, an error dialog window shows up. The process starts after the error is acknowledged by the operator.

A new file is created automatically each month.

Should the execution of the process roll over the month, the traceability feature implements a mechanism to make sure that "Start process information" and "End process information" are saved in the same file.

See the configuration mode for the setting of the traceability parameters.

## 9.5 Running process

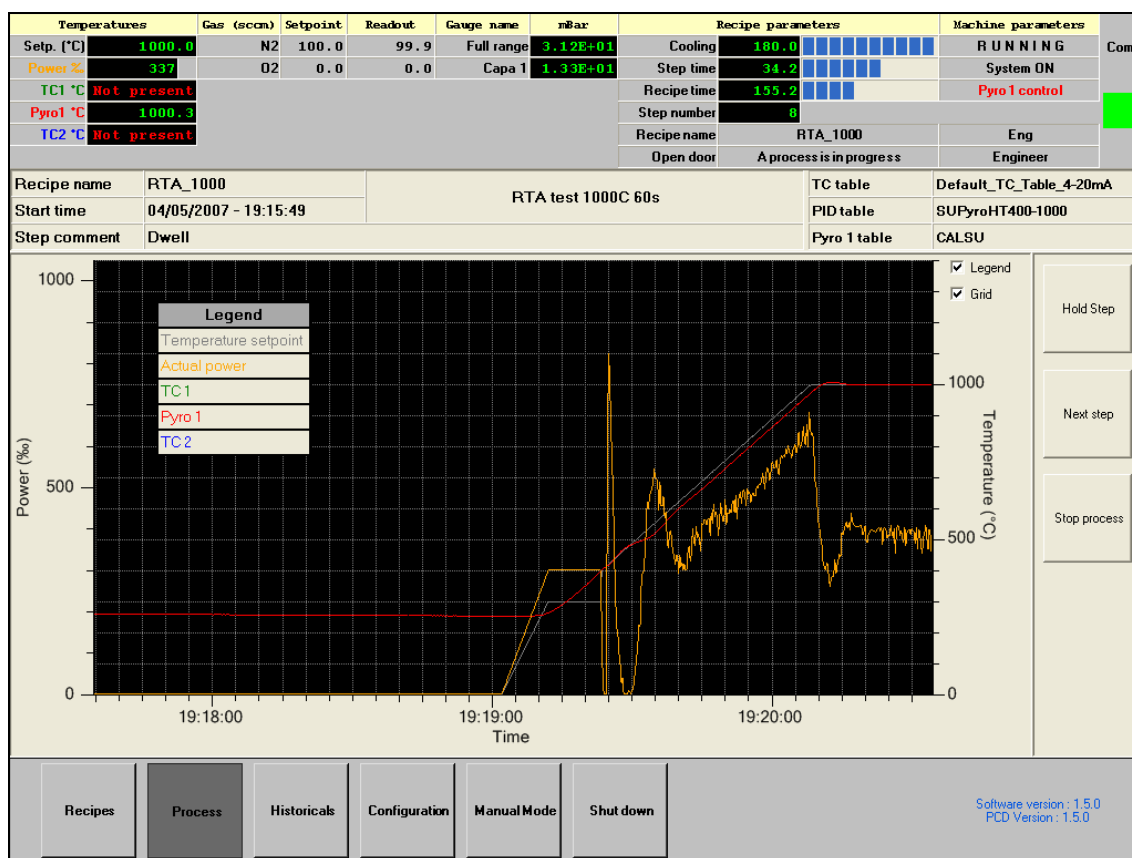
Then the process starts with the "Start Process" button.

Before starting the process the system will check that it can be performed under safe conditions and checks the different interlocks. If the system is in safe conditions the process starts.

During the process, the parameters are displayed on the screen:

The following parameters are displayed in the main system status panel (from left to right):

- Setpoint, power to the lamps, temperatures
- Gas flow setpoints and readouts
- Pressure
- Remaining cooling time
- Remaining time to end of step
- Remaining estimated time to end of process
- Step number
- Heating control mode



The recipe header parameters are displayed on the top of the process information form (from left to right):

- Recipe name
- Starting date and time
- Step comment
- Recipe comment (in the middle)
- Thermocouple calibration table
- PID table
- Pyrometer 1 calibration table
- Pyrometer 2 calibration table (option)

The plotter on the left part of the screen displays different curves:

- Grey: temperature or power setpoint
- Yellow: power to the lamps
- Red: temperature from the pyrometer 1
- Purple: temperature from the pyrometer 2 (if option is installed)
- Green: temperature from the thermocouple 1
- Blue: temperature from thermocouple 2

The X-axis is time and the full scale is 3 minutes. The plotter has slipping capability to display the last 3 minutes of the curves.

The Y axis on the right side of the plotter is the temperature axis (scale: 0°C to 1400°C); the Y axis on the left is the power to the lamps (scale: 0 to 1000‰ of full power).

**It is possible to stop the process at any time by clicking on the "Stop process" button**

The software offers the capability to switch from process window to recipe or synoptic (manual mode) when the process is running. Only visualization is possible in the manual mode during the process.

#### 9.5.1 Function hold

**This function is not enabled in operator mode.**

The hold function can be controlled using the "Hold step" button in the secondary navigation panel at the right side of the screen.

During any step of the process it is possible to activate the hold function. When hold is activated all the parameter setpoints as well as the recipe and step countdowns are frozen.

The operator can continue the process at any time with the same button. When hold is stopped the process continues from where it was frozen.

As a safety the hold function stops automatically after 30 minutes if the operator has not stopped it.

|                                                                                     |                                                                                                                                                                                                                                                        |
|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <b>⚠ CAUTION</b>                                                                                                                                                                                                                                       |
|                                                                                     | <p><b>Function HOLD.</b></p> <p><b>Possible overheating of the furnace</b></p> <p><b>Long duration with high heating power can lead to equipment damage.</b></p> <p><b>The user must check the maximum time during which the step can be held.</b></p> |

### 9.5.2 Function next step

This function is not enabled in operator mode.

The next step function can be controlled using the "Next Step" button in the secondary navigation panel at the right side of the screen.

During the process the operator can decide at any time to go directly to the end of the current step. When the operator activates this function the end of the step condition is immediately set to true and the system starts the following step.

|                                                                                   |                                                                                                                                                                                                                                                                                                                      |
|-----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <b>⚠ CAUTION</b>                                                                                                                                                                                                                                                                                                     |
|                                                                                   | <p><b>Function NEXT STEP.</b></p> <p><b>Possible sudden condition change in the process chamber</b></p> <p><b>Selecting this option can lead to thermal shock, power peak, big pressure changes and gas mixtures.</b></p> <p><b>The user must check that the conditions to switch to the next step are safe.</b></p> |

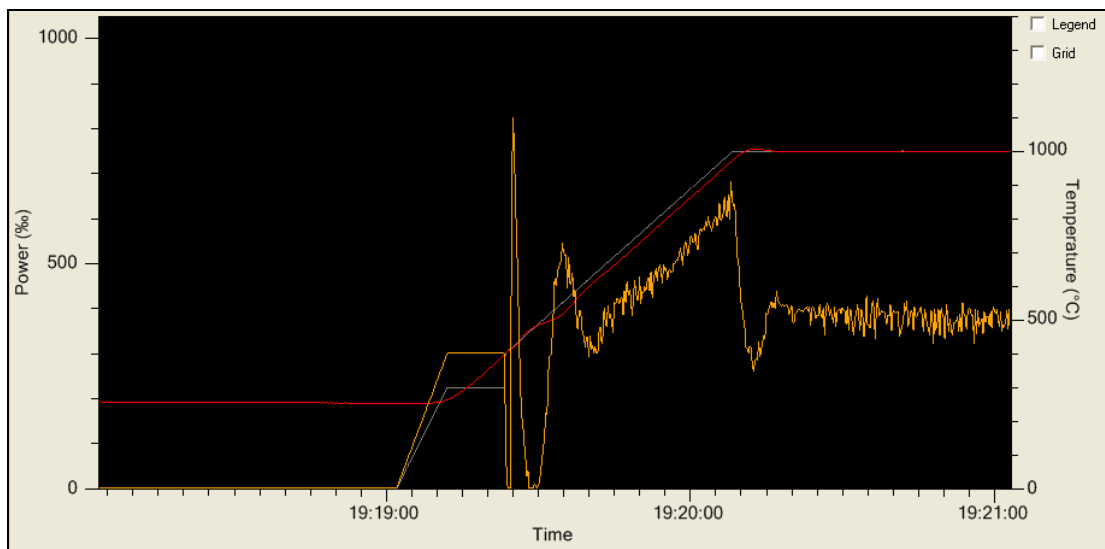
### 9.5.3 Process windows optional features

The user can select an option in order to display the legend of the curves (description of curve for each color). This option is selected by checking the legend box on the right side of the screen.

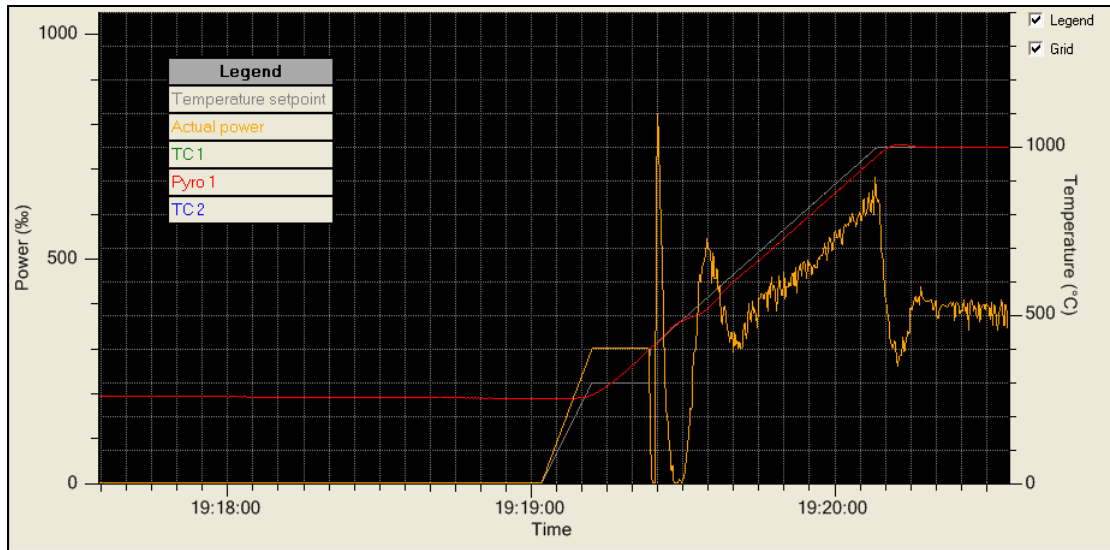
It is also possible to display the grid or not on the plotter.



**Process plotter without grid or legend**

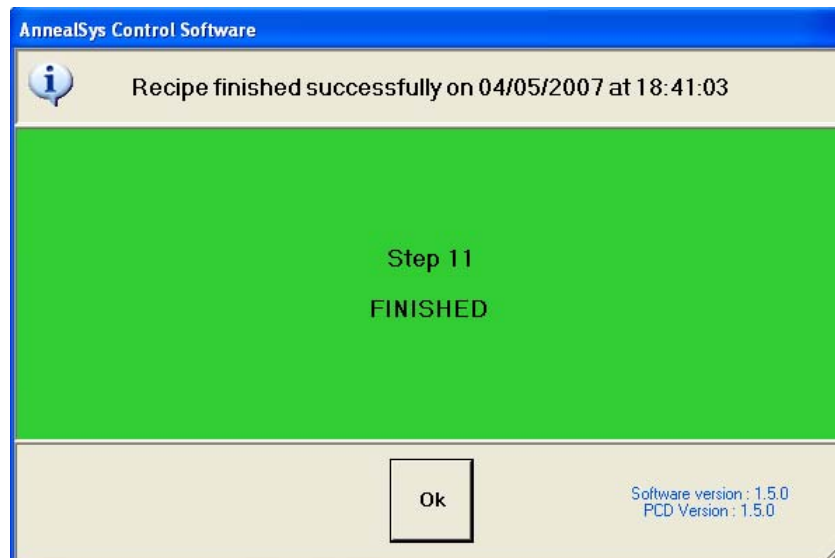


Process plotter with grid and legend



## 9.6 End of process

At the end of the process the computer confirms that the process has been performed successfully:



The system has the following actions:

- Reset all the setpoints (power, temperature, gas and pressure)
- Close all the gas valves
- Close turbo pump valve and process gas valve (Turbo pump option only)
- Start furnace cooling down function for 180 seconds
- Purge of the chamber (open purge valve) during the cooling down

At the end of the cooling down, the cooling and purge are stopped and the door can be unlocked.

Note that the data logging will continue until the end of the cooling period and will be part of the process historical.



## 9.7 Substrate unloading

During the process the gas inside the chamber is expanding and going out through the exhaust gas line. When the substrate and the chamber cool down the pressure inside the chamber will decrease. In order to keep the pressure above the atmospheric pressure the chamber is purged during the cool down period.

If the substrate is unloaded sometime after the end of the cooling period it is possible that the pressure inside the chamber has decreased again and that the pressure inside the process chamber is below the atmospheric pressure. This will make the chamber impossible to open.

In this case it is possible to open the chamber using the "start purge" button on the screen.

| Temperatures |             | Gas (scm) | Setpoint | Readout | Gauge name | mBar     | Recipe parameters |             | Machine parameters |  |     |
|--------------|-------------|-----------|----------|---------|------------|----------|-------------------|-------------|--------------------|--|-----|
| Setp. (%)    | 0.0         | N2        | 0.0      | 0.0     | Full range | 8.14E+00 | Cooling           | Not cooling | Idle               |  | Com |
| Power %      | 1.0         | 02        | 0.0      | 0.0     | Capa 1     | 1.33E+01 | Step time         | 0.0         | System ON          |  |     |
| TC1 °C       | Not present |           |          |         |            |          | Recipe time       | 0.0         | POWER              |  |     |
| Pyro1 °C     | 104.0       |           |          |         |            |          | Step number       | 0           |                    |  |     |
| TC2 °C       | Not present |           |          |         |            |          | Recipe name       | RTA_1000    | Eng                |  |     |
|              |             |           |          |         |            |          | Open door         | Safe        | Engineer           |  |     |

| < Last downloaded recipe >         |                         | < Recipe to download >             |                         |
|------------------------------------|-------------------------|------------------------------------|-------------------------|
| Name                               | RTA_1000                | Name                               | RTA_1000                |
| Date of last download              | 11/05/2007              | Creation date                      | 03/05/2007 - 09:30:21   |
| Hour of last download              | 15:18:17                | Modification date                  | 09/05/2007 - 17:22:22   |
| User name                          | Adm                     | User name                          | Adm                     |
| PID table                          | SUPyroHT400-1000        | PID table                          | SUPyroHT400-1000        |
| Thermocouple calibration table     | Default_TC_Table_4-20mA | Thermocouple calibration table     | Default_TC_Table_4-20mA |
| Pyrometer 1 calibration table (HT) | CALSU                   | Pyrometer 1 calibration table (HT) | CALSU                   |

RTA test 1000C 60s

RTA test 1000C 60s

Door locked

Start purge

**Turbo pump**

State : Stopped

Start

Recipes

Process

Historicals

Configuration

Manual Mode

Shut down

Software version : 1.5.0

PCD Version : 1.5.0

Click on "Door locked" to unlock the process chamber and to open it.

| <b>⚠ WARNING</b>                                                                    |                                                                                                                                                                                                                                                                       |                                                                                       |
|-------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
|  | <p><b>Burn hazard</b></p> <p>After process both substrate and quartz window may still be at high temperature.</p> <p>Wait for sufficient cool down of the substrate for unloading. Wear high temperature gloves to unload the substrate from the process chamber.</p> |  |

**Use tools that are compatible with the substrate temperature to unload it.  
Never use metal tweezers to move or to remove silicon carbide coated graphite susceptor and lid.**

## 10. HISTORICALS

The user can access to the historical mode by the Historical button from the main navigation panel.

This mode gives access to:

- Process historicals
- Alarm list

The secondary navigation panel at the right side of the screen gives access to these two modes.

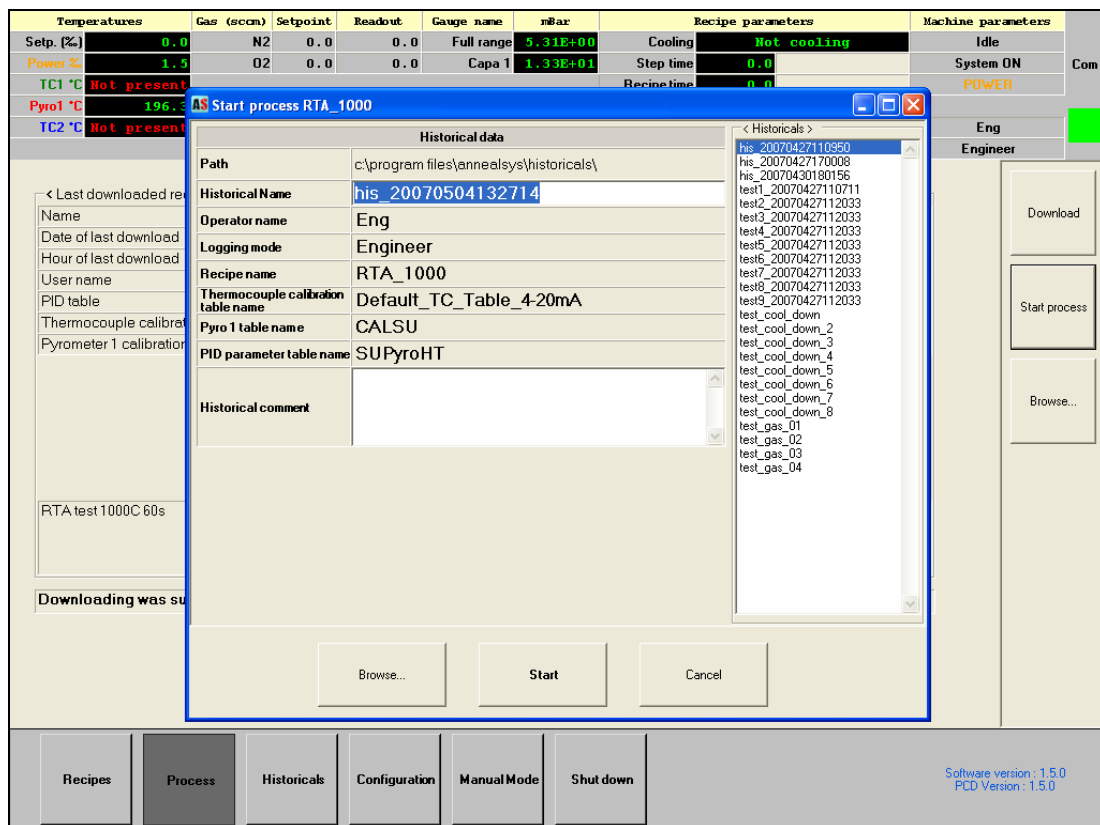
### 10.1 Process historicals

#### 10.1.1 Loading process historical

By default or when the historical mode is selected, the screen will display the historical header that allows selecting the right process historical to be loaded.

When historical mode is first selected or when File button is clicked, the historical selection dialog shows up.

The window contains the main information about the process and allows the user selecting the right file.



The process historicals are saved by default in the folder "c:\Program files\annealsys\historicals" or in the user folder if it has been set in the configuration login mode.

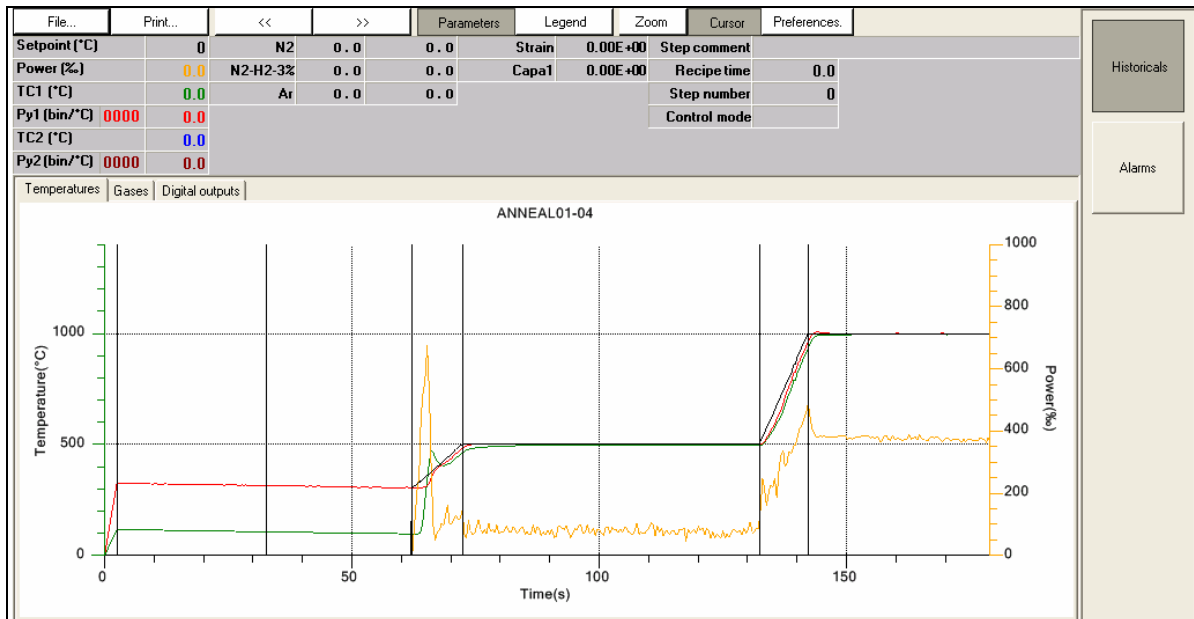
The extension of the historical files is ".his".

It is possible to load process historical files from any folder of the computer using the browse function.

When the process historical is selected it can be loaded by clicking on the "OK" button.

### 10.1.2 Process historical

When the historical has been loaded the curves are displayed with the process values on top. Several plots are available to display different process data. By default the system displays the temperature plot.



The curves are:

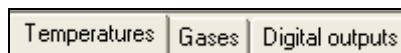
- Black: temperature setpoint (light green for power setpoint)
- Red: Pyrometer 1
- Green: Thermocouple 1
- Purple: Pyrometer 2 (option)
- Blue: Thermocouple 2
- Yellow: Power to the lamps

The parameters are displayed on top of the plotter. The cursor function (see hereafter) allows displaying the value of the parameters of any time of the process historical.

The black vertical lines mark the step limits.

Several functions are available for the plotter and are described hereafter.

Three different plotters are available:

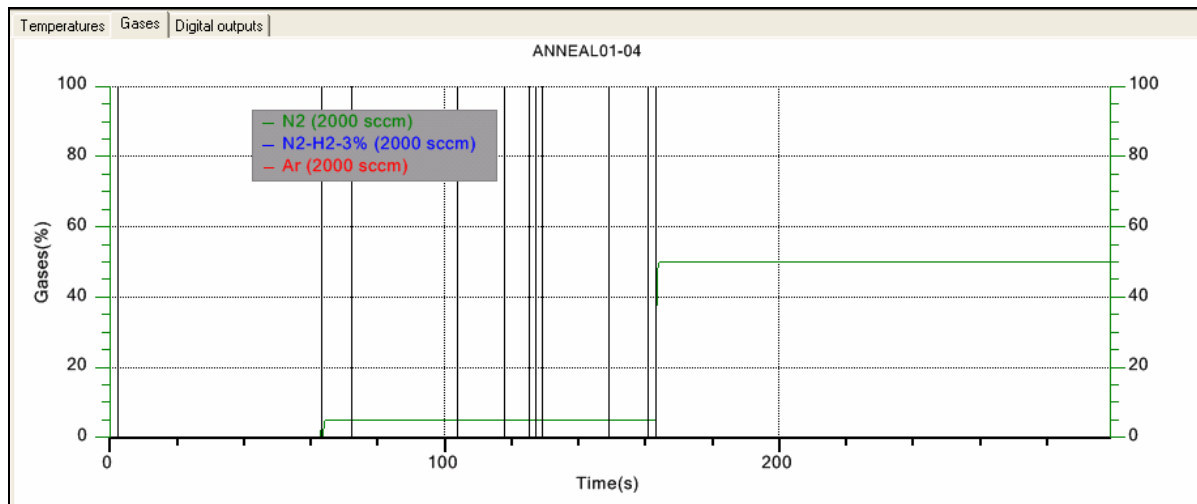


- Temperature: Displays the temperature and power curves function of the time
- Gases: Displays the gas flow curves function of the time
- Digital outputs: Displays the status of the system digital outputs function of the time

The 3 plotters are synchronized and the user can switch from one plotter to the other to display the full status of the system at the selected time: temperatures, gas flows and digital outputs (valves, temperature control mode, etc.)

### 10.1.2.1 Gas plotter

The gas plotter displays the gas flow of each gas line in **percentage of the full scale** of the mass flow controller.

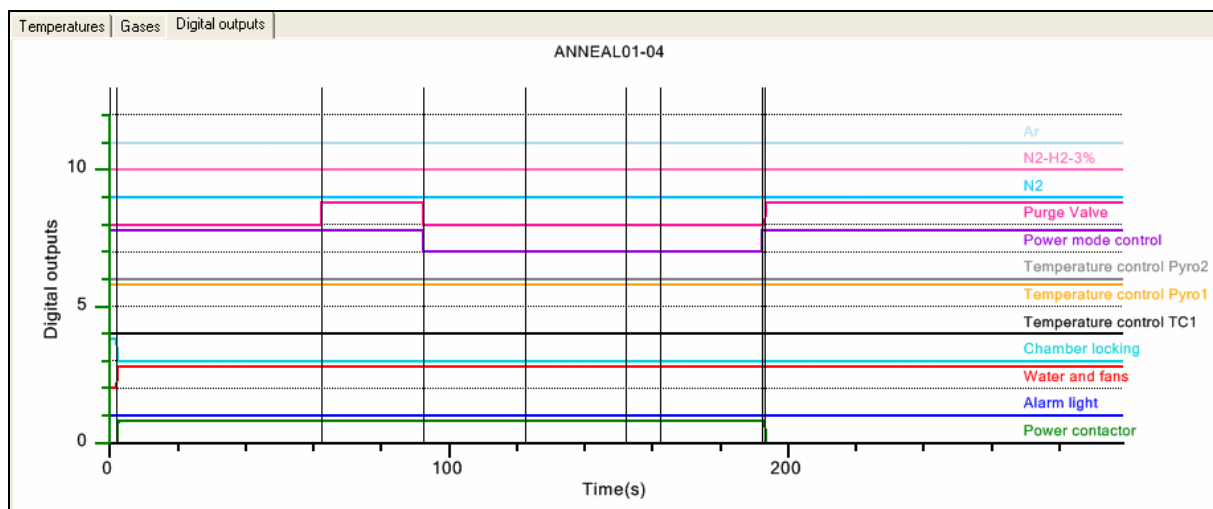


Gas plotter with legend (here only one gas flow: N<sub>2</sub>)

### 10.1.2.2 Digital outputs plotter

The digital outputs plotter displays the status of the digital outputs during the process.

When the digital output is activated there is an up step on the curve.



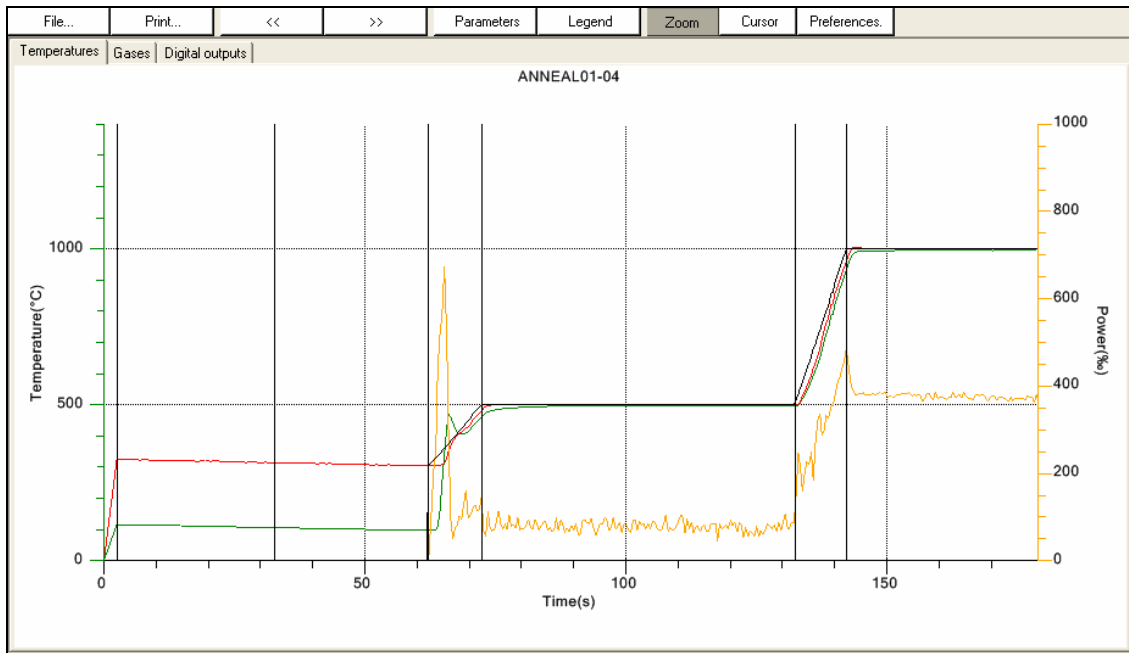
Digital outputs plotter

### 10.1.2.3 Functions of the historical plotter

The function buttons of the plotter are located on top of the plotter:

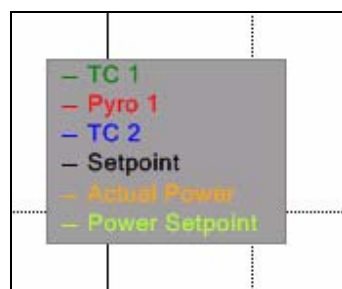


- File: Allows selecting a new historical file.
- Print: To print the historical plot as displayed on the screen
- >>: To move forwards in the process historical
- <<: To move backwards in the process historical
- Parameters: Will enlarge the plotter and hide the parameters (the parameter button turns white):



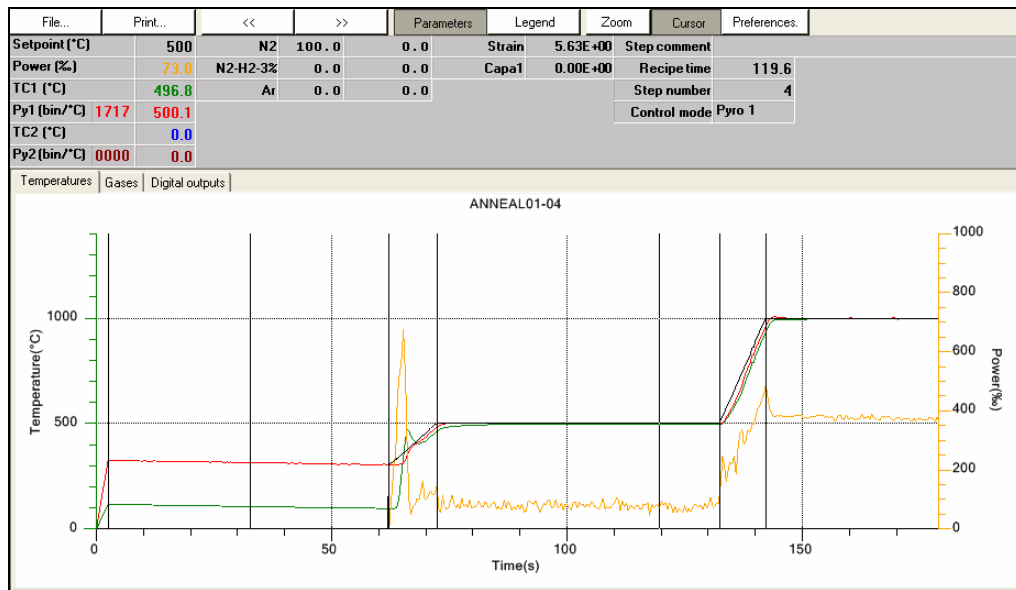
A second click on the parameter button displays again the parameter values (the button turns to grey).

- Legend: Display the legend of the plotter with the curve descriptions:



- Zoom: The zoom is activated by a click on this button (the zoom button turns to grey). When the zoom function is activated the cursor function is disabled (and vice versa). Select the part of the plotter to be enlarged keeping the left button of the mouse pushed. A right click on the mouse will go back to the previous size, click again to come back to the original size. Click on the cursor button to disable the zoom function.

- **Cursor:** The cursor is activated by a click on this button (the cursor button turns to grey). When the cursor is activated the zoom function is disabled. Moving the mouse clicking on the left button allows displaying the values of the process parameters for any time of the process. Parameter values are displayed on top of the plotter.



- **Preferences:** This button gives access to window to set the display parameters of the process historicals.

| General                                          | Temperatures                               | Gases                                  | Digital outputs                               |
|--------------------------------------------------|--------------------------------------------|----------------------------------------|-----------------------------------------------|
| Time span (s)<br>420                             | <input checked="" type="checkbox"/> TC 1   | <input checked="" type="checkbox"/> N2 | <input checked="" type="checkbox"/> PWC       |
| <input checked="" type="checkbox"/> Step markers | <input checked="" type="checkbox"/> Pyro 1 | <input checked="" type="checkbox"/> O2 | <input checked="" type="checkbox"/> ALARM     |
|                                                  | <input checked="" type="checkbox"/> TC 2   |                                        | <input checked="" type="checkbox"/> COOL      |
|                                                  |                                            |                                        | <input checked="" type="checkbox"/> CTRLPOWER |
|                                                  |                                            |                                        | <input checked="" type="checkbox"/> CTRLTC1   |
|                                                  |                                            |                                        | <input checked="" type="checkbox"/> CTRLPYR01 |
|                                                  |                                            |                                        | <input checked="" type="checkbox"/> LDO       |
|                                                  |                                            |                                        | <input type="checkbox"/> DEBUG                |
|                                                  |                                            |                                        | <input checked="" type="checkbox"/> VV        |
|                                                  |                                            |                                        | <input checked="" type="checkbox"/> PUV       |
|                                                  |                                            |                                        | <input checked="" type="checkbox"/> GV1       |
|                                                  |                                            |                                        | <input checked="" type="checkbox"/> GV2       |
|                                                  |                                            |                                        | <input checked="" type="checkbox"/> PGV1      |
|                                                  |                                            |                                        | <input checked="" type="checkbox"/> GIV       |
|                                                  |                                            |                                        | <input checked="" type="checkbox"/> TPV       |
|                                                  |                                            |                                        | <input checked="" type="checkbox"/> TFPV      |

The preferences (Graphics) have the following functions:

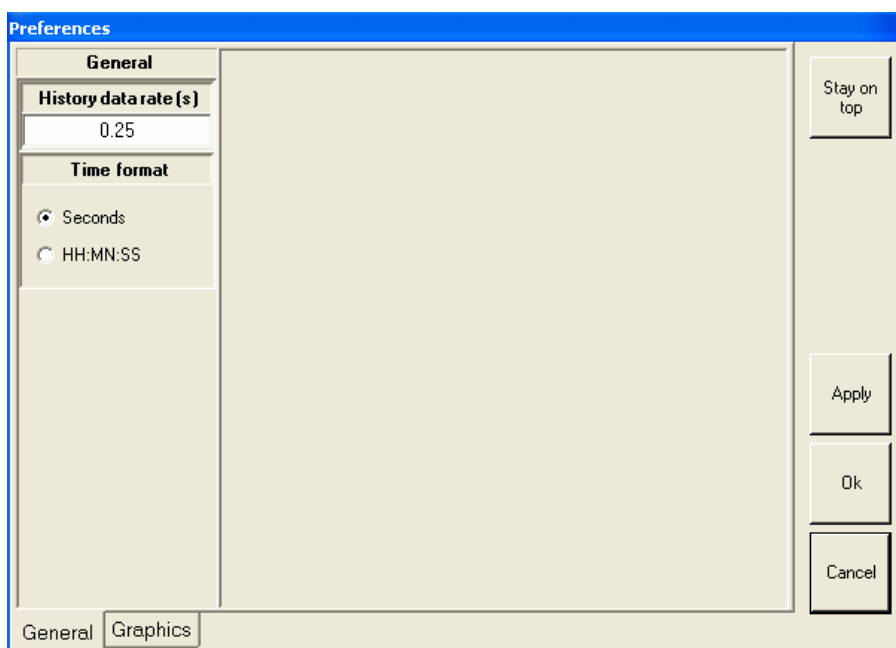
- **Time span:** To select the duration of historical to be displayed on the active part of plotter
- **Step marker:** If the box is checked the vertical black lines (step markers) are displayed
- **Temperature:** Temperature plotter displays corresponding curve if the box is checked
- **Gases:** Gas plotter displays corresponding curve if the box is checked
- **Digital outputs:** Digital output plotter displays corresponding curve if the box is checked

List of the outputs:

The list of outputs may vary from one system to another depending on the optional features that are installed. The list corresponding to the above example is:

| Mnemonic  | Description                                         | Comment           |
|-----------|-----------------------------------------------------|-------------------|
| PWC       | Power contactor                                     |                   |
| ALARM     | Alarm light                                         |                   |
| COOL      | Cooling (water valve and fans)                      |                   |
| CTRLPOWER | Temperature control: power mode                     |                   |
| CTRLTC1   | Temperature control: TC1                            |                   |
| CTRLPYRO1 | Temperature control: Pyrometer 1                    |                   |
| LDO       | Lock door (process chamber)                         |                   |
| DEBUG     | Used for investigation purposes: oscillating output |                   |
| VV        | Vacuum valve                                        |                   |
| PUV       | Purge valve                                         |                   |
| GV1       | Gas valve 1                                         | Gas line 1 option |
| GV2       | Gas valve 2                                         | Gas line 2 option |
| PGV1      | Process gas valve                                   | UHV kit option    |
| GIV       | Gauge isolation valve (for full range gauge)        | Turbo pump option |
| TPV       | Turbo pump valve                                    | Turbo pump option |
| TFPV      | Turbo forepump valve                                | Turbo pump option |

The preferences window has a second sheet (see index on the lower left side corner).



The preferences (General) have the following functions:

- History data rate: To adjust the data logging speed. The default value is 0.25 second. This is also the minimum value. The logging period can be increase to reduce the size of the process historical files for long processes.
- Time format: To select the format of the time which is displayed in Main Status Area as well as in the Recipe Editor.

## 10.2 Alarm list

To view the alarm list click on the "Alarm" button of the secondary navigation bar at the right side of the screen. The number into brackets is the number of alarms.

[illegible]



## 11. CONFIGURATION

The configuration menu allows:

- Setting gas names, mass flow ranges and alarm threshold values
- Selection of vacuum gauges names and ranges
- Setting the temperature alarm parameters
- Setting the high power and power rate limits
- Selection of the pressure unit
- Setting the atmospheric pressure
- Setting of door (reactor) unlocking conditions
- Setting PID parameter tables
- Setting thermocouple calibration tables
- Setting pyrometer calibration tables
- Setting traceability parameters.(option)

**Additional features are enabled in the administrator mode:**

- Passwords administration
- Gas safety interlocks
- System hardware configuration

These additional features are described in the administrator mode (see 11.6).

When the user enters the configuration mode the system displays the PID form by default.

| Bits | Degrees |
|------|---------|
| 840  | 252.0   |
| 900  | 302.0   |
| 1070 | 352.0   |
| 1250 | 402.0   |
| 1470 | 452.0   |
| 1670 | 502.0   |
| 1835 | 552.0   |
| 1955 | 602.0   |
| 2065 | 652.0   |
| 2172 | 702.0   |
| 2275 | 752.0   |
| 2378 | 802.0   |
| 2483 | 852.0   |
| 2587 | 902.0   |
| 2695 | 952.0   |
| 2798 | 1002.0  |
| 2901 | 1052.0  |
| 3004 | 1102.0  |
| 3107 | 1152.0  |
| 3210 | 1202.0  |

The secondary navigation panel at the right gives access to the following functions:

- Configuration
- PID parameter tables
- Thermocouple calibration tables
- Pyrometer calibration tables

## 11.1 Configuration

The configuration form gives access to the following functions:

- Set gas names, mass flow ranges and alarm threshold values
- Selection of vacuum gauges names and ranges
- Setting of temperature alarm parameters
- Setting of high power and power rate limits
- Selection of the pressure unit
- Setting atmospheric pressure
- Setting of door (reactor) unlocking conditions

The screenshot displays the configuration interface for the ANNEALSYS system. It is divided into several sections:

- < Gas lines >**: A table with columns for Gas name, MFC range (sccm), Alarm limit (% of Setpoint), and Stop process. It lists N2 and O2 with MFC ranges of 1000 and alarm limits of 5. There are 10 empty rows below.
- < Pressure >**: A section for gauge settings. It includes a table for Gauge name and Range (Pa) with entries for Full range (100000) and Capa 1 (1333.22). It also has fields for Local atmospheric pressure (Pa) set to 100000 and a Unit dropdown set to mBar.
- < Temperature >**: A section for temperature alarm parameters. It includes fields for Alarm limit (% of Setpoint) set to 5, Alarm delay (s) set to 0, Power high limit (%) set to 100, and Power rate (% / s) set to 0. There is a checkbox for Stop process which is checked.
- < Open door condition >**: A section for door unlocking conditions. It includes a Condition dropdown set to None, Time (s) set to 0, and Temperature (°C) set to 0. There is a checkbox for Force pick and download recipe which is unchecked.
- < Commands >**: A section at the bottom with a Save button.

### 11.1.1 Select gas names, mass flow ranges and alarm threshold values

The system can receive up to 5 mass flow controllers. The configuration mode allows entering the gas name and the mass flow range for each mass flow controller.

The number of mass flow controller settings is limited to the gas number of gas lines actually installed inside the system.

This is a close-up view of the < Gas lines > table from the configuration form. It shows the following data:

| Gas name | MFC range (sccm) | Alarm limit (% of Setpoint) | Stop process                        |
|----------|------------------|-----------------------------|-------------------------------------|
| N2       | 1000             | 5                           | <input checked="" type="checkbox"/> |
| O2       | 1000             | 5                           | <input checked="" type="checkbox"/> |
|          | 0                | 0                           | <input checked="" type="checkbox"/> |
|          | 0                | 0                           | <input checked="" type="checkbox"/> |
|          | 0                | 0                           | <input checked="" type="checkbox"/> |
|          | 0                | 0                           | <input checked="" type="checkbox"/> |
|          | 0                | 0                           | <input checked="" type="checkbox"/> |
|          | 0                | 0                           | <input checked="" type="checkbox"/> |
|          | 0                | 0                           | <input checked="" type="checkbox"/> |
|          | 0                | 0                           | <input checked="" type="checkbox"/> |
|          | 0                | 0                           | <input checked="" type="checkbox"/> |
|          | 0                | 0                           | <input checked="" type="checkbox"/> |

|                                                                                   |                                                                                                                                                                                                                                                                                                                         |
|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <b>⚠ CAUTION</b>                                                                                                                                                                                                                                                                                                        |
|                                                                                   | <b>CONFIGURATION</b><br><b>Mass flow range selection in the configuration mode.</b><br><b>The mass flow configuration will not have any effect on the hardware. The configuration of the mass flow controllers is used to have the correct information for the software function of the hardware that is installed.</b> |

The gas name and the mass flow range should be set according to the hardware that is actually installed in the gas panel.

- Gas name: Up to 12 characters
- Mass flow range: 1 sccm to 50 000 sccm

If no gas name is set then the software will not activate the gas line in recipes, process and manual modes.

It is possible as well to select the alarm threshold value.

- Gas alarm threshold value: 5 to 100%

If the difference between the setpoint and the mass flow controller readout exceeds the limit the system will generate an alarm.

In future version the configuration mode will allow selecting the action in case of gas alarm:

- Message during process and at the end of the process (stop process box is not checked). This feature is not yet implemented.
- Stop process and message at the end of the process (stop process box is checked)

**On the current software version this feature is not implemented.**  
**The process stops in case of alarm. The box is checked and disabled.**

### 11.1.2 Select the pressure gauge ranges

| < Pressure > |            |
|--------------|------------|
| Gauge name   | Range (Pa) |
| Full range   | 100000     |
| Capa 1       | 1333.22    |
|              | 0          |
|              | 0          |

The configuration mode allows selecting the range of the vacuum gauges from 1 Pascal full range to 500 000 Pascal full range.

**The vacuum gauge range values are always selected in Pascal. They must be converted by the user if the ranges of the gauges are not given in Pascal.**

This configuration will allow displaying the right pressure according to the capacitance manometers or constraint gauges actually installed on the system.

One vacuum constraint gauge is installed as a standard feature and the default value for the full range is 100,000 Pascal. Additional capacitance manometers are available as optional features.

***When the system is equipped with a turbo pump the vacuum gauge 1 is a full range gauge. The range of this gauge is directly set in the software and cannot be modified by the user.***

### 11.1.3 Setting the atmospheric pressure

The atmospheric pressure value is used by the system to unlock the process chamber. The value is set to 1000 mbar before shipment of the system.

The chamber is unlocked when the pressure reaches the threshold value less 10 mbar.

If the furnace is installed in altitude the atmospheric pressure is lower and must be adjusted in order to allow unlocking of the chamber when it is purged.

To adjust the atmospheric pressure purge the system in manual mode and open the purge valve to reach atmospheric pressure. Then get the pressure value from the constraint gauge (top of the screen) and enter the value in the configuration mode. For example at 1,500 m the atmospheric pressure is typically 850 mbar.

|                                 |        |
|---------------------------------|--------|
| Local atmospheric pressure (Pa) | 100000 |
|---------------------------------|--------|

The atmospheric pressure is set in Pascal so for 1,500 m it should be set to 85,000 Pa.

### 11.1.4 Select pressure unit

|      |      |
|------|------|
| Unit | mBar |
|      | Pa   |
|      | mBar |
|      | Torr |

The configuration mode allows selecting the following pressure units:

- Pascal: Pa
- Millibar: mbar
- Torr: Torr

The software displays the pressure values with the selected unit and will make the pressure calculations accordingly.

### 11.1.5 Temperature alarm threshold values

|                             |   |                                                  |
|-----------------------------|---|--------------------------------------------------|
| < Temperature >             |   |                                                  |
| Alarm limit (% of Setpoint) | 5 | <input checked="" type="checkbox"/> Stop process |
| Alarm delay (s)             | 0 |                                                  |

This mode allows setting the different temperature alarm threshold values.

#### 11.1.5.1 Temperature band alarm limit

This is the limit to activate the temperature alarm during the process if it has been selected for the ongoing step and if the system is in temperature control mode.

- Temperature band alarm limit 5 to 100%

If the temperature difference between setpoint and current temperature measured by the temperature control sensor (chosen for temperature control for the step) exceeds the threshold value then the alarm is activated.

If the alarm limit is set to 100% the alarm will never occur.

Setting the alarm limit to 0% disables this alarm capability.

In future version the configuration mode will allow selecting the action in case of temperature alarm:

- Message during process and at the end of the process (stop process box is not checked). This feature is not yet implemented.
- Stop process and message at the end of the process (stop process box is checked)

**On the current software version this feature is not implemented.**

**The process stops in case of alarm. The box is checked and disabled.**

It is recommended no to use this alarm on fast ramp rate because of possible higher offset between setpoint and actual temperature during transient phases.

#### 11.1.5.2 Temperature alarm delay

During process the system checks that the difference between setpoint and actual temperature remains under 50% after a delay. This delay avoids unexpected alarm for ramping steps.

- Temperature alarm delay 0 to 10,000 s

This alarm is not related to the previous one.

Setting "Temperature alarm delay" to 0 seconds disables this alarm capability.

### 11.1.6 Temperature control parameters

In addition of the PID parameters the system offers the capability to set the maximum output power and the maximum rate for the temperature controller.

|                      |     |
|----------------------|-----|
| Power high limit (%) | 100 |
| Power rate (% / s)   | 0   |

#### 11.1.6.1 Power high limit

In temperature control mode the output power of the temperature controller (power to the lamp) can reach the maximum value of 100%. For some low temperature processes and especially using a susceptor it may help the temperature control to limit the power output of the controller.

The power high limit is set in %.

The default value is 100%.

#### 11.1.6.2 Power rate

In addition the system also offers to limit the power rate (%/s). This is the maximum possible increase of the power by unit of time at the temperature controller output.

This parameter may have big impact on the temperature control and the PID parameters.

The default value is 0%/s (function is disable).

### 11.1.7 Pressure band alarm limit

To be added to the software in a future version.

### 11.1.8 Condition to unlock reactor (door open condition)

During the process the reactor is locked in order to avoid gas release or contact with hot substrate.

This area allows setting the conditions to fulfill for opening the door. All listed conditions must be verified before the door can be opened.

| < Open door condition > |      |
|-------------------------|------|
| Condition               | Time |
| Time (s)                |      |
| Temperature (°C)        |      |

Note that the temperature condition can be limited in time so that the door does not keep deadlock closed.

In all the cases, the door does not open before the following conditions are verified:

- Process is finished
- Cooling is finished
- Reactor is at atmospheric pressure

#### 11.1.8.1 Condition = None

| < Open door condition > |      |
|-------------------------|------|
| Condition               | None |
| Time (s)                | 0    |
| Temperature (°C)        | 50   |

The door is allowed to open when:

- Process is finished
- Cooling is finished
- Reactor is at atmospheric pressure

#### 11.1.8.2 Condition = Time

| < Open door condition > |      |
|-------------------------|------|
| Condition               | Time |
| Time (s)                | 400  |
| Temperature (°C)        | 50   |

The door is allowed to open when:

- Process is finished
- Cooling is finished
- Reactor is at atmospheric pressure
- Additional "Time" elapses (400 seconds in the above example)

Additional "Time" starts after the cooling time of 180 seconds finishes.

Note that if additional "Time" = 0, then "Time" condition is the same as "None" condition.

### 11.1.8.3 Condition = Temperature

Temperature condition can be set on any temperature sensor present on the system. It can be TC1, TC2, Pyro1 or Pyro2 (option).

| < Open door condition > |     |
|-------------------------|-----|
| Condition               | TC1 |
| Time (s)                | 400 |
| Temperature (°C)        | 200 |

The door is allowed to open when:

- Process is finished
- Cooling is finished
- Reactor is at atmospheric pressure
- Actual temperature from the chosen sensor is less than the configured "Temperature"  
**OR** "Time" timer elapses.

"Time" timer starts after the cooling time of 180 seconds finishes.

With "Temperature" condition, "Time" parameter is used as the maximum time to wait for the actual temperature to be under "Temperature" threshold.

#### **Important notice:**

When the temperature condition is selected to unlock the process chamber, the user must check that the temperature that is set can be measured by the selected sensor. Otherwise the chamber will only unlock when the additional time elapses.

The typical temperature ranges for the sensors are:

- |                               |                            |
|-------------------------------|----------------------------|
| • Thermocouple K type:        | Room temperature to 1000°C |
| • Low temperature Pyrometer:  | 150°C to 1100°C            |
| • High temperature pyrometer: | 400°C to 1400°C            |

If the high temperature pyrometer is selected with a temperature of 200°C, then this condition will never be fulfilled. The chamber will unlock when the additional time after the standard 180 seconds cooling elapses.

### 11.1.9 Force pick and download

|                                                         |
|---------------------------------------------------------|
| <input type="checkbox"/> Force pick and download recipe |
|---------------------------------------------------------|

When the option "Force pick and download" is **enabled**, the operator must pick and download the recipe before starting a process. The selected recipe clears from the listbox whenever:

- the operator navigates to another page
- the process is started
- the selected recipe changes

When the option "Force pick and download" is **disabled**, consecutive "Starts" are possible.

## 11.2 Traceability (option)

This software feature is optional and not provided with the standard software version.

The following area appears in the configuration window.

| < Traceability >                           |                   |
|--------------------------------------------|-------------------|
| <input checked="" type="checkbox"/> Enable |                   |
| Item 1                                     | Wafer number      |
| Item 2                                     | Lot number        |
| Item 3                                     | Project reference |

### 11.2.1.1 Enable

Enables or disables the traceability feature. The traceability parameters will be required before starting the process if the function is enabled.

### 11.2.1.2 Items 1, 2 and 3

Up to three items can be configured. They represent the header fields shown in traceability area. None of them is mandatory.

When the item is left empty, corresponding field is not displayed and the operator does not have to fill corresponding entry field.

When the item is filled in with a text, this text is displayed in the corresponding header. Filling the entry field becomes mandatory.

### 11.2.1.3 Traceability example

Let's assume the configuration of the previous figure.

The following fields **must** be filled in before a new process can be started:

- Historical name (always required)
- Wafer number
- Project name
- Project reference

The corresponding parameters will be saved in the traceability file.

### 11.2.1.4 Traceability files

The traceability files are located in the directory **C:\Program Files\Annealsys** and have the following name:

- Traceability\_YYYYMM.csv (*where YYYY is the year and MM is the month*).

The files have "Read only" attribute.

A new file is created each month.

See file management for file reading information (see 18.2).



### 11.3 Thermocouple calibration table

| Bits | Degrees |
|------|---------|
| 819  | 0.0     |
| 957  | 50.0    |
| 1096 | 100.0   |
| 1232 | 150.0   |
| 1365 | 200.0   |
| 1498 | 250.0   |
| 1636 | 300.0   |
| 1774 | 350.0   |
| 1914 | 400.0   |
| 2056 | 450.0   |
| 2196 | 500.0   |
| 2339 | 550.0   |
| 2480 | 600.0   |
| 2622 | 650.0   |
| 2760 | 700.0   |
| 2899 | 750.0   |
| 3035 | 800.0   |
| 3173 | 850.0   |
| 3305 | 900.0   |
| 3439 | 950.0   |

The thermocouple calibration tables are made of couples of digital and temperature values from standard K thermocouple converters used in the system. The thermocouple default calibration table has been adjusted in our facilities using a K thermocouple calibrator. This table must not be changed.

|  |                                                                                                                                                                                           |
|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <b>⚠ CAUTION</b>                                                                                                                                                                          |
|  | <b>Thermocouple calibration</b>                                                                                                                                                           |
|  | <b>The default calibration table is for K thermocouple.</b>                                                                                                                               |
|  | <b>Only K type thermocouples are used in the AS-One except special modification. The default thermocouple calibration table should not be changed without the agreement of Annealsys.</b> |
|  | <b>A different calibration table may lead to wrong thermal behavior of the furnace.</b>                                                                                                   |

For any special requirement for the thermocouple table contact Annealsys.

The user has anyway the capability to create new thermocouple calibration tables if he needs to adjust the calibration for a special process.

Each calibration table has a comment of 50 characters and up to 30 lines of 2 values:

- Thermocouple digital signal (0 to 4095)
- Thermocouple temperature in °C (0°C to 1200°C)

There is no limitation to the number of thermocouple calibration tables.

The available functions for thermocouple calibration table management are:

- New: to create a new table
- Browse: to open an existing table in a different folder than the default one
- Delete: to delete an existing table
- Print: to print the table that is displayed
- Save as: to save a table under a new name
- Save: to save table after creation or modification

The extension of the thermocouple calibration table files is ".tcc".

See the thermocouple calibration paragraph for further information on the calibration procedure.

#### 11.4 Pyrometer calibration tables

The pyrometer calibration tables are made of couples of digital and temperature values obtained during the calibration procedure.

|                                                                                   |                                                                                                                                                                                                                            |
|-----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <b>⚠ CAUTION</b>                                                                                                                                                                                                           |
|                                                                                   | <b>Pyrometer calibration</b>                                                                                                                                                                                               |
|                                                                                   | <b>A special care must be taken in creating or modifying the pyrometer calibration tables.</b>                                                                                                                             |
|                                                                                   | <b>In pyrometer control mode, the temperature of the substrate and the thermal behavior of the furnace depend on the calibration parameters. Wrong values in the table can lead to unexpected behavior of the furnace.</b> |
| <b>Refer to pyrometer calibration paragraph for further information.</b>          |                                                                                                                                                                                                                            |

< Pyrometer table >

|            |               |
|------------|---------------|
| Table name | CALSI         |
| User name  | Administrator |
| Emissivity | 1             |

☐ Low Type

**Comment**  
Pyrometer on center

< Commands >

New

Browse...

Delete

Print

Save as

Save

| Bits | Degrees |
|------|---------|
| 840  | 252.0   |
| 900  | 302.0   |
| 1070 | 352.0   |
| 1250 | 402.0   |
| 1470 | 452.0   |
| 1670 | 502.0   |
| 1835 | 552.0   |
| 1955 | 602.0   |
| 2065 | 652.0   |
| 2172 | 702.0   |
| 2275 | 752.0   |
| 2378 | 802.0   |
| 2483 | 852.0   |
| 2587 | 902.0   |
| 2695 | 952.0   |
| 2798 | 1002.0  |
| 2901 | 1052.0  |
| 3004 | 1102.0  |
| 3107 | 1152.0  |
| 3210 | 1202.0  |

Config  
PID  
TC table  
Pyro table

Each calibration table has up to 30 lines of 2 values:

- Digital signal value (0 to 4095)
- Pyrometer temperature in °C (0°C to 1400°C)

It is also possible to enter:

- A comment of 50 characters
- The type of pyrometer: high temperature or low temperature pyrometer (information only)
- Emissivity of the pyrometer (0.00 to 1) (information only)

There is no limitation to the number of pyrometer calibration tables.

The available functions for pyrometer calibration table management are:

- New: to create a new table
- Browse: to open an existing table in a different folder than the default one
- Delete: to delete an existing table
- Print: to print the table that is displayed
- Save as: to save a table under a new name
- Save: to save table after creation or modification

The extension of the pyrometer calibration table files is ".cal".

See the pyrometer calibration paragraph for further information on the calibration procedure.

### 11.5 PID parameter tables

The PID parameters table contains the temperature control parameters for the digital temperature controller.

< Table Information >

|                |         |
|----------------|---------|
| Table name     | SITCPID |
| User name      | Eng     |
| Digital filter | 0       |
| Offset         | 0       |

Comment

Silicon wafer TC control

Commands

New

Browse...

Delete

Print

Save as

Save

SITCPID

|                             | Slot 1 | Slot 2 | Slot 3 | Slot 4 | Slot 5 | Slot 6 | Slot 7 | Slot 8 |
|-----------------------------|--------|--------|--------|--------|--------|--------|--------|--------|
| High temperature limit (°C) | 110    | 210    | 310    | 410    | 510    | 620    | 820    | 1200   |
| Proportional (%)            | 6      | 6.5    | 5.5    | 4.9    | 5.8    | 8      | 9.7    | 9.1    |
| Integral (s)                | 3.06   | 2.44   | 2.5    | 2.36   | 2.51   | 2.57   | 1.95   | 1.69   |
| Derivative (s)              | 0.76   | 0.61   | 0.62   | 0.59   | 0.62   | 0.64   | 0.48   | 0.42   |
| Response time               | 1      | 1      | 1      | 1      | 1      | 1      | 1      | 1      |

Each PID parameter table **must have** 8 columns of 5 parameters:

- High temperature limit (°C): 0 to 1400
- P parameter (%): 0.00 to 100.00
- I parameter (s): 0.01 to 360.00
- D parameter (s): 0.00 to 360.00
- Control response parameter (Slow=0, Medium=1, Fast=2): 0 to 2

and 2 additional parameters:

- Digital filter time constant (s): 0 to 10
- Offset (°C): -100 to 100

It is also possible to enter:

- A comment of 50 characters

There is no limitation to the number of PID tables.

The available functions for PID table management are:

- New: to create a new table
- Browse: to open an existing table in a different folder than the default one
- Delete: to delete an existing table
- Print: to print the table that is displayed
- Save as: to save a table under a new name
- Save: to save table after creation or modification

The extension of the PID table files is ".pid".

Refer to the autotuning procedure for further information on the way to make the PID tables.

## 11.6 Administrator mode

The administrator mode gives access to the following additional features:

- Passwords administration
- System hardware configuration
- Gas safety interlocks

The screenshot displays the Administrator mode interface with the following sections:

- < Gas lines >**: A table with columns: Gas name, MFC range (sccm), Alarm limit (% of Setpoint), and Stop process.
 

| Gas name | MFC range (sccm) | Alarm limit (% of Setpoint) | Stop process                        |
|----------|------------------|-----------------------------|-------------------------------------|
| H2       | 1000             | 5                           | <input checked="" type="checkbox"/> |
| O2       | 1000             | 5                           | <input checked="" type="checkbox"/> |
|          | 0                | 0                           | <input checked="" type="checkbox"/> |
|          | 0                | 0                           | <input checked="" type="checkbox"/> |
|          | 0                | 0                           | <input checked="" type="checkbox"/> |
|          | 0                | 0                           | <input checked="" type="checkbox"/> |
|          | 0                | 0                           | <input checked="" type="checkbox"/> |
|          | 0                | 0                           | <input checked="" type="checkbox"/> |
|          | 0                | 0                           | <input checked="" type="checkbox"/> |
|          | 0                | 0                           | <input checked="" type="checkbox"/> |
|          | 0                | 0                           | <input checked="" type="checkbox"/> |
- < Pressure >**: A table with columns: Gauge name and Range (Pa).
 

| Gauge name                      | Range (Pa) |
|---------------------------------|------------|
| Full range                      | 100000     |
| Capa 1                          | 1333.22    |
|                                 | 0          |
|                                 | 0          |
| Local atmospheric pressure (Pa) | 100000     |
| Unit                            | mBar       |
- < Temperature >**: A table with columns: Alarm limit (% of Setpoint), Alarm delay (s), Power high limit (%), and Power rate (% / s).
 

| Alarm limit (% of Setpoint) | Alarm delay (s) | Power high limit (%) | Power rate (% / s) |
|-----------------------------|-----------------|----------------------|--------------------|
| 5                           | 0               | 100                  | 0                  |

 A checkbox labeled "Stop process" is checked.
- < Open door condition >**: A table with columns: Condition and Time (s).
 

| Condition | Time (s) |
|-----------|----------|
| None      | 120      |
|           | 0        |
- < Commands >**: A "Save" button.

On the right side, there is a vertical navigation bar with buttons: Config, Hardware, Password, PID, TC table, and Pyro table.

Additional buttons are available in the secondary navigation bar when the software is in administrator mode.

The "Hardware" button allows setting the gas safety interlocks and hardware configuration.

The "Password" button, allows password administration.

### 11.6.1 Passwords administration

The password administration requires the administrator to enter again his login and password.

The screenshot shows a dialog box titled "Type login and password". It contains two input fields: "Login" and "Password". Below the input fields is an "Ok" button.

The administrator has access to the password management:

| Passwords and access management     |          |                                              |               |                              |
|-------------------------------------|----------|----------------------------------------------|---------------|------------------------------|
| <input checked="" type="checkbox"/> | Login    | <input checked="" type="checkbox"/> Password | Level         | Data path                    |
| <input type="checkbox"/>            | Adm      | aaaa                                         | Administrator | >                            |
| <input type="checkbox"/>            | Ope      | oooo                                         | Operator      | >                            |
| <input type="checkbox"/>            | Eng      | eeee                                         | Engineer      | >                            |
| <input checked="" type="checkbox"/> | User_001 | password                                     | Operator      | C:\Program Files\AnnealSys > |

< Commands >

This function allows entering and modifying the logins and passwords of the users.

The administrator can also assign a private folder to each user called "Data path". Recipes and historicals files are stored in this specific folder.

The buttons at the right of data path entry fields open a folder browser dialog window from which the data path can be chosen. If the folder name does not exist, then it is created at logging time.

An empty data path defaults to "C:\Program files\Annealsys" folder.

It is possible to enter an unlimited logins with their associated password and data path.

If a user login is not saved in the configuration file the access to the software will not be allowed.

A password should have 4 to 10 characters.

If the checkbox near "Password" subtitle box is checked, the passwords are displayed. Otherwise the screen displays stars.

Use the "Add user" button to add a new user. Then enter login, password and select the access level and data path.

To delete a user, check the box on in front of the login on the left side and press "Delete user(s)". The user will be erased from the list.

### 11.6.2 System hardware configuration

| User configuration                            |     |
|-----------------------------------------------|-----|
| Converters voltage                            | 400 |
| <input checked="" type="checkbox"/> Pyro 1 HT |     |
| <input type="checkbox"/> Pyro 2 HT            |     |

| System configuration        |           |
|-----------------------------|-----------|
| System name                 | AS-ONE    |
| Model                       | ASONE_150 |
| Temperature controller type | RKC       |
| Turbo pump type             | PFEIFFER  |
| Number of current sensors   | 3         |
| Number of MFM               | 0         |
| Number of MFC               | 2         |
| MFC type                    | BROOKS    |
| Vaporizer type              | WITHOUT   |

| Check currents parameters |       |
|---------------------------|-------|
| Item                      | Value |
| Threshold (%)             | 5     |
| Delta (%)                 | 7     |
| Time (s)                  | 0     |
| Power threshold (%)       | 0     |

| Gas interlocks |                                     |                          |                                     |                                     |
|----------------|-------------------------------------|--------------------------|-------------------------------------|-------------------------------------|
|                | Air                                 | Pressure                 | H2                                  | O2                                  |
| H2             | <input checked="" type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/>            | <input checked="" type="checkbox"/> |
| O2             | <input type="checkbox"/>            | <input type="checkbox"/> | <input checked="" type="checkbox"/> | <input type="checkbox"/>            |

< Commands >

Save

The system hardware configuration allows:

- Setting the power block voltage
- Setting the pyrometer type
- Checking the system configuration
- Setting the lamp failure detection parameters
- Restoring the PLC program
- Setting the gas interlocks

#### 11.6.2.1 Setting the power block voltage

The lamps are supplied with power blocks that runs in power control mode. In order to avoid lamp breakage hardware jumpers on the power blocks set the voltage maximum value.

|                    |     |
|--------------------|-----|
| Converters voltage | 400 |
|--------------------|-----|

This value is set at Annealsys before shipment of the system and should not be modified without the prior written agreement of Annealsys.

This value is saved in the configuration mode for user information and for the current calculation.

#### 11.6.2.2 Setting the pyrometer type

|                                               |
|-----------------------------------------------|
| <input checked="" type="checkbox"/> Pyro 1 HT |
| <input type="checkbox"/> Pyro 2 HT            |

The pyrometer type, low or high temperature, is set in this form. This is for information only. It allows the user checking that the pyrometer tables are made for the right type of pyrometer. This can be checked when downloading the recipe.

There is no automatic check of the pyrometer table type by the software.

### 11.6.2.3 Checking the system configuration

| System configuration        |           |
|-----------------------------|-----------|
| System name                 | AS-ONE    |
| Model                       | ASONE_150 |
| Temperature controller type | RKC       |
| Turbo pump type             | PFEIFFER  |
| Number of current sensors   | 3         |
| Number of MFM               | 0         |
| Number of MFC               | 2         |
| MFC type                    | BROOKS    |
| Vaporizer type              | WITHOUT   |

The actual system configuration with the optional features that are installed is set at Annealsys facilities and is saved in the PLC controller.

When the PC software starts and if a new version is installed, the hardware configuration is automatically updated from the PLC.

The user has no access to modify the hardware configuration but it can be checked using the configuration mode.

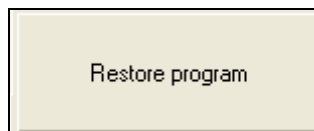
### 11.6.2.4 Setting the lamp failure detection parameters

| Check currents parameters |       |
|---------------------------|-------|
| Item                      | Value |
| Threshold (%)             | 5     |
| Delta (%)                 | 7     |
| Time (s)                  | 0     |
| Power threshold (%)       | 0     |

The current parameters are used for the lamp failure detection. These parameters are set at Annealsys facilities and should allow the utilization of the system in most of the applications.

It may happen sometime that the parameters should be adjusted in order to avoid some unexpected alarm. Please contact Annealsys in case of requirement for modification of these parameters.

### 11.6.2.5 Restoring the PLC program



When the system starts the PC check the integrity of the PLC program. In case of PLC program is not running (for example because of battery failure), the PLC program is automatically restored by the PC.

In case the automatic restoration program would not be successful the user has the capability to activate the PLC program restoration.

This function should normally never been used by the customer. Contact Annealsys if you need to restore the PLC program by this way.

## 11.6.2.6 Gas safety interlocks

|                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <b>⚠ WARNING</b>                                                                                                                                                                                                                                                                                                                                                                                                    |
|                                                                                   | <p><b>PROCESS GASES INTERLOCKS</b></p> <p>The interlocks are set by Annealsys to avoid dangerous gas operations. The gas safety interlocks must not be modified without the prior written approval of Annealsys.</p> <p>In spite of these gas interlocks, the user must take all the necessary precautions to prevent hazardous mixtures occurring inside the process chamber during the process step sequence.</p> |

The gas interlocks are part of the hardware settings. This module is located at the right side of the window.

| Gas interlocks |                                     |                          |                                     |                                     |
|----------------|-------------------------------------|--------------------------|-------------------------------------|-------------------------------------|
|                | Air                                 | Press<br>ure             | H2                                  | O2                                  |
| H2             | <input checked="" type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/>            | <input checked="" type="checkbox"/> |
| O2             | <input type="checkbox"/>            | <input type="checkbox"/> | <input checked="" type="checkbox"/> | <input type="checkbox"/>            |

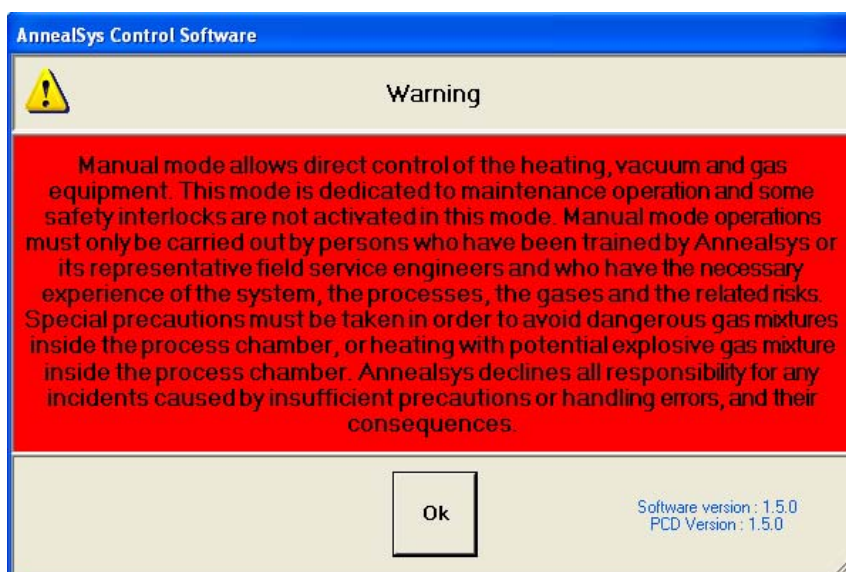
For each gas the interlock will have the following functions **if the box is checked**:

| Gas 1    | Air                                                                         | Comment                                                                                                                        |
|----------|-----------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| Air      | Gas cannot be introduced if the chamber has not been previously pumped down | Activated if risk when gas is mixed with air<br>Example: H <sub>2</sub> and air (possible explosion when heating)              |
| Pressure | Gas cannot be introduced if pressure>10 Torr                                | If risk to have gas at high pressure<br>Example: dangerous gases                                                               |
| Gas 1    | No effect                                                                   |                                                                                                                                |
| Gas 2    | Gas 1 cannot be introduced simultaneously with gas 2                        | Activated if risk when mixture between 2 gases<br>Example: H <sub>2</sub> and O <sub>2</sub> (possible explosion when heating) |
| Gas 3    | Gas 1 cannot be introduced simultaneously with gas 2                        | Activated if risk when mixture between 2 gases<br>Example: H <sub>2</sub> and O <sub>2</sub> (possible explosion when heating) |

Contact Annealsys customer support if you need any further information.



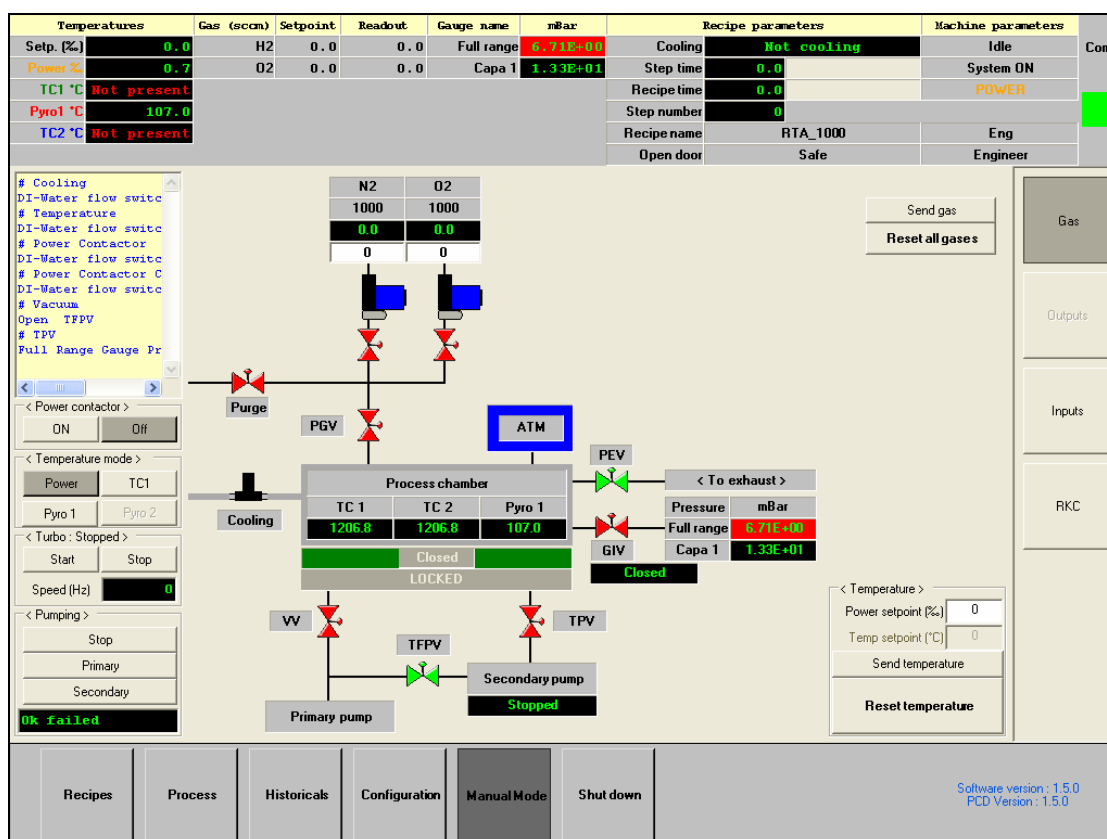
## 12. MANUAL MODE



The above message is displayed each time the user enters the manual mode to warn about the potential risks of utilization of this mode. Read the message and acknowledge by clicking on "OK"

The safety software module is not activated in the manual mode in order to allow easier maintenance and checking operations. Anyway hardware interlocks are active and will prevent most of the serious damage to the system. Some software interlocks are still activated as well.

When the manual mode is selected the screen displays:

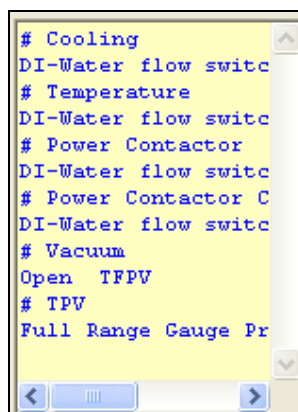


*In the above example the system has optional features.  
The synoptic and display change function of the actual system configuration.*

The secondary navigation buttons at the right side propose the following options:

- Gas for manual control of the different sub-systems (default form displayed on the screen)
- Outputs (Access reserved to Administrator and Annealsys for maintenance purposes)
- Inputs to displays the status of the logic inputs of the system
- RKC : for autotuning of the temperature controller

The yellow frame at the upper-left side gives information on interlocks and some actions. It allows an easier control of the system in manual mode:



The manual mode allows running the following operations:

- Manual control of cooling
- Manual control of vacuum equipment
- Manual control of gas panel
- Manual heating
- Adjustment of PID parameters

## 12.1 Gas form

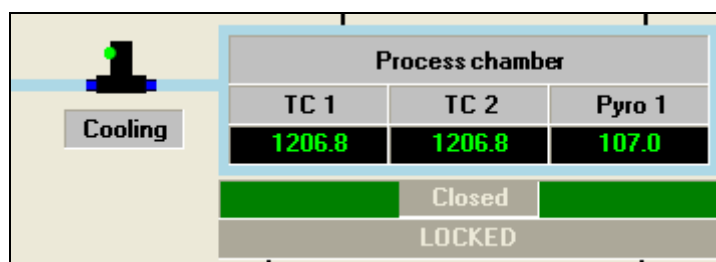
The gas form is the default one displayed when the user selects the manual mode. It displays a synoptic of the machine and the status of the valves and main sub-systems.

It gives access to the control of the different sub-systems:

- Cooling
- Reactor locking and unlocking
- Heating
- Gas panel control
- Vacuum equipment

### 12.1.1 Manual control of cooling

Click on water valve to open and close it. A green indicator is on when the water valve is switched on. The water line is colored in blue if the water flow controller detects a sufficient flow.



The lamp cooling fan of the infrared furnace is switched on when the water valve is switched on.

### 12.1.2 Reactor locking and unlocking

The operator can close the chamber by clicking on "UNLOCKED". The machine will activate the system to lock the chamber.

When the chamber is locked, the message on the button turns to "LOCKED". The chamber can be opened by clicking again on the same button.

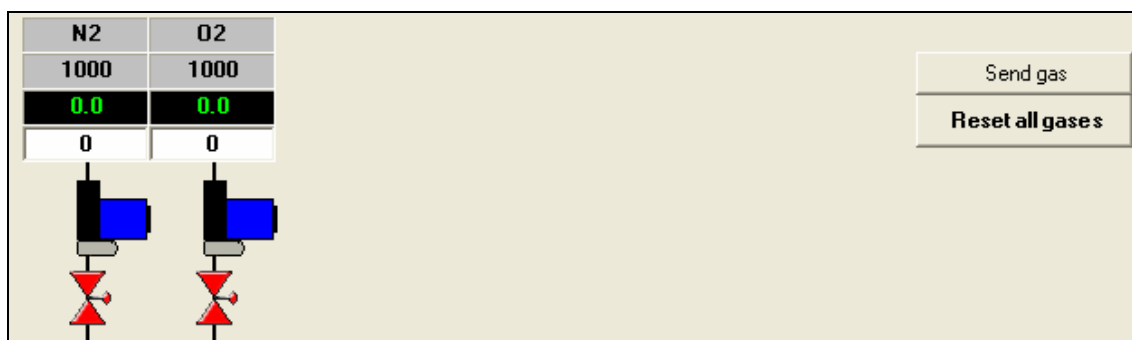
**If the chamber is under vacuum it is not possible to open the chamber.  
The chamber must be purged previously to open.**

### 12.1.3 Gas panel control

|                                                                                   |                                                                                                                                                                                                                                                                                                                                                       |
|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <p style="text-align: center;"><b>⚠ WARNING</b></p> <p><b>GAS HAZARD</b></p> <p><b>Gas and air mixing interlock.</b></p> <p>Some gases should not be mixed with air or with other gases. The user must check gases compatibility and avoid any dangerous mixture and especially any potentially explosive gas mixture inside the process chamber.</p> |
|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

The screen displays the gas lines that are installed and that have been selected in the configuration mode.

The chamber must be locked to allow flowing gases.



On the top of each line the mass flow controller range is indicated (1000, and 1000 sccm in the example).

The mass flow setpoint can be set in the small white box. The setpoint value is not applied and the user must click on the "Send gas" button to apply the setpoint to the mass flow controller.

The gas valve of the corresponding gas line will be opened as soon as a setpoint different from zero is applied to the mass flow controller. The valve will turn from red to green when it is open.

The mass flow reading is displayed in green over black.

If some interlocks have been set to avoid some gas mixtures, they will be active in the manual mode.

Click on "Reset all gases" to reset all mass flow controller setpoints and to close all the process gas valves.

**Note: If the system is equipped with UHV kit, then the PGV valve must be open by clicking on it in order to allow gas flowing to the process chamber.**

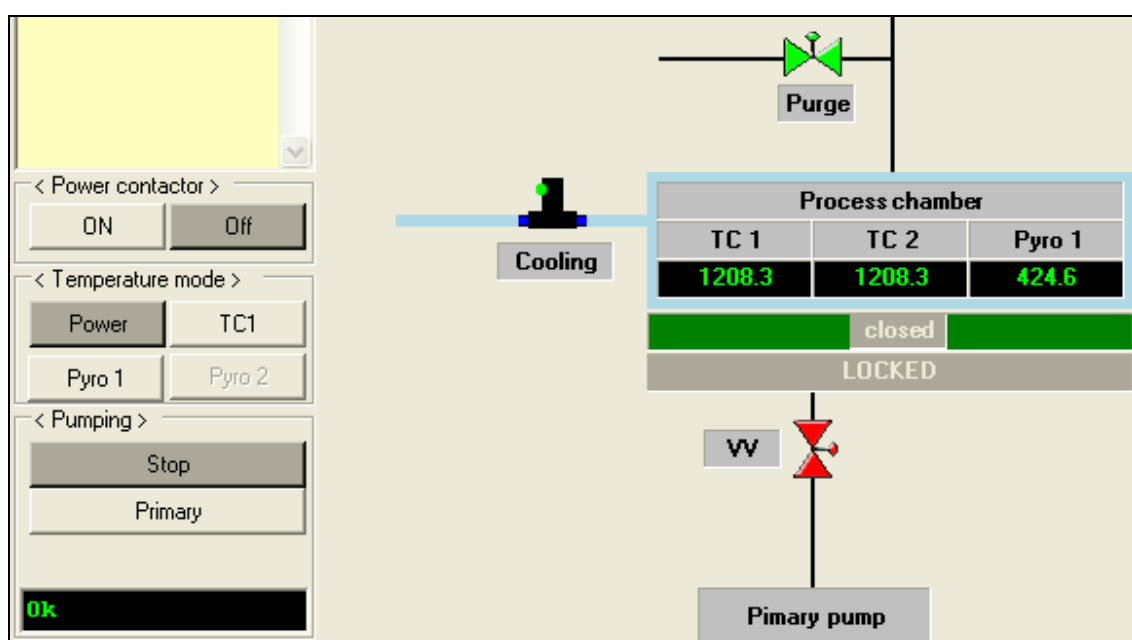
|                                                                                   |                                                                                                                                                                                                              |
|-----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <b>⚠ WARNING</b>                                                                                                                                                                                             |
|                                                                                   | <p><b>GAS HAZARD</b></p> <p><b>Purge chamber before opening.</b></p> <p><b>Pump down and purge the process chamber after flowing process gases in order to be sure to open it under safe conditions.</b></p> |

#### 12.1.4 Purge valve (PUV)

Click on the purge valve (PUV) to open and close it. The valve is green when it is opened and red when it is closed.

#### 12.1.5 Vacuum equipment control

The user can control the vacuum equipment with the buttons at the lower left side of the manual mode gas form.



There is a frame < Pumping > at the lower left corner of the form. The pumping control buttons are:

- "Stop": close the vacuum valve (VV)
- "Primary" : open the vacuum valve (VV)

The vacuum valve turns to green color when it is opened. It is red color when it is closed.

The different vacuum gauges are displayed on the synoptic depending on the configuration (constraint gauge and one capacitance manometer in the example on the above picture).

The system indicates if the chamber is under vacuum or not. The chamber can be filled to atmospheric pressure using the purge valve (PUV).

### 12.1.6 Optional vacuum equipment control

The AS-One can receive the following optional vacuum equipment:

- Pressure control
- UHV kit
- Turbo pumping unit

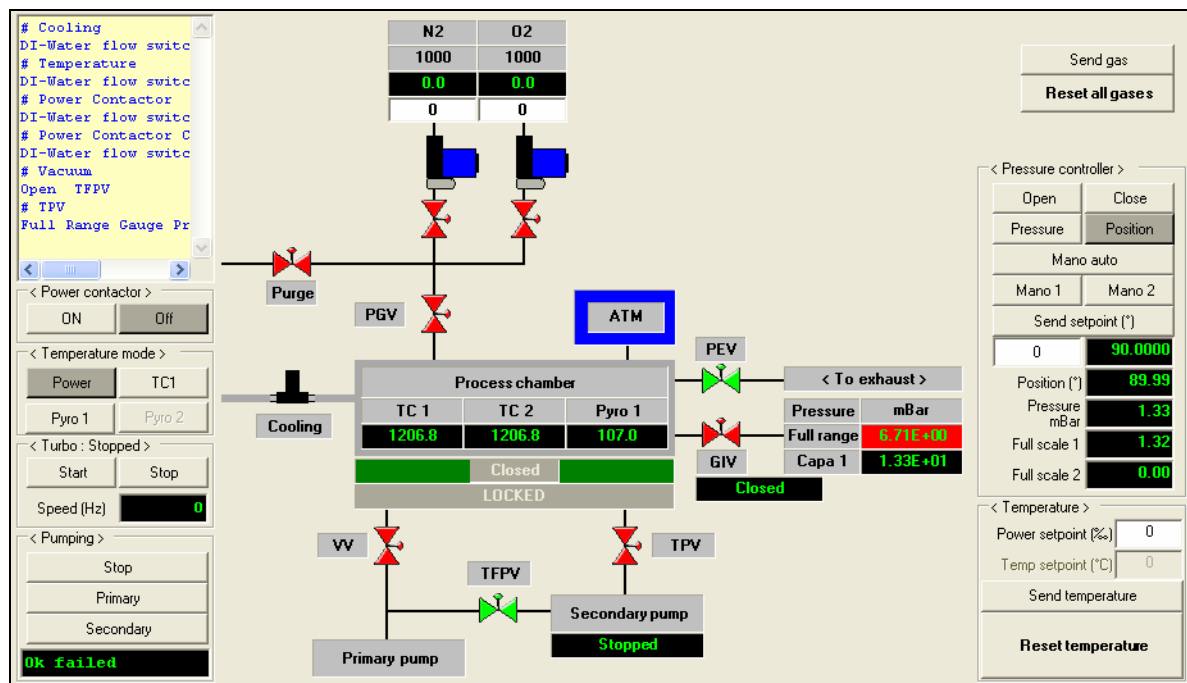
The pressure control is a throttle valve with integrated controller. The manual mode offers to control the valve in pressure or angle mode.

The UHV kit consists of 2 gas isolation valves on reactor gas inlet (PGV) and reactor gas outlet (PEV) and a pressure switch to detect the atmospheric pressure.

The turbo pump comes as a kit with a full range gauge and several valves that allows using the system at different pressures.

All the options are fully controlled by the system.

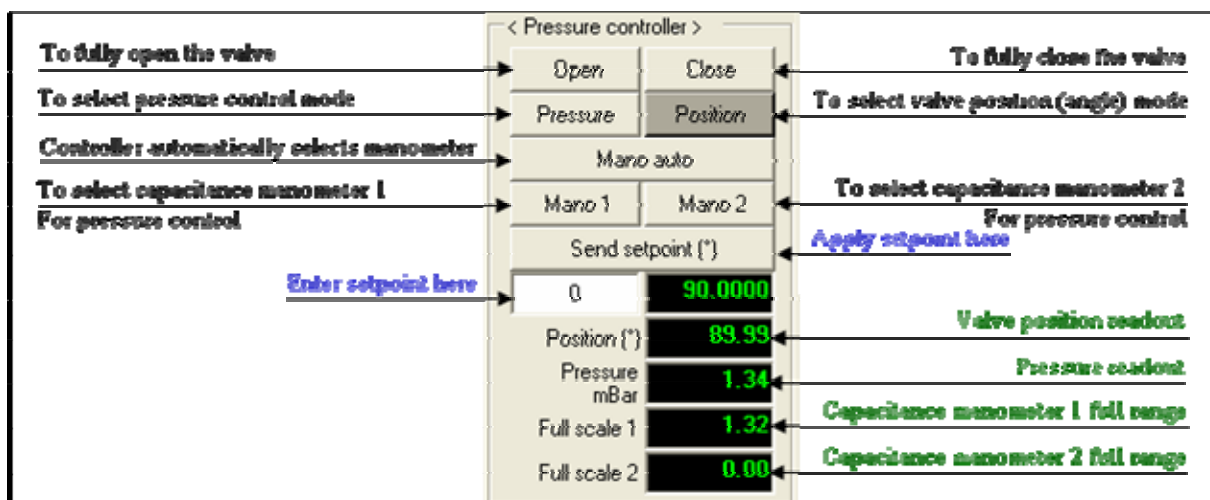
When the AS-One is equipped with the additional vacuum options the manual control form is displayed as here after:



The user has full access to the different features but some sub-systems are automatically controlled for safety reasons.

### 12.1.6.1 Pressure control

The throttle can be pressure or angle (position) controlled. The operator can also fully close or open the valve with the corresponding buttons. Up to 2 capacitance manometers can be connected to the same controller.



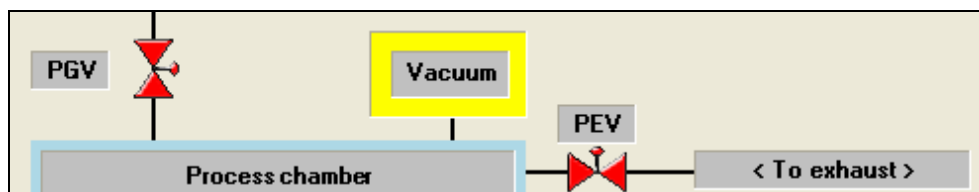
The pressure control can be set as indicated on the above picture.

### 12.1.6.2 UHV kit

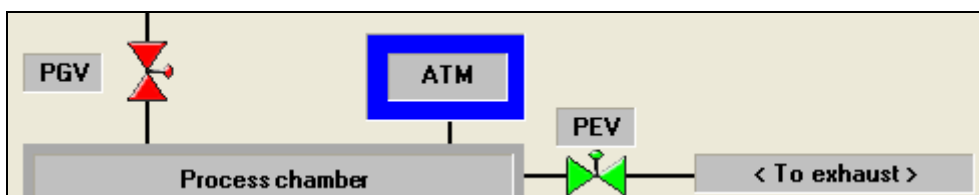
The user can control the PGV valve (Process gas valve) by clicking on the valve icon.

The PEV valve (Purge exhaust valve) is automatically controlled by the pressure switch.

The threshold value of the pressure switch is set to detect the atmospheric pressure. The system indicates "Vacuum" if the pressure is below the atmospheric pressure. In this case PEV is closed.



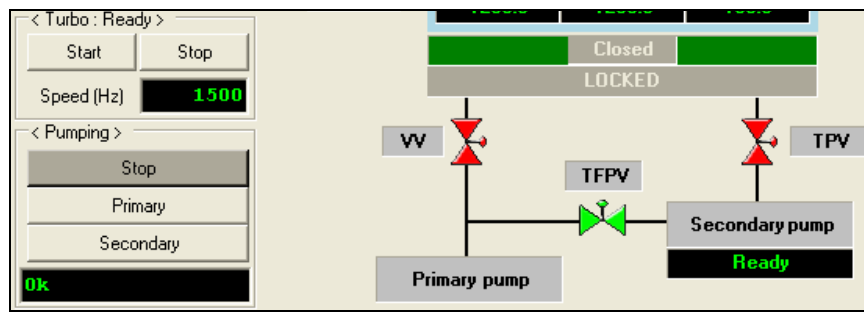
It displays "ATM" if the pressure inside the process chamber is above the atmospheric pressure. In this case PEV is open.



### 12.1.6.3 Turbo pump

|                                                                                     |                                                                                                                                                                                                                                                                                             |
|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <b>⚠ CAUTION</b>                                                                                                                                                                                                                                                                            |
|                                                                                     | <p><b>Turbo pump</b><br/> <b>Risk of turbo pump damage with gas flow.</b><br/> <b>Purge or process gases must not be introduced inside the process chamber when the system is under turbo pumping.</b><br/> <b>Introducing gases under turbo pumping may lead to turbo pump damage.</b></p> |

When the turbo pump is installed the manual mode allows starting and stopping the turbo pump and running vacuum sequences.



The turbo pump can be started and stopped using the "Start" and "Stop" buttons in the turbo pump frame.

The computer displays the rotating speed of the pump in Hz when the turbo pump is running. (The speed function is not available for Turbovac 50 turbo pump). When the turbo pump reaches the nominal speed, the information "Ready" is displayed on the screen.

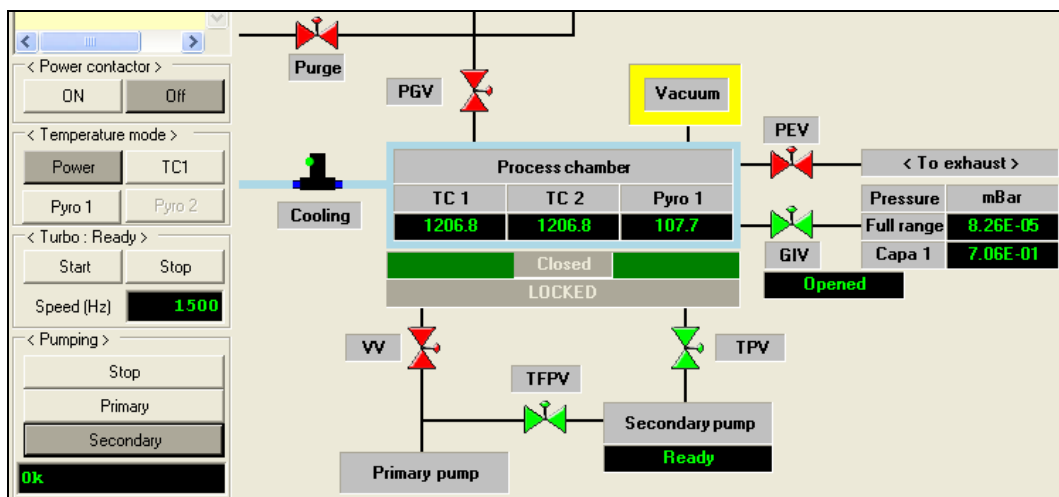
The vacuum sequences are controlled using the buttons of the pumping frame "Stop", "Primary" and "Secondary" (Turbo). The system will manage the valves accordingly.

**Note: Some safety conditions must be fulfilled to run the vacuum sequences. They are described hereafter.**

The gauge isolation valve (GIV) will open if one of the following conditions is fulfilled:

- The rough pump vacuum valve (VV) is opened **and** all the gas valves (from the process gas lines) are closed for more than 30 seconds. The GIV valve opens if these 2 conditions are fulfilled simultaneously for 30 seconds
- The turbo pump valve is opened

The gauge isolation valve automatically closes if one of the process gas valves opens.

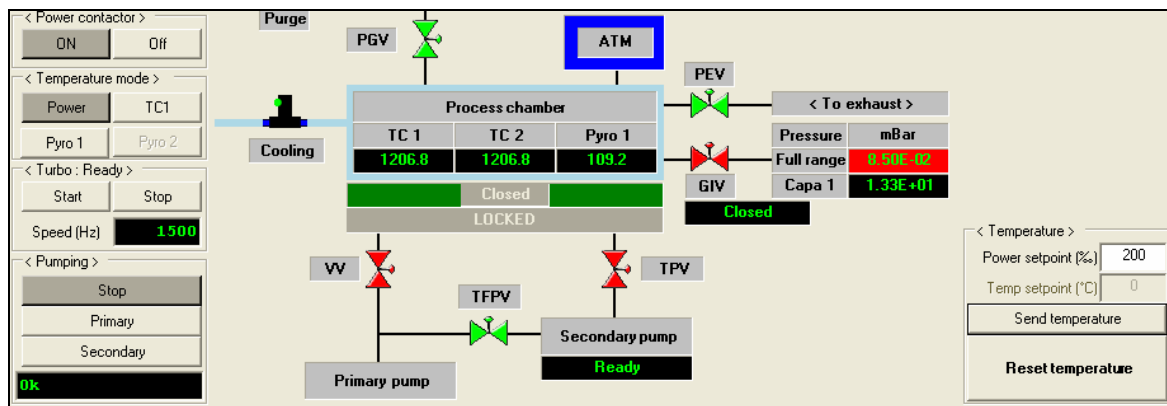


Before a turbo pumping step the user must run a primary pumping with the following conditions:

- The primary pumping must be long enough in order to reach a pressure < 200 Pascal.
- No process gases should be introduced during this step
- The step duration must be at least 35 seconds to allow opening the GIV valve
- The turbo pump must have reached the nominal speed

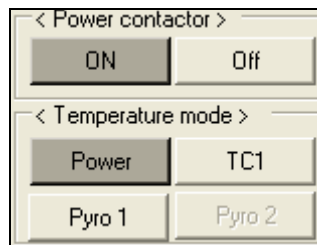
If these conditions are not fulfilled the turbo pump valve will not open.

### 12.1.7 Manual heating



There are 2 frames to run the manual heating. At the left side of the screen the frame gives access to the following functions:

- Power contactor: "On" or "Off"
- Temperature control mode: "Power", "TC1", "Pyro1" or "Pyro2" (option)



Before to close the power contactor, the cooling must be started.

The following conditions must be fulfilled to close the power contactor:

- System on
- No overheating
- Water cooling (all flow switch detects sufficient flow)
- Chamber is locked
- No external failure
- The system is in power mode
- Setpoint = 0

The system can be switched to "TC1", "Pyro1" or "Pyro2" (option) after the power contactor has been switched on.

At the lower right side the user can set the power or temperature setpoint depending on the selected heating mode.

|  |                                                                                                                                                                                                                                                                            |
|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <b>⚠ CAUTION</b>                                                                                                                                                                                                                                                           |
|  | <p><b>Power mode.</b></p> <p><b>Possible overheating of the substrate and the furnace</b></p> <p><b>Do not set power over 30% when start heating. Higher setpoint will heat to temperature over 1100°C and can melt the silicon wafer and damage the thermocouple.</b></p> |



When a setpoint is set in the box it can be applied by clicking on "Send Temperature" button.

|                    |     |
|--------------------|-----|
| < Temperature >    |     |
| Power setpoint (%) | 200 |
| Temp setpoint (°C) | 0   |
| Send temperature   |     |
| Reset temperature  |     |

The heating setpoint can be reset at any time by clicking on "Reset temperature".

The heating will stop if there is no sufficient cooling or in case of overheating.

|                                                                                   |                                                                                                                                                                                                                                                                                                                                |                                                                                     |
|-----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
|  | <p style="text-align: center;"><b>⚠ WARNING</b></p> <p><b>Burn hazard</b><br/>After heating both substrate and quartz window may still be at high temperature.<br/>Wait for sufficient cool down of the substrate for unloading. Wear high temperature gloves to load and unload the substrate inside the process chamber.</p> |  |
|-----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|

## 12.2 Inputs

AS-One

At any time it is possible to display the system logic inputs status by clicking on the "Inputs" button.

| Digital inputs | State | Digital inputs | State | Digital inputs | State | Digital inputs | State |
|----------------|-------|----------------|-------|----------------|-------|----------------|-------|
| SYSTEM_ON      | ON    | TFPVC          | OFF   |                |       |                |       |
| TS             | ON    | VFPE           | ON    |                |       |                |       |
| WFS1           | OFF   |                |       |                |       |                |       |
| RPS            | ON    |                |       |                |       |                |       |
| CAPS           | ON    |                |       |                |       |                |       |
| VPF            | ON    |                |       |                |       |                |       |
| EXTF1          | ON    |                |       |                |       |                |       |
| MMA            | OFF   |                |       |                |       |                |       |
| DOC            | ON    |                |       |                |       |                |       |
| VVO            | OFF   |                |       |                |       |                |       |
| VVC            | ON    |                |       |                |       |                |       |
| TPVO           | OFF   |                |       |                |       |                |       |
| TPVC           | ON    |                |       |                |       |                |       |
| GIVO           | OFF   |                |       |                |       |                |       |
| GIVC           | ON    |                |       |                |       |                |       |
| TFPVO          | ON    |                |       |                |       |                |       |

| Mnemonic  | Digital input           | Status = ON                                                                                                |
|-----------|-------------------------|------------------------------------------------------------------------------------------------------------|
| System ON | System on               | System powered ON by green push button on front panel                                                      |
| TS        | Thermal switches        | Lamp reflector thermal switch is closed and thermal switch of the lamp fans area is closed: No overheating |
| WFS1      | Water flow switch 1     | Sufficient water flow in circuit 1: Single circuit for AS-One                                              |
| RPS       | Reactor pressure switch | UHV kit option: Atmospheric pressure when ON                                                               |
| CAPS      | Compressed air pressure | Sufficient air pressure for system operations                                                              |
| VPF       | Vacuum pump failure     | Vacuum pump option:<br>Vacuum pump circuit breaker or contactor is OFF                                     |
| EXF1      | External failure 1      | External failure loop is closed (No alarm)                                                                 |
| MMA       |                         | Annealsys use only                                                                                         |
| DOC       | Door closed             | Chamber is closed                                                                                          |
| VVO       | Vacuum valve open       | Valve is opened if ON                                                                                      |

**Additional valve statuses (open and closed) are added for the turbo pump option:**

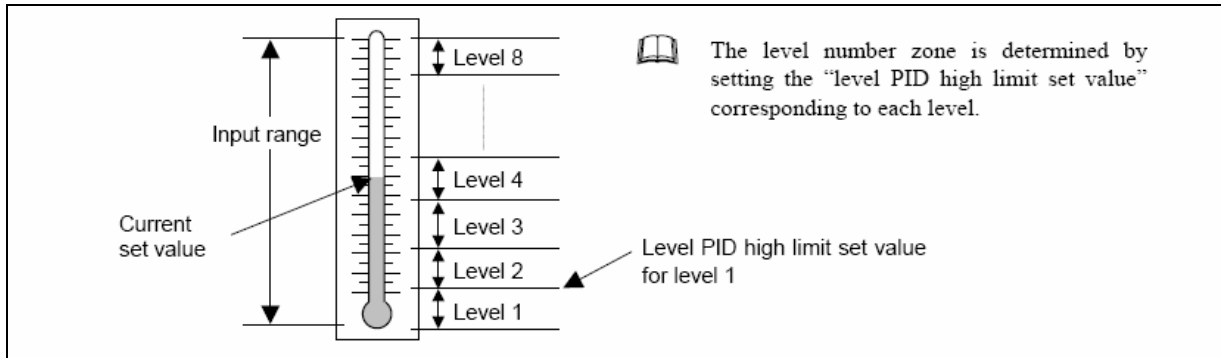
| Mnemonic | Digital input                                               | Status = ON                                                                           |
|----------|-------------------------------------------------------------|---------------------------------------------------------------------------------------|
| VVC      | Vacuum valve closed                                         | Valve closed if ON                                                                    |
| TPVO     | Turbo pump valve open                                       | Valve open if ON                                                                      |
| TPVC     | Turbo pump valve closed                                     | Valve closed if ON                                                                    |
| GIVO     | Gauge isolation valve open                                  | Valve open if ON                                                                      |
| GIVC     | Gauge isolation valve closed                                | Valve closed if ON                                                                    |
| TFPVO    | Turbo forepump valve open                                   | Valve open if ON                                                                      |
| TFPVC    | Turbo forepump valve closed                                 | Valve closed if ON                                                                    |
| VFPE     | Vacuum forepump failure<br>(Same than VPF if only one pump) | Turbo pump option. Pump OFF if OFF<br>Vacuum pump circuit breaker or contactor is OFF |

### 13. SETTING THE TEMPERATURE CONTROL PARAMETERS

#### 13.1 Description of temperature control parameters

The Annealsys RTP furnaces use a state of the art **fast digital PID temperature controller**. The software gives full access to the temperature control parameters and makes the tuning of the controller very easy.

The PID function is divided into up to 8 levels. For each temperature level the system uses a different set of PID parameters. Each set of parameters is used up to the high temperature level.



For each temperature level it is possible to set 5 parameters:

- High temperature limit: Range: 0 to 1200°C
- P: proportional action: Range: 0 to 1000
- I: integral action: Range: 0.01 to 360.00
- D: derivative action: Range: 0.00 to 360.00
- Control response: Values: slow, medium, fast

##### 13.1.1 High temperature limit

This item is the high limit value of each level area.

The temperature scale is from 0 to 1200°C for the thermocouple and it is from 0°C to 1400°C for the high temperature pyrometer. The temperature controller is set with a 1200°C full scale. A proportional ratio is applied to the high level temperature depending on the pyrometer calibration. This is explained in the procedure for setting the temperature control parameters. (Chapter 13.2)

##### 13.1.2 PID parameters

The PID parameters are usual parameters used for the close loop control.

- P = proportional action
- I = integral parameter
- D = derivative parameter

##### 13.1.2.1 Proportional

The proportional action is also called gain. The temperature controller compares the difference between the setpoint value and the temperature readout. The output control is proportional to the deviation (setpoint – temperature readout).

This proportional action is active within a zone around the setpoint called the proportional band. When the temperature enters the proportional band, the output becomes gradually smaller and the temperature stabilizes somewhere within the proportional band.

If there is only a proportional parameter, the actual temperature will not stabilize exactly on the setpoint. It usually becomes stable with some deviation called offset.

### 13.1.2.2 Integral action

The degree of integral action is expressed as integral time in seconds. The purpose of the integral action is to automatically compensate any steady state offset inherent with a proportional controller.

The integral action moves or resets the proportional band up or down depending on the offset.

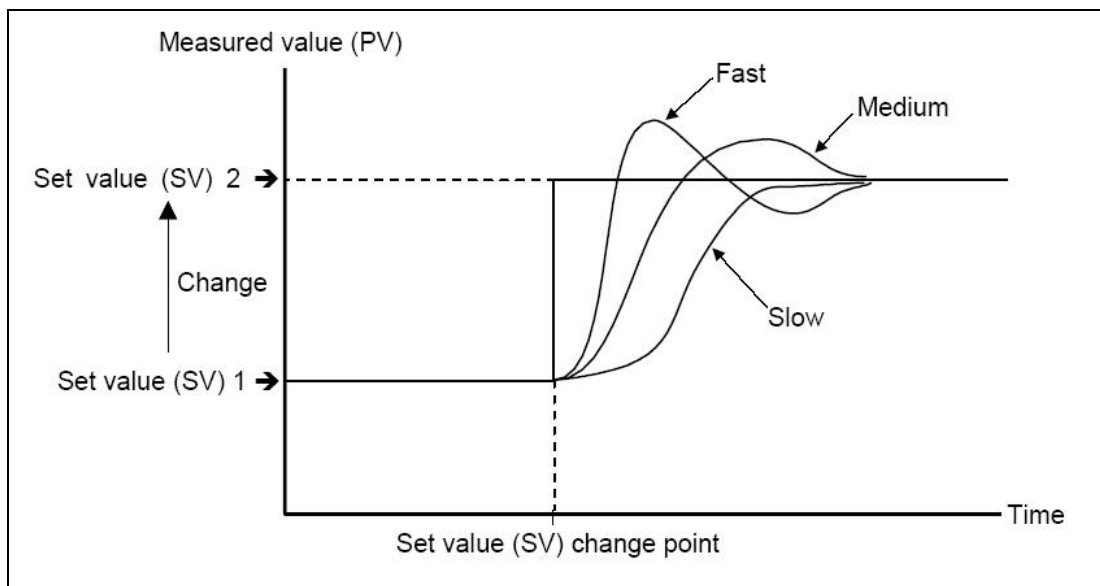
### 13.1.2.3 Derivative action

Derivative action is also called rate. The degree of derivative action is expressed by derivative time. This action on the controller is adjustable in seconds. The controller measures the rate of the temperature increase and moves the proportional band to minimize overshoot. The output change is directly proportional to the rate of change in the process value.

### 13.1.3 Control response

The control response time is set by the user. This parameter allows setting the speed of the control and will mainly impact on the overshoot. If "fast" is selected the temperature control will be fast and an overshoot will occur at the end of the ramping up. If the overshoot is critical the parameter should be set to slow.

For the AS-One system the controller is usually set with a control response time to "medium". The "slow" value may reduce the power peak when switching from a temperature control mode to another.



### 13.1.4 Offset parameter

The offset parameter is set for the whole range of temperature.

It allows compensating temperature offset between setpoint and actual temperature measured by the sensor used for the temperature control.

Offset should be reduced if the actual temperature is below setpoint and increased if the actual temperature is over the setpoint.

#### Example:

If the setpoint is 850°C and the actual temperature during process is 848°C and if the current offset is zero, then the offset must be set to - 2.

### 13.2 Setting the temperature control parameters

The manual mode gives access to the setting of the temperature control parameters clicking on the "RKC" button.

The screenshot displays the ANNEALSYS RTP interface. On the left, a table shows temperature control parameters for Channel 1 and three temperature levels. The main area contains three sections: RKC control, Temperature control mode, and Power contactor. The RKC control section has buttons for Auto, Manual, Remote, Local, Run, and Stop. The Temperature control mode section has buttons for Power, TC1, Pyro 1, and Pyro 2. The Power contactor section has buttons for ON and Off. Below these sections are buttons for << Copy <<, Save as ..., and Download PID table ... A Refresh checkbox is also present. A log window at the bottom shows a series of 'Uploading ...' and 'Upload Ok' messages with timestamps.

| Channel 1                 |          |
|---------------------------|----------|
| Temperature readout (°C)  | 4.50     |
| Actual power (%)          | 0.0      |
| Setpoint(°C)              | 0.00     |
| Power (%)                 | 0.0      |
| Autotuning                | Autotune |
| P (%)                     | 4.00     |
| I (s)                     | 13.64    |
| D (s)                     | 3.41     |
| Control response(0,1,2)   | 1        |
| Digital filter (s)        | 0.20     |
| Offset(add to measure °C) | 3.00     |

| Temperature level 1 parameters |        |
|--------------------------------|--------|
| High temperature limit (°C)    | 280.00 |
| P (%)                          | 4.00   |
| I (s)                          | 13.64  |
| D (s)                          | 3.41   |
| Control response(0,1,2)        | 1      |

| Temperature level 2 parameters |         |
|--------------------------------|---------|
| High temperature limit (°C)    | 1200.00 |
| P (%)                          | 3.00    |
| I (s)                          | 3.61    |
| D (s)                          | 0.90    |
| Control response(0,1,2)        | 1       |

| Temperature level 3 parameters |         |
|--------------------------------|---------|
| High temperature limit (°C)    | 1200.00 |

The user can start the cooling and switch on the power contactor in the RKC mode. They must be started before heating.

The RKC mode is used to set the temperature control parameters. It has different functions:

- Perform heating in power mode or temperature control mode
- Display the temperature values
- Display and set the values of the temperature control parameters

The actual temperature values are displayed in the main parameter band on the top of the screen.

### 13.3 Action buttons

The manual heating is controlled using the action buttons in the middle of the form.

The screenshot displays the ANNEALSYS RTP interface, focusing on the RKC control section. It shows buttons for Auto, Manual, Remote, Local, Run, and Stop. The Temperature control mode section has buttons for Power, TC1, Pyro 1, and Pyro 2. The Power contactor section has buttons for ON and Off. The Cooling section has buttons for ON and Off.

#### 13.3.1 Cooling

To start and stop the cooling.

#### 13.3.2 Power contactor

To close the power contactor to allows heating. Cooling must be start prior switching the power contactor.

### 13.3.3 Auto - Manual

When in automatic mode the temperature control runs in closed loop. In manual mode the system is in open loop and runs in power mode (setpoint is applied directly to the lamps).

When the "Auto" button is in dark grey the system is in temperature control mode (auto). A click on the "Manual" button will switch to the open loop (power mode) control.

When the "Manual" button is in dark grey the system is in power mode (manual). A click on the "Auto" button will switch to the temperature control mode (auto).

### 13.3.4 Local - Remote

The temperature controller may accept the control from two different sources. In Local mode, the control is given to the PC. In Remote mode, the control is given to the PLC. Running a process requires the temperature regulator be in Remote mode.

The "Local" button switches the temperature controller from remote to local mode. **The local mode must be used for running the auto tuning.**

When the "Local" button is in dark grey the system is in local mode (direct control on the temperature controller). A click on the "Remote" button will switch back to the remote mode (control through the PLC).

When the "Remote" button is in dark grey the system is remote control. A click on the "Local" button will switch to the local mode (direct control on the temperature controller).

### 13.3.5 Run - Stop

This button allows starting and stopping the heating.

When the "Run" button is in dark grey the system is in heating mode and the temperature controller output or power setpoint will be applied immediately to the lamps. A click on "Stop" stops the heating immediately.

When the "Stop" button is in dark grey the system is in stop mode and there is no heating. A click on "Start" immediately applies setpoint (temperature or power) to the furnace.

|                                                                                     |                                                                                                                                                                                                                                                                                                                         |
|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <p style="text-align: center;"><b>⚠ CAUTION</b></p> <p><b>Power mode.</b></p> <p><b>Possible overheating of the substrate and the furnace</b></p> <p><b>Do not set power over 30% when start heating. Higher setpoint will heat to temperature over 1100°C and can melt the silicon wafer and the thermocouple.</b></p> |
|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### 13.3.6 Temperature mode

The temperature mode offers 4 options:

- Power
- TC1
- Pyro1
- Pyro2 (Option)

The power mode is the default mode when entering the RKC mode. It is not use for autotuning but can be used for pre-heating of the substrate.

The other buttons are for selecting the sensor that will be used for the temperature control. The PID parameters depend on the sensor. The user must do a calibration table for each different temperature sensor that is used for the temperature control.

### 13.4 Temperature control values

The temperature control values are the parameters used for the temperature control. When the user enters the RKC mode the system uploads the current values from the temperature controller (if refresh box is checked).

| Channel 1                      |              |
|--------------------------------|--------------|
| Temperature readout (°C)       | 4.50         |
| Actual power (%)               | 0.0          |
| Setpoint (°C)                  | 0.00         |
| Power (%)                      | 0.0          |
| Autotuning                     | Autotune PID |
| P (%)                          | 4.00         |
| I (s)                          | 13.64        |
| D (s)                          | 3.41         |
| Control response(0,1,2)        | 1            |
| Digital filter (s)             | 0.20         |
| Offset(add to measure °C)      | 3.00         |
| Temperature level 1 parameters |              |
| High temperature limit (°C)    | 280.00       |
| P (%)                          | 4.00         |
| I (s)                          | 13.64        |
| D (s)                          | 3.41         |
| Control response(0,1,2)        | 1            |
| Temperature level 2 parameters |              |
| High temperature limit (°C)    | 1200.00      |
| P (%)                          | 3.00         |
| I (s)                          | 3.61         |
| D (s)                          | 0.90         |
| Control response(0,1,2)        | 1            |
| Temperature level 3 parameters |              |
| High temperature limit (°C)    | 1200.00      |

There are 2 columns for the temperature control parameters. On the left side white boxes where the user can enter parameters and on the right side a column with the values uploaded from the controller.

When the RKC mode starts the values are uploaded from the temperature controller and automatically copied from the right column to the left column.

There are 3 buttons and one check box on the screen.

|                                             |            |             |                        |
|---------------------------------------------|------------|-------------|------------------------|
| <input checked="" type="checkbox"/> Refresh | << Copy << | Save as ... | Download PID table ... |
|---------------------------------------------|------------|-------------|------------------------|

They have the following functions:

- When the "Refresh" box is checked the parameters are unloaded from the temperature controller continuously about every 4 seconds. If the box is unchecked they are no more uploaded
- "<<Copy<<" copies the value from the right column to the left column.
- "Save as" allows saving all the values from the right column (uploaded from the temperature controller) directly to a PID table when the autotuning has been performed.
- "Download PID table" downloads the values from the 8 temperature level groups (for each level: high temperature limit, P, I, D and control response) to the temperature controller. This button will not download the filter and the offset. **Previously** set these parameters in the upper frame of the parameter column and use the arrow keys to download filter and offset to the temperature controller.

### 13.5 How to run the furnace in manual heating

The system offers to run in power mode or in temperature control mode using one of the temperature sensors. The cooling and the power contactor can be switched on in the RKC mode.

The heating control mode is selected in the RKC form:

|                    |        |                                 |        |                        |     |
|--------------------|--------|---------------------------------|--------|------------------------|-----|
| <b>RKC control</b> |        | <b>Temperature control mode</b> |        | <b>Power contactor</b> |     |
| Auto               | Manual | Power                           | TC1    | ON                     | Off |
| Remote             | Local  | Pyro 1                          | Pyro 2 | <b>Cooling</b>         |     |
| Run                | Stop   |                                 |        | ON                     | Off |

The setpoints are set using the corresponding boxes of on top the parameter column.

The setpoint is applied by clicking on the corresponding arrow key.

|                          |     |                  |   |                                  |
|--------------------------|-----|------------------|---|----------------------------------|
|                          |     | <b>Channel 1</b> |   |                                  |
| Temperature readout (°C) |     | 512.20           |   |                                  |
| Actual power (%)         |     | 0.0              |   |                                  |
| Setpoint (°C)            | 500 | 500.00           | > | Click on arrow to apply setpoint |
| Power (%)                | 0.0 | 0.0              | > |                                  |

#### 13.5.1 To enter a power setpoint:

Set the system in manual mode: Click on "Manual" button to have "Manual" in dark grey.

Set the system in local mode: Click on "Local" button to have "Local" in dark grey.

Enter the power setpoint in the power setpoint box and press the corresponding arrow key to apply the value. The heating starts immediately with the power setpoint applied to the power blocks of the furnace.

**It is recommended to increase the setpoint step by step especially if there is a susceptor in the process chamber.**

|                                                                                     |                                                                                                                                                                                                                                                                            |
|-------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <b>⚠ CAUTION</b>                                                                                                                                                                                                                                                           |
|                                                                                     | <p><b>Power mode.</b></p> <p><b>Possible overheating of the substrate and the furnace</b></p> <p><b>Do not set power over 30% when start heating. Higher setpoint will heat to temperature over 1100°C and can melt the silicon wafer and damage the thermocouple.</b></p> |

To stop heating enter zero for the power setpoint value and click arrow key to apply the value. The heating can also be stopped using "Stop" button.



### 13.5.2 To enter a temperature setpoint

Set the system in automatic control mode: Click on "Auto" button to have "Auto" in dark grey.

Set the system in local control: Click on "Local" button to have "Local" in dark grey.

When running in temperature control mode it is possible to select the sensor used for the temperature control clicking on the corresponding button.

The temperature setpoint is entered in the "Setpoint (°C)" box. Then press the corresponding arrow key to apply the setpoint.

**It is recommended to increase the setpoint step by step especially if there is a susceptor in the process chamber.**

|                                                                                   |                                                                                                                                                                                                                                         |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <b>⚠ CAUTION</b>                                                                                                                                                                                                                        |
|                                                                                   | <p><b>Power mode.</b></p> <p><b>Possible overheating of the substrate and the furnace</b></p> <p><b>High temperature gap in the setpoint can lead to big power amount as the system will not ramp to reach the new temperature.</b></p> |

It is possible at any time to change the temperature setpoint and to switch from a sensor to the other (thermocouple or pyrometer).

## 13.6 Procedure to adjust the temperature controller parameters

### 13.6.1 Autotuning procedure

**The conditions to run the autotuning should be as close as possible to the process conditions. First set the chamber pressure and gas flow in the manual mode is needed.**

The autotuning must be performed in temperature control mode and using the right temperature sensor.

Procedure:

- Set the system in temperature control mode: Click on "Auto" button to have **"Auto" in dark grey** (temperature control mode)
- Set the system in local control: Click on "Local" button to have **"Local" in dark grey**
- **Check "Refresh" box** to have continuous update of the PID parameters

The PID parameters are selected function of the setpoint. For each process it is recommended to run the autotuning for each dwell setpoint value. (See the examples hereafter).

- Select the temperature sensor to be used for the control
- Enter a setpoint and press arrow key to apply the setpoint
- Wait for temperature to be close to setpoint
- Click on "Autotuning"

|                            |                  |        |                         |
|----------------------------|------------------|--------|-------------------------|
|                            | <b>Channel 1</b> |        |                         |
| Temperature readout [ °C ] | 505.00           |        |                         |
| Actual power [ % ]         | 0.0              |        |                         |
| Setpoint [ °C ]            | 500              | 500.00 | > ← Apply setpoint here |
| Power [ % ]                | 0.0              | 0.0    | >                       |
| Autotuning                 | Autotune         | PID    | ← Enter setpoint here   |

**Autotuning can be stopped by clicking on the "PID" button**

The temperature controller will immediately start the autotuning and the "Autotuning" button turns to dark grey and is disabled until the autotuning is completed. The controller applies 3 power peaks at full power and analyses the responses to run the autotuning. The autotuning usually takes 10 seconds to 2 minutes (It can be much longer at low temperature with a susceptor).

**Important notice:** The temperature controller is set for a temperature range of 0°C to 1200°C with a linear signal. The furnace allows calibration tables for both thermocouple and pyrometer. The actual temperature for a given signal of the pyrometer or of the thermocouple is not linear and may not have the same scale than the temperature controller. For example it can go up to 1300°C for the pyrometer.

The autotuning can only be performed in local mode using directly the signal input of the temperature controller without using an existing calibration table. The furnace will not compensate the real signal non linearity or scale.

In consequence when a temperature setpoint is set for the autotuning, the temperature controller will not control the temperature at the setpoint level but at an **actual temperature that can be read in the parameters readout frame**. The user must enter a setpoint that will give a temperature level of the requested autotuning temperature as described on the figure hereunder.

| Temperatures |             | Gas (scm) | Setpoint |
|--------------|-------------|-----------|----------|
| Setp. (°C)   | 0.0         | N2        | 0.0      |
| Power %      | 41.0        | O2        | 0.0      |
| TC1 °C       | Not present |           |          |
| Pyro1 °C     | 666.6       |           |          |
| TC2 °C       | Not present |           |          |

Read actual temperature

| Channel 1                |              |
|--------------------------|--------------|
| Temperature readout (°C) | 512.20       |
| Actual power (%)         | 0.0          |
| Setpoint (°C)            | 500 500.00 > |
| Power (%)                | 0.0 0.0 >    |
| Autotuning               | Autotune PID |

Enter setpoint here

In the example the temperature setpoint is set to 500°C to get a pyrometer signal of 666.6°C. The 500°C will then be the high temperature level for a process dwell to be performed at 666.6°C.

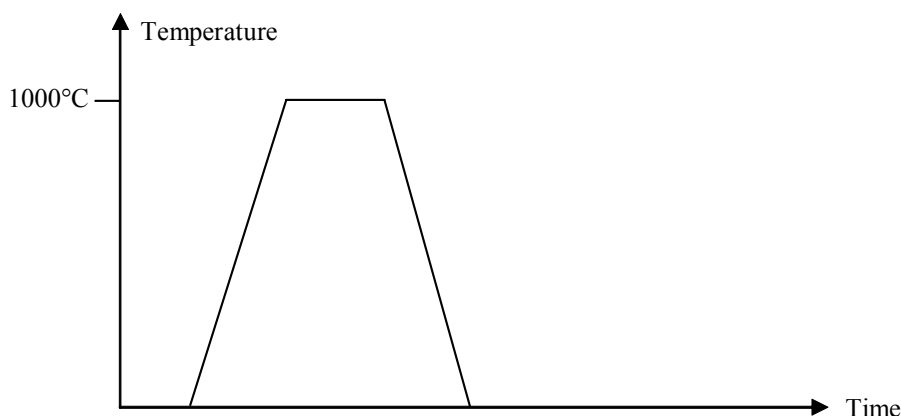
At the end of the autotuning the "Autotuning" button is enabled again. It is now possible to run the autotuning for another temperature. When all temperature level autotunings have been performed it is possible to save the data into a PID table using the "Save as" function.

For silicon application the PID parameters change a lot at low temperature (below 400°C) and in the range of 600°C to 700°C. Silicon optical and thermal behavior changes a lot in this temperature range. For this reason it is recommended to have 1 or 2 PID set parameters for low temperature and low ramp rates. Then depending on the process one set of parameters may be enough for the remaining temperature range (see examples hereafter).

During the process when the system switches from one set of parameters to another one (when it reaches a high temperature level) it is possible to see some changes in the temperature control behavior and some power peaks. It is recommended for some processes to limit the number of levels and in general to check that the PID parameters values do not change too much from a level to the next one.

The lower high temperature level is level 1. The 8 temperature levels must receive parameters. If only some of them are needed for the process, the remaining ones should be completed for higher temperature level (or 1200°C) and using the last set of parameters (from higher used level). The system will not accept a high temperature value if the high temperature of the next level is lower.

### 13.6.2 Example 1

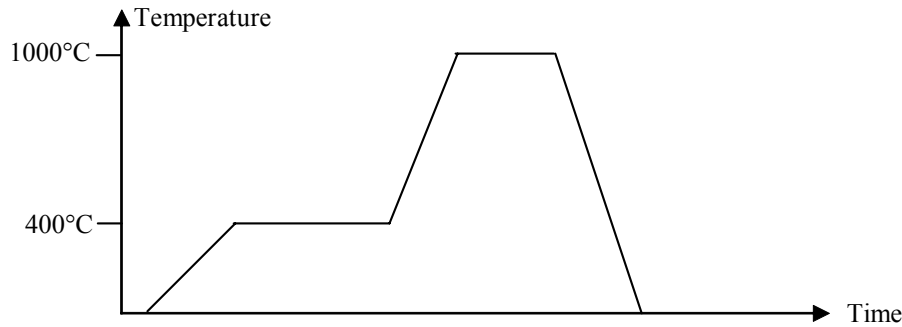


In this example it may be better to have only one set of parameters running an autotuning at 1000°C.

The procedure will be:

- Set gas flow and reactor pressure in gas manual mode if needed
- Check that the system is in control mode: button should display "Auto" in dark grey
- Set the system in local control: Click on "Local" button to have "Local" in dark grey
- Set the controller in run mode "Run" button shall be in dark grey
- Check the refresh box to have continuous uploading of the temperature controller parameters
- Select the temperature control sensor (thermocouple or pyrometer)
- Enter 1200°C as high temperature for temperature zones 8 to 2
- Enter 938°C as setpoint to get 1000°C as actual temperature and apply setpoint
- Run autotuning
- Enter 938°C as high temperature level for high temperature level 1
- Set the control response to medium (enter 1) for each level
- Set digital filter: 0.1 to 0.2 usually for the pyrometer. (It must be set to zero for thermocouple)
- Now all the temperature control parameters are available and can be copied into a PID table using the "Save as" button to create a new table, or go to the configuration mode to modify an existing table.
- Run a process with this new PID table and adjust the offset if necessary to have the same value for the temperature readout and the setpoint.

### 13.6.3 Example 2



For this example it is recommended to run autotuning at:

- 200°C
- 400°C
- 1000°C

The procedure will be:

- Set gas flow and reactor pressure in gas manual mode if needed
- Check that the system is in control mode: button should display "Auto" in dark grey
- Set the system in local control: Click on "Local" button to have "Local" in dark grey
- Set the controller in run mode "Run" button shall be in dark grey
- Check the refresh box to have continuous uploading of the temperature controller parameters
- Select the temperature control sensor (thermocouple or pyrometer)
- Enter 1200°C as high temperature for temperature zones 8, 7, 6, 5 and 4
- Enter 1000°C as high temperature for temperature zone 3
- Enter 400°C as high temperature for temperature zone 2
- Enter 200°C as high temperature for temperature zone 1
- Enter 187°C as setpoint to get 200°C as actual temperature and press enter
- Run autotuning
- Enter 187°C as high temperature level for Temperature zone 1
- Enter 382°C as setpoint to get 400°C as actual temperature and press enter
- Run autotuning
- Enter 382°C as high temperature level for Temperature zone 2
- Enter 938°C as setpoint to get 1000°C as actual temperature and press enter
- Run autotuning
- Enter 938°C as high temperature level for Temperature zone 3
- Set the control response to medium (enter 1) for each level
- Set digital filter 01 to 0.2 usually for the pyrometer. (It must be set to zero for thermocouple)
- Now all the temperature control parameters are available and can be saved into a PID table using the "Save as" button to create a new table or go to the configuration mode to modify an existing table.
- Run a process with this new PID table and adjust the offset if necessary to have the same value for the temperature readout and the setpoint.

## 13.7 Exiting from the RKC mode

When exiting from the RKC mode the system automatically returns to the auto, remote and run modes. The temperature control is set to power mode.

Power contactor and cooling must be switched off by the user in RKC mode or in manual gas mode.

## 14. SYSTEM SHUT DOWN

The system shut-down is performed using the following procedure:

- Shut down software
- Switch-off the system
- Shut down the computer
- Equipment isolation
- Verify isolation

### 14.1 Shut-down software

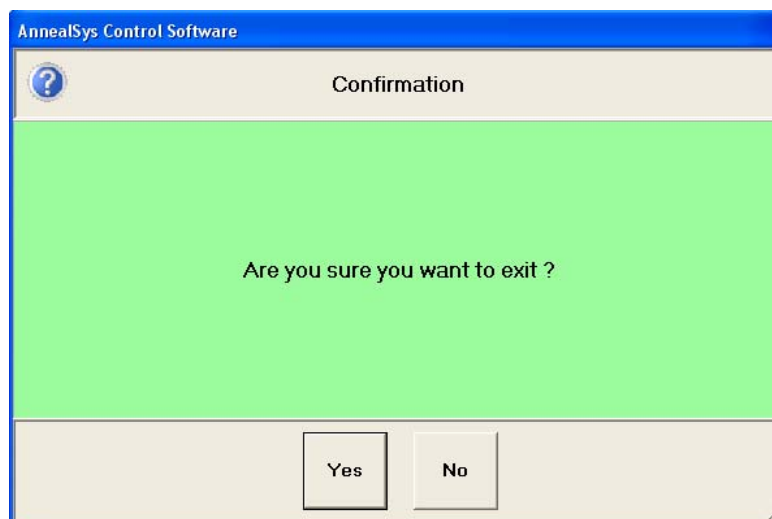
If the system is equipped with a turbo pump stop the turbo pump using the manual mode before system shut-down. If the software needs to be temporary shut-down but not the system, then the turbo pump can remain running.

|                                                                                   |                                                                                                                                                                                                                                                                                                                                      |
|-----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <b>⚠ CAUTION</b>                                                                                                                                                                                                                                                                                                                     |
|                                                                                   | <b>Turbo pump control</b><br>If the furnace is equipped with a turbo pump and if the furnace is going to be stopped, then the turbo pump must be stopped using the manual mode before switching off the furnace.<br>The PC software can be shut-down without stopping the turbo pump if the furnace is not going to be switched off. |

The software can be shut-down using the "Shut-down" button in the main navigation bar or by a right click on the mouse.

The PC software can be stopped even if the machine is running. Nevertheless it is recommended to stop both PC and machine.

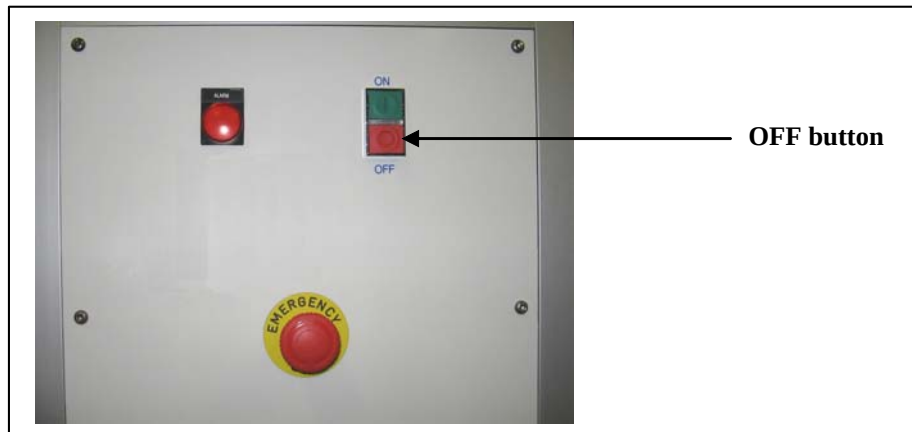
The system will ask for shut-down confirmation:



Confirm by "Yes" to exit.

## 14.2 Switch-off the system

Switch-off the system by pressing the red OFF button on the front panel.



## 14.3 Shut down computer

The computer is usually supplied by the sockets installed on the system backside. When the general circuit breaker is switched off these sockets are de-energized.

The computer must be shut down before switching off the general circuit breaker and isolating the system.

## 14.4 Equipment isolation 3x220V version

The equipment can be isolated using the general circuit breaker on the backside panel.

For the 3x220V version: Turn the handle on the left side (white part on top the circuit breaker is ON).



**Circuit breaker ON**



**Circuit breaker OFF**

Lockout of the circuit breaker:

The circuit breaker can be lockout when it is in off position. Pull the small plastic slider to open the hole for the installation of a padlock.



#### 14.5 Equipment isolation 3x400V+N version

The equipment can be isolated using the general circuit breaker on the backside panel.

For the 3x400V+N version: Push the handle down.



**Circuit breaker ON**



**Circuit breaker OFF**

When the circuit breaker is in OFF position the locking device that is provided with the system can be installed with a padlock to lock the general circuit breaker in isolating position.

Insert the 2 metallic end of the locker inside the small groove in the black part on the circuit breaker handle and under the white enclosure as shown on the picture. Then push the orange plastic part and install a padlock between the orange part and the metal ring.

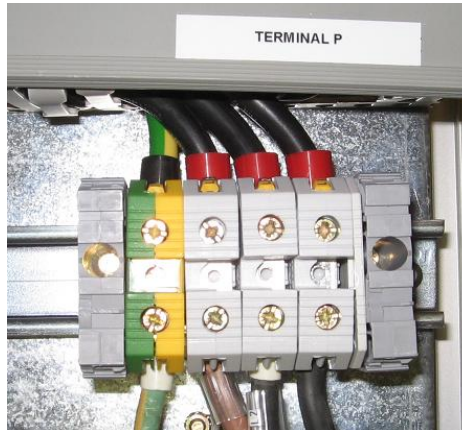
#### 14.6 Stored energies

The power blocks include power capacitors but they are equipped with resistors to dissipate stored energy. It takes less than 2 seconds to relieve the stored energy. The power blocks can be dismantled without risk after the system has been isolated.



#### 14.7 Verify isolation

In addition of the general circuit breaker it is recommended to isolate the equipment supply line and to check that there is no more energy by measuring the voltage on the supply cable terminal.



#### **⚠ WARNING**

**Electrical shock hazard**

**If the power line is not isolated the supply cable terminal and wired to the main circuit breaker will remain energized.**

**The service operations must be carried out by trained and authorized technicians only.**

If the power line remains energized the isolation of the system from the general circuit breaker can be checked on control circuit breaker and power contactor inlets.



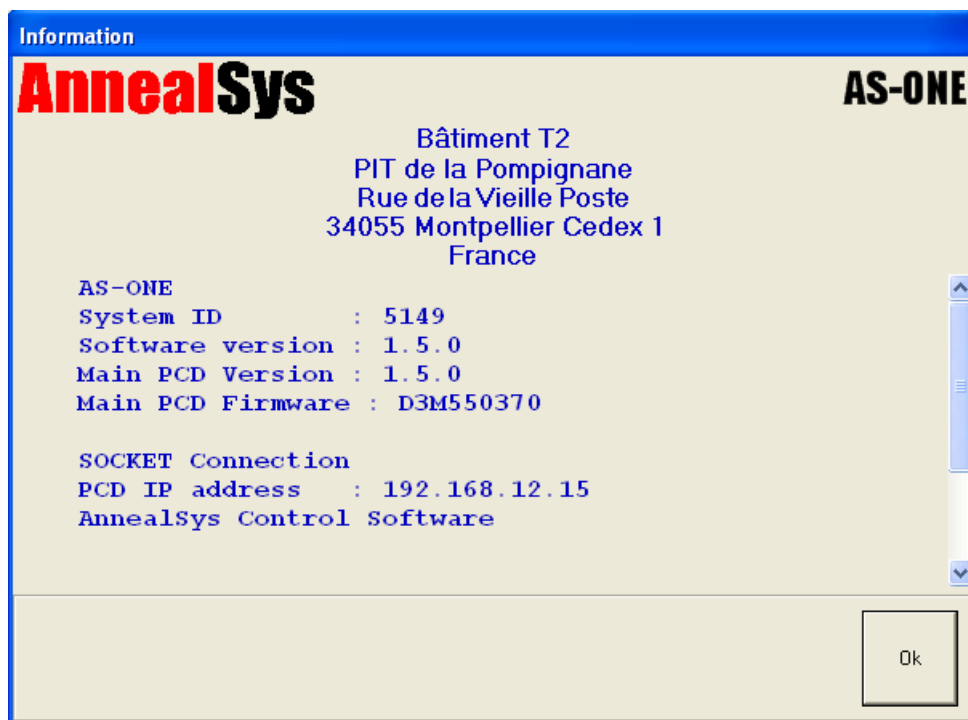
## 15. HELP

F1 key gives access to the help mode.

The "help" window can also be displayed by a right click on the mouse when the cursor is on the main parameter band on top of the screen or on the main navigation band on the bottom of the screen.

- System type
- System ID (design number)
- PC software version
- PLC software version
- PLC firmware version
- PCD IP address
- Current user information.

This information is useful for maintenance and must be provided to Annealsys for remote diagnostic on the system.



## 16. DIAGNOSTICS

The software has diagnostic capabilities that allow getting additional system information for troubleshooting.

The "diagnostics" window can be displayed by a right click on the mouse when the cursor is on the main parameter band on top of the screen or on the main navigation band on the bottom of the screen.

There are different windows that can be selected by checking the different tabs at the bottom of the diagnostic window:

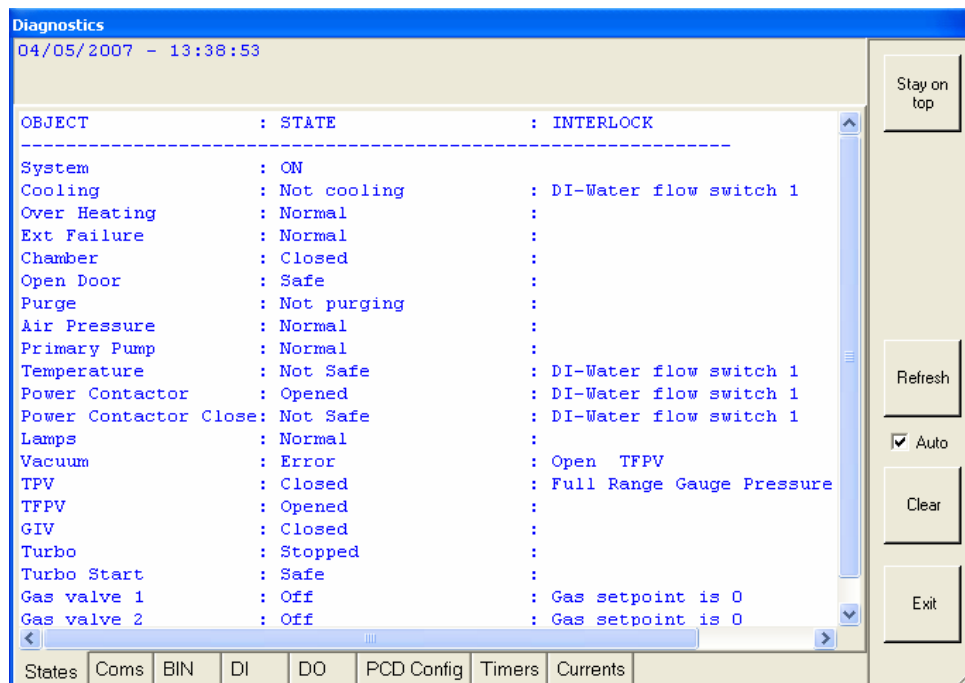
- States: general diagnostics information and interlock status
- Coms: digital communication status
- BIN: Binary values of the analog values (converted by the PLC)
- DI: Digital inputs status
- DO: Digital outputs status
- PCD Config: PLC and hardware configuration
- Timers: Current values of the timers
- Currents: values of currents from the different heating zones, lamp age

Button functions:

- **Refresh** button: clicking this button refreshes the window.
- **Auto** check box: when checked, the window refreshes continuously.
- **Clear** button: resets internal diagnostics data and clears the window.
- **Stay on stop** button: when pushed, diagnostics window keeps the top of the screen so that it is always visible.

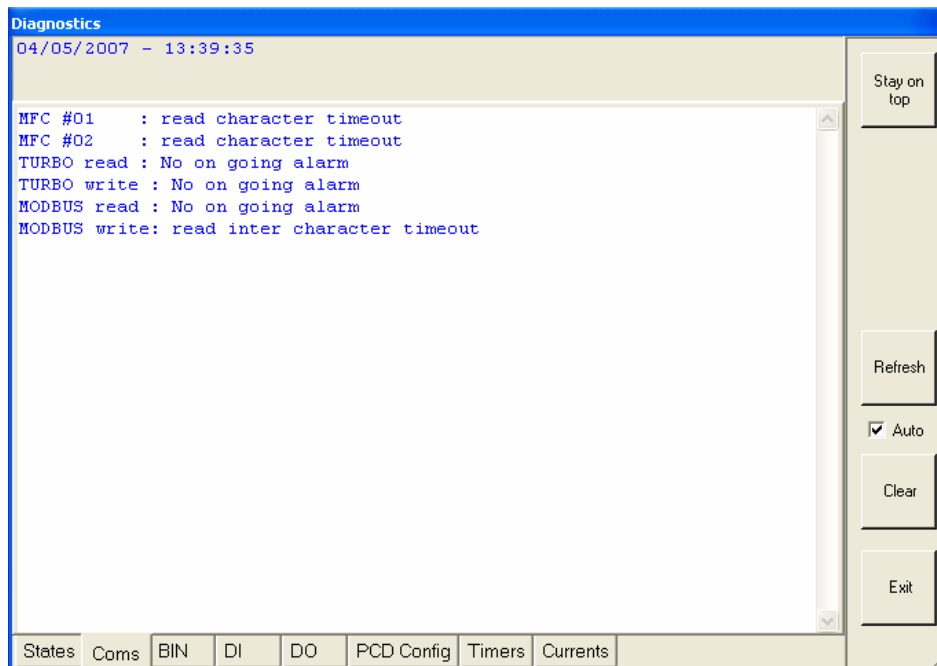
### 16.1 States

This window provides general information on the system status and the interlocks.



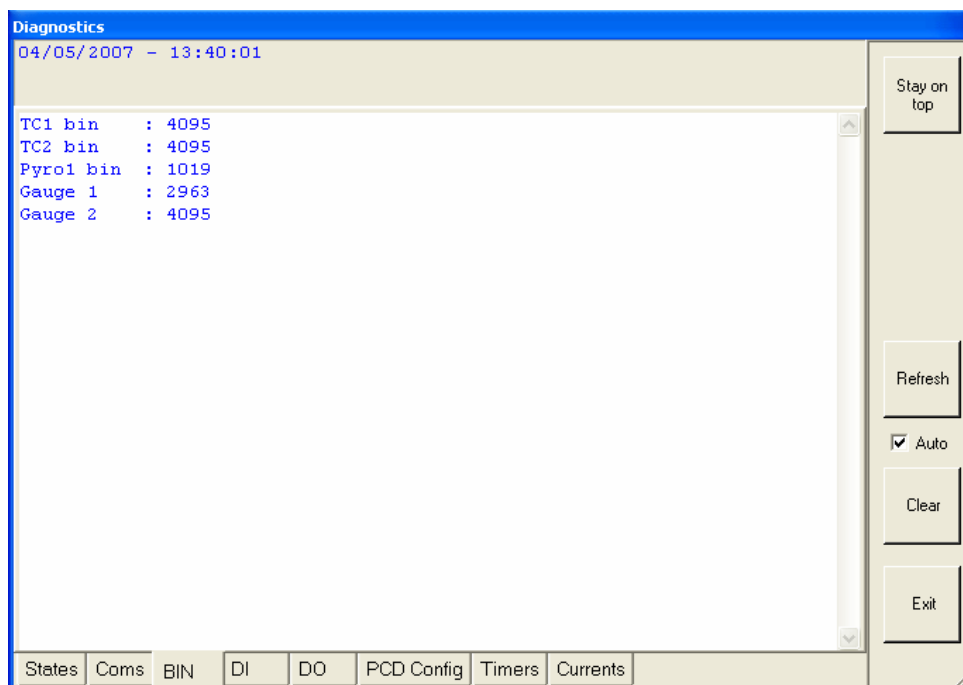
## 16.2 Coms: digital communication status

This window gives the status of the different communication interfaces with mass flow controllers, temperature controller, pressure controller and turbo pump.



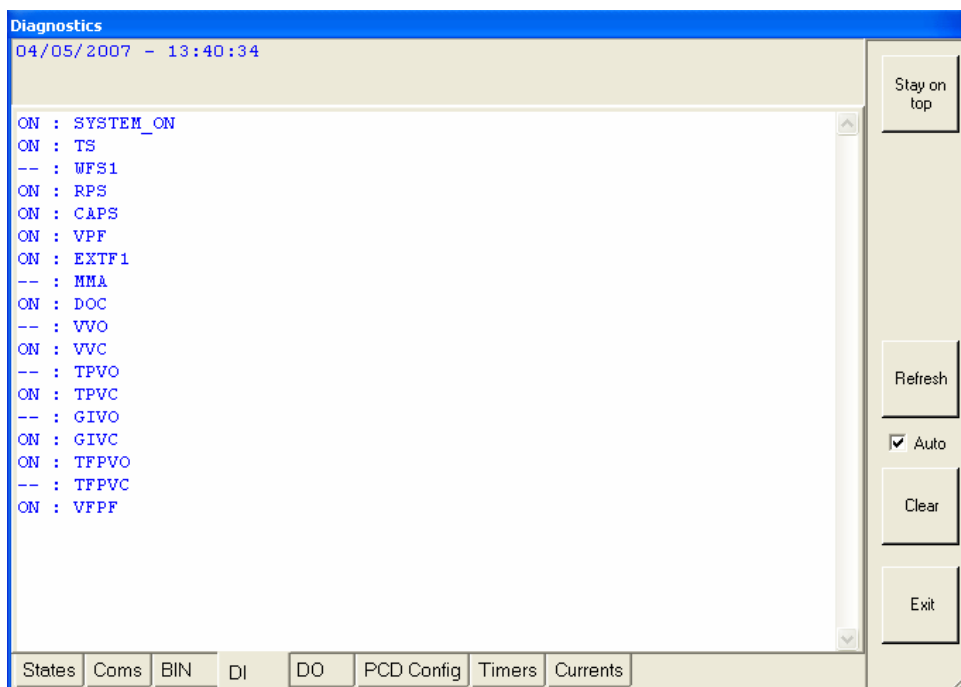
## 16.3 BIN: Binary values of the analog values (converted by the PLC)

This window gives the binary values of the analog sensors as they are converted by the PLC. It can be used to check the signal inputs from these sensors for troubleshooting.



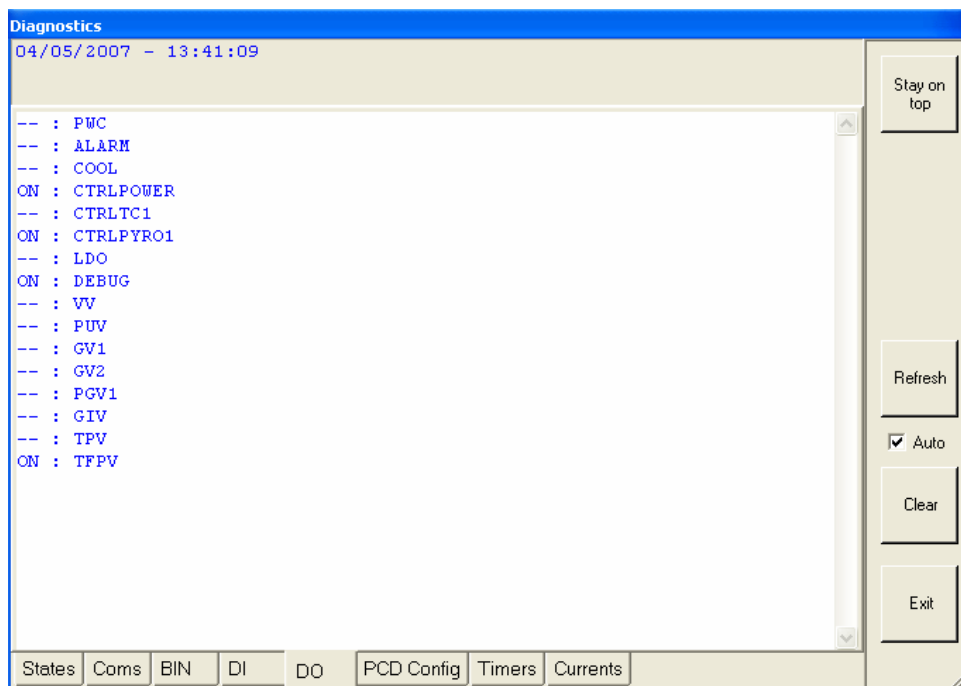
#### 16.4 DI: Digital inputs status

This window displays the status of the digital inputs of the PLC. See paragraph 12.2 for further details on the digital inputs.



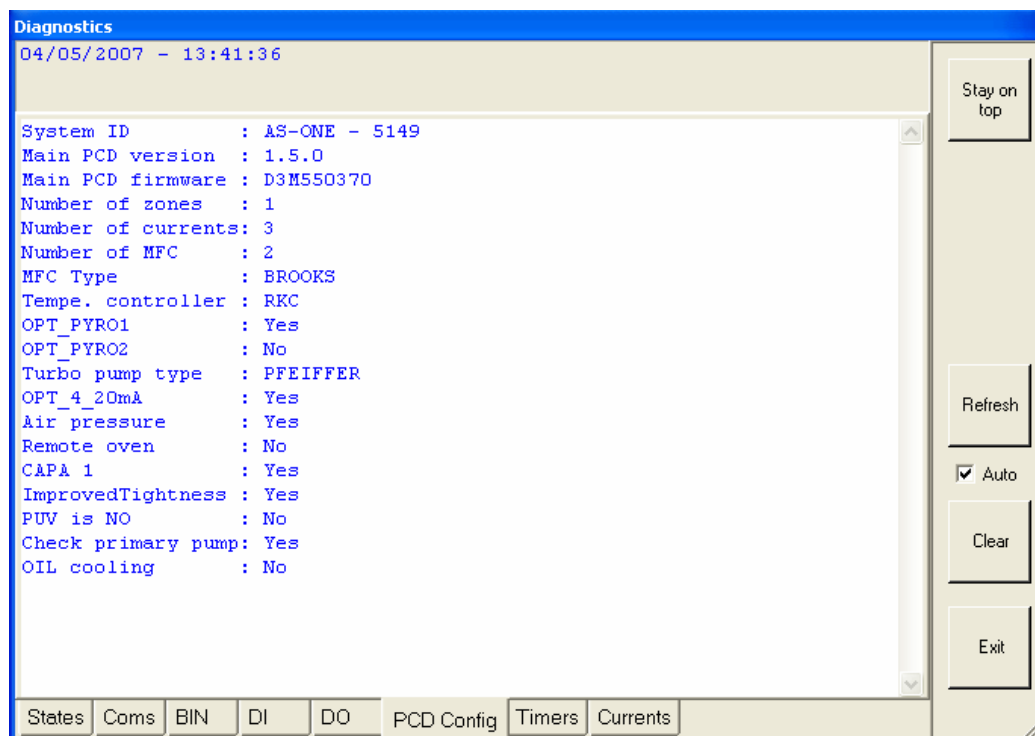
#### 16.5 DO: Digital outputs status

This window displays the status of the digital outputs of the PLC. See paragraph 10.1.2.3 for further details on the digital inputs.



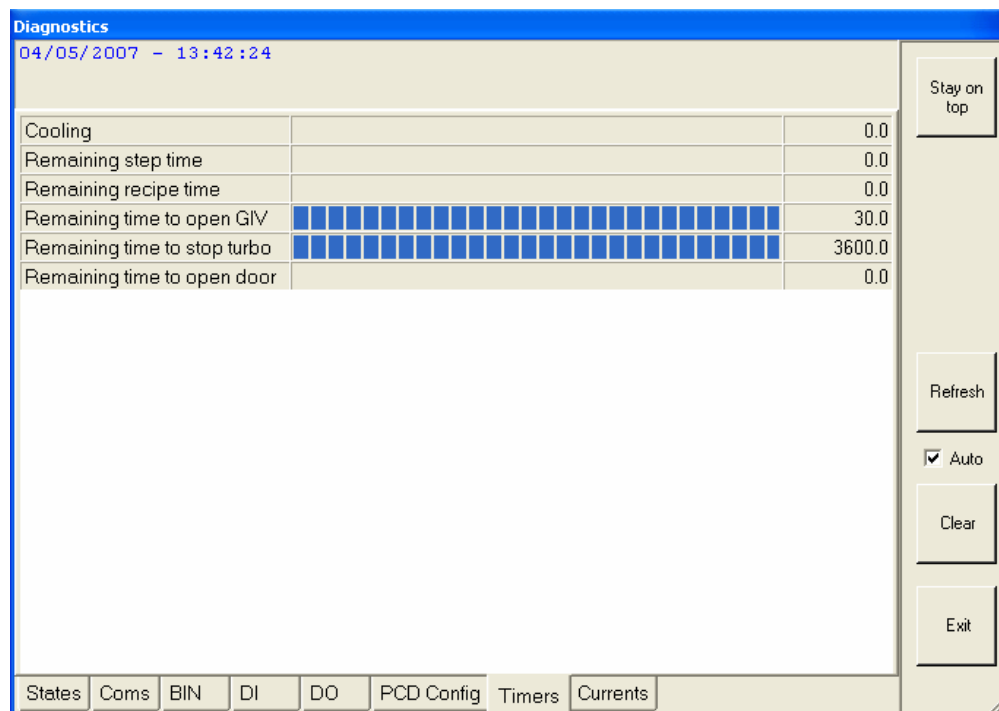
## 16.6 PCD Config: PLC and hardware configuration

This window provides information on the system ID, PCD software and firmware versions as well as the hardware configuration of the system.



## 16.7 Timers: Current values of the timers

This window gives the current values of various timers.

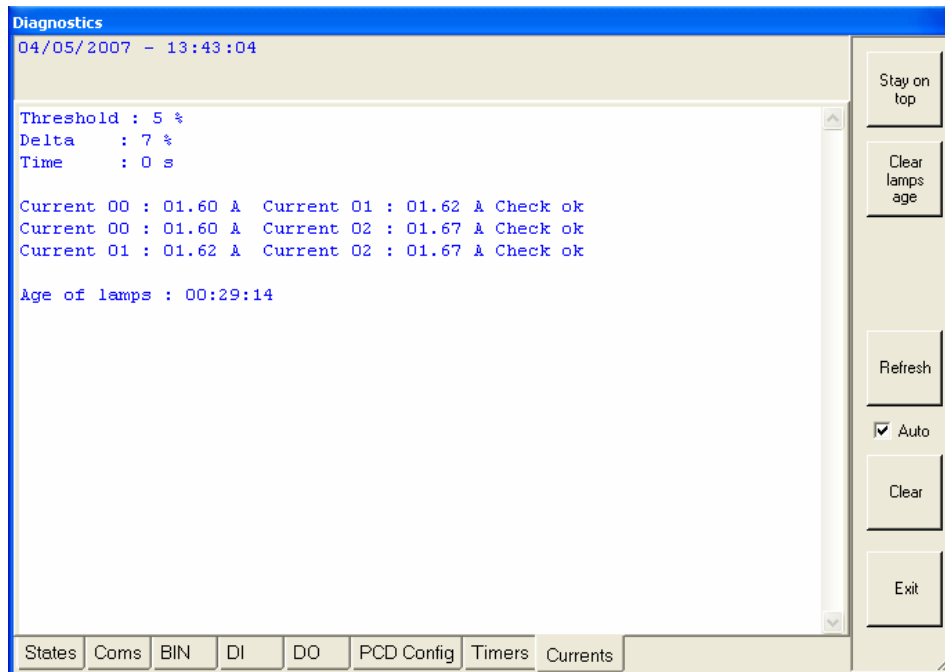


### 16.8 Currents: values of currents from the different heating zones, lamp age

This window displays the values of the currents from the different heating zones.

It also displays the age of the lamps. This is the time during which a power setpoint of more than 3% has been applied to the lamps since the last reset. This function can be used for preventive maintenance to change the lamps. Age is in hour, minutes and seconds.

The age of the lamp can be reset using the button "Clear lamps age". This button is only available in administrator mode.



## **17. SOFTWARE INSTALLATION AND UPGRADES**

### **17.1 Software installation**

The furnace is delivered with a PC on which the software has been installed by Annealsys before shipment.

The software can be reinstalled using the CD that is provided with the system. The installation procedure is available on the CD.

### **17.2 Software upgrades**

Annealsys also provides software upgrades by email, by CD or that can be uploaded from the Internet. The installation procedure is provided with the software upgrade.

The PLCs are equipped with Flash memories. This allows easy PLC upgrade of the PLC software. The PLC software can be automatically upgraded from the PC by new software package provided by Annealsys.

Feel free to make comments on software utilization or to inform us of software bugs. All the Annealsys systems use the same software base and this is part of our quality organization.

## **18. SOFTWARE FILE MANAGEMENT**

### **18.1 Software and data folders**

The software and the default folders are installed in: "c:\Program Files\annealsys".

This folder contains 6 folders:

- "Bin" contains the software program and associated dll
- "Data" contains user related files
- "Tables" contains calibration, PID and all tables parameters
- "Historicals" is the default folder to save the process historicals
- "Recipes" is the default folder to save the recipe files
- "System" contains the configuration and system files

#### **User data path:**

When a user private data path is configured by the administrator, the following folders are used:

- "<UserDataPath>\Data" contains the user related files (preferences),
- "<UserDataPath>\Historicals" contains the process historicals,
- "<UserDataPath>\Recipes" contains the recipe files.

Note that:

- Configuration files are always located in "C:\Program Files\Annealsys\System" folder,
- Tables files are always located in "C:\Program Files\Annealsys\Tables" folder.

## 18.2 Traceability

The traceability is an optional feature of the software.

A new traceability file is automatically created each month.

The traceability files are located in the directory **C:\Program Files\Annealsys** and have the following name:

- Traceability\_YYYYMM.csv (*where YYYY is the year and MM is the month*).

The files can be opened with Excel. The list separator is the comma.

### 18.2.1.1 Accessing CSV files with MS Excel

To access CSV files (Comma Separated Values) with MS Excel, Control Panel -> Regional Options have to be set as follows:

- Dot "." as decimal separator
- Comma "," as list separator.

### 18.2.1.2 File fields description

A traceability record is made of the following fields:

- Historical name: Fully qualified name of the file holding the historical data collection.
- Item 1: Item 1 as typed by the operator. Note that this field can be either a number or a text.
- Item 2: Item 2 as typed by the operator. Note that this field can be either a number or a text.
- Item 3: Item 3 as typed by the operator. Note that this field can be either a number or a text.
- Recipe name: Name of the recipe used to process material.
- User name: Name of the logged user when the process was started.
- User mode: Current access level when the process was started.
- Start process date: Starting date of the process.
- Start process time: Starting time of the process.
- End process date: Ending date of the process.
- Start process time: Ending time of the process.
- Alarm: Description of the alarm, if any, that caused the process to end.



## 19. SAFETY INTERLOCKS

### 19.1 EMO

The system is equipped with an EMO push button on the front panel.

This emergency switch shuts down power of the system.

### 19.2 PLC watch dog

The PLC is equipped with a watch dog interlock that controls an internal contact.

When the operator closes the rear panel general circuit breaker, the PLC runs a diagnostic routine that takes several seconds. When the initialization cycle is completed the alarm indicator turns to red and the system can be switched on using the green push button. The alarm can be reset after powering up the system by clicking on the alarm indicator on the software screen. Otherwise it will be automatically reset when the first process is performed.

Each time the system is switched off (by the red push button); the alarm indicator turns to red to inform the user of a system power off.

If the PLC diagnostic routine or if there is any diagnostic problem of the PLC when the system is running then the watch dog interlock is activated and the power is automatically shut-down by hardware interlock.

### 19.3 Hardware interlocks

There are some hardware interlocks that automatically switch off the power contactor in case of alarm. When power contactor is off there is no more power to the lamps. Other sub-systems (including cooling) are still powered on.

The hardware interlocks that are necessary to operate the power contactor are:

- Thermal switch on the lamp reflector (overheating interlock)
- Thermal switch inside the furnace cover (overheating interlock)
- Compressed air pressure (air pressure interlock)
- Water flow switch (insufficient water flow interlock)

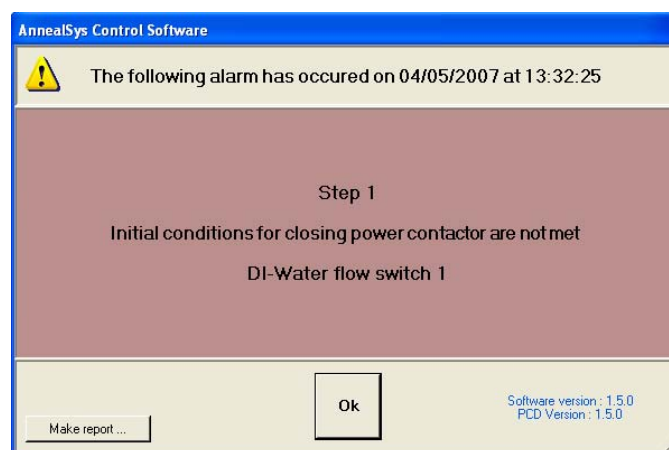
These interlocks are also software interlocks and are described hereafter in more details.

### 19.4 Software interlocks

During process the system checks safety interlocks that are listed in the table hereafter.

If any utilization condition is not fulfilled the system generates an alarm, the process stops and the system runs a cooling and a purge sequence to clean the process chamber.

The screen displays an alarm message:

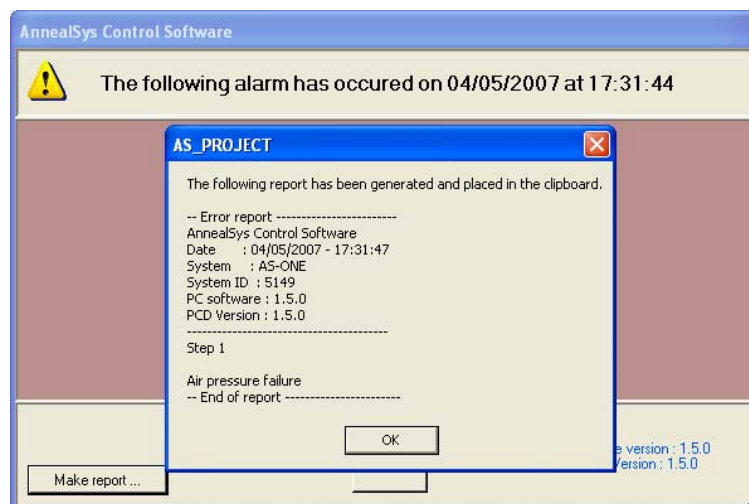


The user must press OK to acknowledge the alarm when the alarm message is displayed.

The problem must be fixed prior to any new process.

Note that the Alarm light is switched on when an alarm occurs. This Alarm light is switched off as soon as the operator acknowledges the alarm by clicking on OK button.

The user can automatically generate an alarm report that is automatically copied in the clipboard. This report can be saved in another file or copy into an email to Annealsys customer support.



The alarms are saved in an alarm list that can be displayed by the user (see: 10.2).

[illegible]

The interlocks of the system are listed and described in the next pages.

| Alarm                                                                                      | Sensor                                                                    | Possible causes                                                                                                                                                                                    | Comment                                                                                                                                                                                                                                                                                                                                                                                   |
|--------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Chamber is open                                                                            | Chamber closed sensor<br>(Mechanical sensor)                              | Sensor adjustment not correct<br>Mechanical problem to close the chamber                                                                                                                           | Interlock that prevents process to start if the process chamber is not perfectly closed<br>When chamber is closed it is locked before starting process.                                                                                                                                                                                                                                   |
| Cooling was expected                                                                       | Water flow controller<br>(Flow controller on water circuit before outlet) | The water supply tap is closed<br>The water pressure is not sufficient to get sufficient water flow<br>The water flow controller is not correctly adjusted<br>There is a water leak in the circuit | This alarm stops the process.<br>The system is cooled down and the chamber is purge for 3 minutes.                                                                                                                                                                                                                                                                                        |
| Improper end of step condition was found                                                   | PLC Sanity check                                                          | Unknown end of step condition detected by the PLC during the process.                                                                                                                              | This alarm is likely not to happen since the integrity and consistency of recipes are checked<br>1) at recipe edition;<br>2) at recipe download<br>This alarm stops the process.<br>The system is cooled down and the chamber is purge for 3 minutes.                                                                                                                                     |
| End of step condition found at step time 0                                                 | Software interlock                                                        | Unexpected end of step condition detected by the PLC during the process                                                                                                                            | This may happen if the end of step condition is already fulfilled when the step starts.<br>Example: actual T=357°C and end of step condition is T>200°C.<br>This alarm stops the process.<br>The system is cooled down and the chamber is purge for 3 minutes.                                                                                                                            |
| Initial conditions to close the power contactor are not met.<br>(Alarm detail is provided) | Software interlock                                                        | The process has been launched when the system was not ready or some facilities are not available<br>It may come from remaining status after manual heating mode.                                   | The conditions to close the power contactor are:<br>- System on<br>- No overheating<br>- Water cooling (Flow switch detects sufficient flow)<br>- Chamber is locked<br>- The system is in power mode<br>- Setpoint = 0<br>This interlock is in addition to the hardware interlocks.<br>This alarm stops the process.<br>The system is cooled down and the chamber is purge for 3 minutes. |

| Alarm                                                | Sensor                                        | Possible causes                                                                                                                             | Comment                                                                                                                                                                                                                                                                                                                    |
|------------------------------------------------------|-----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Gas n band alarm<br>(n = gas line number)            | Gas line n mass flow controller               | Gas line not supplied with gas<br>Problem with mass flow controller                                                                         | Active only if it has been activated in the recipe for the current step<br>Alarm occurs if the difference between the real gas flow and the setpoint is more than the threshold value set in the configuration mode.<br>This alarm stops the process.<br>The system is cooled down and the chamber is purge for 3 minutes. |
| Gas n mixture interlock<br>(n = gas line number)     | Software interlock                            | Request to introduce gas n in the process chamber with another gas for which a mixture interlock has been set in the hardware configuration | Should usually not happen as the mixture interlocks are also checked during recipe edition.<br>This alarm stops the process.<br>The system is cooled down and the chamber is purge for 3 minutes.                                                                                                                          |
| Gas n-Air mixture interlock<br>(n = gas line number) | Software interlock                            | Request to introduce gas n in the process chamber although the process chamber has not been pumped down.                                    | Should usually not happen as the mixture interlocks are also checked during recipe edition.<br>This alarm may happen only if an Air-Gas mixture interlock has been set in the hardware configuration<br>This alarm stops the process.<br>The system is cooled down and the chamber is purge for 3 minutes.                 |
| Gas n-Pressure interlock                             | Software interlock                            | Request to introduce gas n in the process chamber although the pressure in process chamber is over 10 mbar.                                 | This alarm may happen only if an Pressure mixture interlock has been set in the hardware configuration<br>This alarm stops the process.<br>The system is cooled down and the chamber is purge for 3 minutes.                                                                                                               |
| Host requested to stop                               | Click on "Stop process" button during process | The user has stopped the process before its end                                                                                             | This action stops the process.<br>The system is cooled down and the chamber is purge for 3 minutes.                                                                                                                                                                                                                        |
| Recipe overflow                                      | Software interlock                            | The last step is reached.                                                                                                                   | Process is completed but the user is warned that the recipe has reached and executed the step number 100.<br>This alarm stops the process.<br>The system is cooled down and the chamber is purge for 3 minutes.                                                                                                            |

| Alarm                                                          | Sensor                                                                       | Possible causes                                                                                                                                                                | Comment                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|----------------------------------------------------------------|------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Air pressure failure                                           | Compressed air pressure sensor                                               | Compressed air pressure facility is down or the supply valve is closed.                                                                                                        | This alarm stops the process.<br>The system is cooled down and the chamber is purge for 3 minutes.                                                                                                                                                                                                                                                                                                                                                                                              |
| Lamp failure #n<br>(where n is the number of the heating zone) | Lamps current sensors                                                        | Broken lamp filament<br>Fuse of one power block is burned<br>Problem with one power block                                                                                      | More information about lamp currents is available in Diagnostics window, tab Current<br>This alarm stops the process.<br>The system is cooled down and the chamber is purge for 3 minutes.                                                                                                                                                                                                                                                                                                      |
| Overheating                                                    | Thermal switch on lamp reflector or thermal switch in lamp fans area is open | Water flow is not sufficient<br>Too long process at high temperature<br>Insufficient air cooling due to fan problem                                                            | Temperature exceed 90°C on reflector or in fans area<br>This alarm stops the process.<br>The system is cooled down and the chamber is purge for 3 minutes.                                                                                                                                                                                                                                                                                                                                      |
| Pyrometer 1 failure                                            | Pyrometer 1                                                                  | Pyrometer disconnected<br>Wiring problem<br>Pyrometer problem                                                                                                                  | The system receives no signal from the pyrometer 1<br>This alarm stops the process.<br>The system is cooled down and the chamber is purge for 3 minutes.                                                                                                                                                                                                                                                                                                                                        |
| Temperature delay                                              | Software interlock                                                           | No substrate in the process chamber<br>Wrong PID parameters<br>Problem with temperature control<br>Pyrometer failure<br>Lamps supply failure<br>Temperature controller failure | Temperature band alarm<br>Active only if it has been activated in the recipe for the current step and in the configuration for settings<br>The alarm occurs if the PLC detects more than 50% difference between setpoint and current temperature measured by the temperature control sensor for a period longer than the alarm delay.<br>The alarm delay is set in the configuration mode<br>This alarm stops the process.<br>The system is cooled down and the chamber is purge for 3 minutes. |
| Temperature is out of range                                    | Software interlock                                                           | Alarm activated on ramp or during unstable phase<br>Wrong PID parameters<br>Temperature control problem<br>Process problem                                                     | Temperature band alarm<br>Active only if it is been activated in the recipe for the current step<br>This alarm stops the process.<br>The system is cooled down and the chamber is purge for 3 minutes.                                                                                                                                                                                                                                                                                          |

| Alarm                 | Sensor                                                                                                        | Possible causes                                                                                                         | Comment                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-----------------------|---------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Thermocouple 1 broken | Thermocouple 1                                                                                                | Thermocouple is not installed<br>Thermocouple is not connected to the cable<br>Thermocouple is broken<br>Wiring problem | Failure if thermocouple signal reaches 1200°C<br>Only active when the system is under thermocouple temperature control mode<br>Possible alternative: use other control mode<br>This alarm stops the process.<br>The system is cooled down and the chamber is purge for 3 minutes.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| External failure      | External failure loop with connector on system rear panel                                                     | The external failure loop is open or connector disconnected                                                             | This interlock is used to connect an external interlock (exhaust control, gas sensor, etc...)<br>This alarm stops the process.<br>The system is cooled down and the chamber is purge for 3 minutes.<br>If the ballast interlock is implemented the external failure is used for ballast interlock see hereunder                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Ballast interlock     | External failure loop with connector on system rear panel connected to external pump ballast safety interlock | Pump ballast failure<br>The external failure loop is open or connector disconnected                                     | External Failure 1 is suited for Ballast Interlock.<br>When a Ballast Failure occurs (broken wire), an internal Flag is raised. When the Flag is raised:<br>a) in case of recipe execution:<br>- recipe is stopped<br>- cooling starts for 3 minutes<br>- purge is not activated<br>b) in case of recipe is not executing:<br>- purging is not possible.<br>- purge is stopped if it is running.<br>The Flag must be cleared manually with "Diagnostics-->States-->Clear" button.<br><b>NOTE: If the Ballast Failure condition still exists, the Flag is not cleared.</b><br>Running a recipe and purging the chamber is not possible until the Flag is cleared<br>The alarm message displayed on Ballast Failure can be configured in:<br>File : CONFIG.INI, Chapter : [ExternalFailure],<br>Item : Message1<br>(PC program must be restarted) |

**Alarms with yellow background are for options and will be active only if the option is installed.**

| Alarm                                                                                                                                      | Sensor                                                                                                   | Possible causes                                                                                                                                                                                      | Comment                                                                                                                                                                                                                                                                                                                                                 |
|--------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Primary pump failure (Option)                                                                                                              | Primary vacuum pump circuit breaker and contactor                                                        | Pump circuit breaker is open<br>Pump contactor is open<br>Pump motor is overloaded                                                                                                                   | No interlock if the pump is not supplied by the Annealsys RTP furnace<br>This alarm stops the process.<br>The system is cooled down and the chamber is purge for 3 minutes.                                                                                                                                                                             |
| <b>Vacuum failure (Options)</b><br><u>Alarm details:</u><br>Vacuum valve closed switch<br><br>Forepump failure<br><br>Turbo pump not ready | Vacuum valves position sensors<br><br>Forepump circuit breaker and contactor<br>Turbo pump communication | Insufficient air pressure<br>Problem on valve control<br>Problem on valve<br><br>Forepump not running or forepump problem<br>Turbo pump was not started<br>Turbo pump overload<br>Turbo pump problem | System indicates the valve that has the problem.<br>This interlock is used with the turbo pump option to avoid any wrong operation of the vacuums valves.<br>Check forepump operation to allow turbo pump running.<br>Check turbo pump status<br><br>This alarm stops the process.<br>The system is cooled down and the chamber is purge for 3 minutes. |

*Alarms with yellow background are for options and will be active only if the option is installed.*

## 20. TEMPERATURE CONTROL

The system is provided with a state of the art fast digital PID temperature controller. The closed loop system with thermocouple or optical pyrometer provides repeatable and accurate temperature control. The system can also be used in power mode (open loop) for pre-heating steps of special applications.

- Optical pyrometer: 1 (2 in option)
- Thermocouple: 2 K type
- Control: Fast digital PID temperature controller

Standard lead free thermocouples can be used from ambient temperature up to 1000°C and only with non-active gases and under none reducing atmosphere. When they are used under other conditions (with H<sub>2</sub> for example), their lifetime is much shorter. For special process atmosphere Annealsys can provide sheathed thermocouples.

The system is provided with one optical pyrometer for high or low temperature control range. The pyrometer measures the temperature from the backside of the substrate. Thanks to the special design of the reactor, the pyrometer only receives signal from the wafer and not from the lamps. This is the guaranty for an accurate temperature measurement.

Pyrometer ranges:

- High temperature pyrometer range (silicon wafer): 400°C to 1400°C
- Low temperature pyrometer range (silicon wafer): 150°C to 1100°C

The pyrometer requires a calibration made according to the K thermocouple. The calibration tables are saved on the computer hard disk. It is possible to save different tables for different substrate materials and to associate them to process recipes.

The system can receive a second pyrometer as an optional feature.

### 20.1 Thermocouples

One thermocouple can be used for temperature control (CONT cable). The thermocouples give a few millivolts signal for a measured temperature. This voltage is amplified by a special converter.

A second thermocouple can be used for temperature measurement (READ cable).

Threads are mounted in a stainless steel sheath and connected to a compensated plug. Tightness is obtained by an epoxy paste. Two stainless steel feedthroughs with Viton O-ring located in the bedplate allow easy installation of the thermocouples.

**Note: The lifetime of lead free thermocouple is dramatically reduced in reducing atmosphere. The welding has some oxide that protects it and that will be removed in reducing atmosphere. The lifetime can be as short as few process cycles. For reducing atmosphere or any atmosphere with thermocouple resistance issue Annealsys can provide sheathed thermocouple. The Inconel sheath offers a good resistance to different type of gas atmospheres.**

Sheathed thermocouples must be used for temperature control of susceptors.

### 20.2 Optical pyrometer

System is provided with one optical pyrometer for high or low temperature control range. The pyrometer looks at the backside of the substrate and only receives signal from the wafer (or susceptor) and not from the lamps.

For accurate and repeatable temperature control, check the pyrometer calibration regularly. If the system is used with a new type of substrate a pyrometer calibration must be performed.

An additional pyrometer can be provided as an optional feature.



## 21. THERMOCOUPLE CALIBRATION

|                                                                                   |                                                                                                                                                                                                                                                                                          |
|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <b>⚠ CAUTION</b>                                                                                                                                                                                                                                                                         |
|                                                                                   | <p><b>Thermocouple calibration table.</b><br/> A wrong thermocouple calibration table can lead to unexpected thermal behavior of the system.<br/> Always take great care to enter the calibration table parameters. The calibration must only be carried out by trained technicians.</p> |

For each recipe the operator can associate a thermocouple calibration table. This feature can be used for special applications.

The system is provided with K type thermocouples and K type thermocouple converters as standard features. The K type thermocouple calibration table is set at Annealsys facilities using a thermocouple calibrator. It should usually not be modified by the operator.

For special applications and requirements about thermocouple calibration table contact Annealsys.

## 22. PYROMETER CALIBRATION

|                                                                                    |                                                                                                                                                                                                                                                                                                                                                              |
|------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <b>⚠ CAUTION</b>                                                                                                                                                                                                                                                                                                                                             |
|                                                                                    | <p><b>Pyrometer calibration</b><br/> A special care must be taken in creating or modifying the pyrometer calibration tables.<br/> The modification of the configuration parameters and especially of the calibration tables may have a big impact on the system thermal behavior.<br/> These operations must only be carried out by trained technicians.</p> |

### 22.1 Why doing a pyrometer calibration?

Pyrometer calibration is an essential operation. A good temperature measurement will result from the quality of the calibration.

The pyrometer is a sensor that receives infrared radiation from the substrate and converts it into an electric signal. This signal is then converted in a digital value by the PLC. The quantity of radiation received by the pyrometer depends on:

- Substrate nature (Silicon, GaAs, Graphite, etc.)
- Doping of the substrate
- Nature of layers on substrate surface (oxide)
- Temperature of substrate

The calibration table allows the system to know the relationship between the pyrometer signal and the substrate temperature. Many calibration tables can be saved on the computer hard disk. For each recipe it is possible to associate a calibration table depending on the substrate to be used and the process conditions.

Usually the calibration does not change for the same type of substrate. If calibration changes check that:

- The pyrometer window is clean
- The pyrometer emissivity has not been changed
- The type of substrate is the same (with no additional layers on it)
- Process parameters have not been changed

## 22.2 Materials

Some substrates, especially GaAs, may have their optical properties definitely changed if they reach some temperature levels. The calibration must be always performed with a substrate with the same "thermal past" than the substrates that have to be processed in the system with the same calibration table.

Note that Silicon and GaAs are transparent for infrared at low temperature. For this reason the thermocouple can see the lamps through the wafer at low temperature. This may impact on thermocouple temperature measurement below 600°C when using high power. Calibration at low temperature should be performed using low ramp rates and low heating power.

Graphite is not transparent at infrared radiation even at low temperature. When using susceptor it is possible to run low temperature pyrometer calibration down to room temperature.

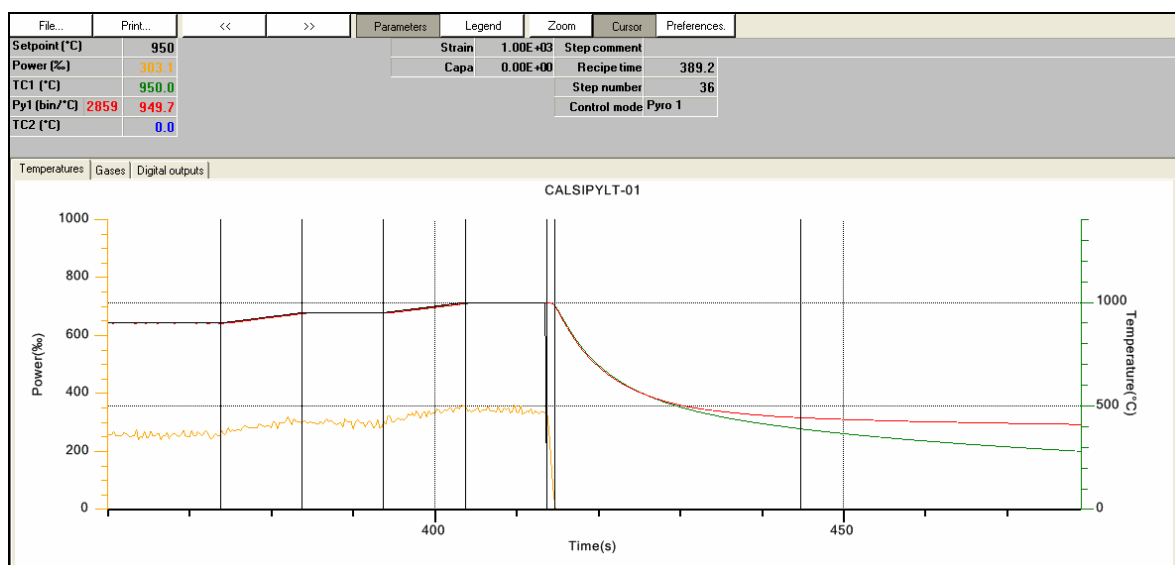
## 22.3 Thermocouple range and pyrometer calibration

It is not possible to use the thermocouple up to 1400°C because it will shortly break if it is used over 1000°C and melt over 1200°C. The thermocouple should usually not be used over 1000°C to have a good lifetime.

There are big emissivity changes of semiconductor materials below 800°C but the emissivity is relatively stable over 1000°C. One can consider that the calibration curve is linear over 1000°C and up to 1400°C.

## 22.4 Pyrometer calibration procedure

1. Install a substrate in the chamber with a thermocouple in contact with the backside center for a wafer. For susceptor calibration install a sheathed thermocouple inside the edge hole of the susceptor and use edge pyrometer.
2. Connect the thermocouple to the cable CONT.
3. Prepare a recipe with temperature dwell every 50°C from 300°C to 1000°C. The **thermocouple temperature control** mode must be selected for each step. The process should be performed under air at atmospheric pressure without any gas flow. The furnace is delivered with standard calibration recipe CALTC that can be used for this purpose.
4. Run the process and save the process historical.
5. Display the process historical.
6. Move the cursor on the first dwell and read the temperature from the thermocouple and the pyrometer digital signal



7. Repeat this operation for the other dwells in order to get all the temperature / digital value couples for the calibration table.

|              |            |                                 |
|--------------|------------|---------------------------------|
| TC1 (°C)     | 950.0      | ← Temperature from thermocouple |
| Py1 (bin/°C) | 2859 949.7 | ↑ Digital signal from pyrometer |

8. Enter the values in the calibration table (see the procedure hereafter).
9. Prepare a recipe using the pyrometer temperature control mode for each step. Start from 300°C with dwells of 50°C up to 1000°C for high temperature pyrometer and from 50°C for low temperature pyrometer.
10. Run the process and save the process historical.
11. Display the process historical.
12. Make adjustment of the values is required.

## 22.5 Generate the pyrometer calibration table

The Annealsys software has the capability to switch from the historical mode to the configuration with the pyrometer calibration table form.

When the calibration process historical is displayed and when you are ready to start the calibration table, select the configuration mode and the pyrometer calibration to display the following screen:

| Bits | Degrees |
|------|---------|
| 840  | 252.0   |
| 900  | 302.0   |
| 1070 | 352.0   |
| 1250 | 402.0   |
| 1470 | 452.0   |
| 1670 | 502.0   |
| 1835 | 552.0   |
| 1955 | 602.0   |
| 2065 | 652.0   |
| 2172 | 702.0   |
| 2275 | 752.0   |
| 2378 | 802.0   |
| 2483 | 852.0   |
| 2587 | 902.0   |
| 2695 | 952.0   |
| 2798 | 1002.0  |
| 2901 | 1052.0  |
| 3004 | 1102.0  |
| 3107 | 1152.0  |
| 3210 | 1202.0  |

The user can modify an existing table selected from the default list (with arrow key) or from any folder using the "Browse" button, or use the "Save as" function, or create a new table.

The emissivity is only set for information into the calibration table and has no effect on the actual pyrometer emissivity. The emissivity of the pyrometer is set by Annealsys before shipment of the system and should usually not be modified. A pyrometer hand terminal is required to modify the emissivity.

As the emissivity is a key parameter of the calibration it is strongly recommended to enter this value in the calibration table. If the emissivity has not been changed just copy the emissivity set by Annealsys.

Check the low type box if the system is equipped with a low temperature pyrometer and if the table will be used for a low temperature pyrometer (if 2 pyrometers are installed for example).

The user can enter a short comment for the calibration table.

The software allows saving several pyrometer calibration tables. A new table must be created each time a new substrate type is used.

It is possible to enter up to 30 couples of temperature / digital values in the calibration table. The software will make a linear interpolation between each couple of value to make the temperature calculation on the whole range.

When the pyrometer calibration has been selected the easiest way to make the pyrometer calibration table is to:

- go in the historical mode
- move the cursor to the right segment
- take the thermocouple temperature and the pyrometer digital values (950 and 2859 in the example curve displayed above)
- enter these value in the pyrometer calibration table
- repeat this operation for all the segments

The calibration table must respect the following rules:

- The first line of the calibration must always have "819", "0.0" values
- Both temperature and digital signal must increase at each line compare to the previous line.

There is no need to perform the calibration on the whole temperature range if the process is to be performed in a narrow temperature range. Nevertheless calibration at low temperature may provide better control over ramps.

If the calibration cannot be performed on the full temperature range because of the thermocouple range limitation, the calibration table can be extended by the user by linear extrapolation of the last segment.

For example if the calibration has been performed up to 1000°C and the last 2 lines of the table are:

| Bits | Temperature |
|------|-------------|
| 2703 | 1048        |
| 2838 | 1098        |

The table can be extended to higher temperature using the following formula:

- To 1200°C:  $\text{Bits}_{1200} = \text{ENT}[(2838-2703)/(1098-1048) * (1200-1098)] + 2838 = 3113$
- Then enter "3113" and "1200" in the next line of the table
- To 1300°C:  $\text{Bits}_{1300} = \text{ENT}[(2838-2703)/(1098-1048) * (1300-1098)] + 2838 = 3383$
- Then enter "3383" and "1300" in the next line of the table
- To 1400°C:  $\text{Bits}_{1400} = \text{ENT}[(2838-2703)/(1098-1048) * (1400-1098)] + 2838 = 3653$
- Then enter "3653" and "1400" in the next line of the table

## 22.6 Pyrometer emissivity

The pyrometer emissivity is set by Annealsys in our facilities. Usually there is no need to change the emissivity.

To change the emissivity the special pyrometer hand terminal must be used.

Contact Annealsys for further information.

## 23. TROUBLESHOOTING

### 23.1 Impossible to unlock the chamber

The chamber must reach the atmospheric pressure to be unlocked. The chamber pressure is usually measured by a constraint vacuum gauge that measures the absolute pressure.

Depending on the altitude of the system the atmospheric pressure can change and if it is not set to the right value the chamber will not be unlocked.

The atmospheric pressure can be set in the configuration mode. Set it to the right value so that the chamber can be unlocked when the pressure of the chamber reaches the atmospheric pressure

### 23.2 Impossible to open the chamber after process

After heating in process or manual mode it may be impossible to open the chamber.

During heating the gas inside the chamber is heated by the wafer and it expands. This increases the pressure inside the chamber and some gas will go out the chamber through the exhaust line.

During cooling down the gas cools down as well and the pressure inside the chamber decreases. As the exhaust line is provided with a check valve (one way valve), no gas will enter the chamber. This lower pressure inside the chamber will keep it close.

To release the chamber purge it for few seconds.

### 23.3 Process running but no heating

It may happen that there is no heating when the process is performed. The heating setpoint is increasing and the setpoint curve seems normal but there is no heating.

The problem may come from the status of the temperature controller that has not been set back to the remote, auto and run modes.

To solve the problem enter the RKC mode and exit confirming the return of the temperature controller to remote, auto and run modes. Then run the process again.

### 23.4 Mass flow hardware or communication problem

The system runs with digital mass flow controllers. These mass flow controllers are controlled by the PLC using a digital communication. If one of the mass flow controller is not working, disconnected or dismantle it generates error message in the communication. As a consequence the software routine slows down and the communication for the control and data logging is slower. The furnace is still safe but the control is not as fast as usual.

In order to avoid the software routine slow down the communication with a mass flow controller can be easily disable by setting the mass flow range to zero in the configuration mode.

When the mass flow is repaired then set the full range again to the right value.

## **24. SOFTWARE INSTALLATION AND BACKUP**

### **24.1 Software installation**

The furnace is delivered with a PC on which the software has been installed by Annealsys before shipment.

The software can be reinstalled using the CD that is provided with the system. The installation procedure is available on the CD.

### **24.2 Software upgrades**

Annealsys also provides software upgrades by email, by CD or that can be uploaded from the Internet. The installation procedure is provided with the software upgrade.

The PLCs are equipped with Flash memories. This allows easy PLC upgrade of the PLC software. The PLC software can be automatically upgraded from the PC by new software package provided by Annealsys.

Feel free to make comments on software utilization or to inform us of software bugs. All the Annealsys systems use the same software and this is part of our quality organization.

### 24.3 PLC Software backup

On the AS-One the PLC software is into a RAM memory that is supplied by a battery with a lifetime of about 2 years. For this reason, it is recommended to replace the RAM battery of the PLC every two years (see technical manual for procedure).

There is a backup of the PLC software in a Flash memory inside the PLC. In case of battery failure or any software corruption, the system will automatically restore the software from the Flash memory to the RAM memory so that the system can operate under safe conditions. This procedure is fully automatic and does not require any action from the user.

The user can anyway restore the PLC program using the configuration mode and administrator login.

The PLC software can also be loaded from the CD with an automatic procedure using the system PC.

Contact Annealsys for further support.

### 24.4 PC software backup

The PC software backup is on the CD that is delivered with the system.

Contact Annealsys for further support.

Revision history

| Revision History      |                                                                                                                                                                                                                                                                                                                                                                   |                   |          |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|----------|
| <b>Document Title</b> | AS-One Software user's manual                                                                                                                                                                                                                                                                                                                                     |                   |          |
| <b>Filename</b>       | ASOne_Software_Manual_V1-4-6-0.doc                                                                                                                                                                                                                                                                                                                                |                   |          |
| <b>Version Number</b> | 1.4.6.0                                                                                                                                                                                                                                                                                                                                                           |                   |          |
| <b>Author</b>         | Franck Laporte                                                                                                                                                                                                                                                                                                                                                    |                   |          |
| Version               | Brief Description of Changes                                                                                                                                                                                                                                                                                                                                      | Date of Issue     | Initials |
| 1.3.3.0               | Creation for new software version with major modifications                                                                                                                                                                                                                                                                                                        | April 20, 2006    | FLA      |
| 1.3.4.0               | Modifications in the administrator mode                                                                                                                                                                                                                                                                                                                           | May 8, 2006       | FLA      |
| 1.3.4.1               | Corrections of the 1.3.4.0 version                                                                                                                                                                                                                                                                                                                                | May 23, 2006      | ADI      |
| 1.3.5.0               | RKC window layout change and others minor modifications.                                                                                                                                                                                                                                                                                                          | June 28, 2006     | ADI      |
| 1.4.0.0               | Improvement of users administration.                                                                                                                                                                                                                                                                                                                              | September 6, 2006 | ADI      |
| 1.4.0.1               | Minor corrections                                                                                                                                                                                                                                                                                                                                                 | Not released      | FLA      |
| 1.4.0.2               | Internal software modifications                                                                                                                                                                                                                                                                                                                                   | November 6, 2006  | FLA      |
| 1.4.5.0               | The user can set the atmospheric pressure in the configuration<br>Temperature controller maximum output power and ramp rate can be set in the configuration                                                                                                                                                                                                       | January 20, 2007  | FLA      |
| 1.4.6.0               | Automatic PLC software restoration is case of RAM battery failure                                                                                                                                                                                                                                                                                                 | January 29, 2007  | FLA      |
| 1.5.0.0               | New software version 1.5.0<br>New default logins to avoid confusion with access modes<br>Modification of RKC mode<br>Additional diagnostics information including lamp age<br>Conditions to unlock the process chamber at end of the process                                                                                                                      | May 15, 2007      | FLA      |
| 1.5.2.0               | 6.7: A "Print screen" function is implemented<br>9.1: Recipe profile can be viewed from Download window<br>11.1.9: Force pick and download option can be enabled or disabled in the configuration mode.<br>19.4: Ballast Interlock has been implement as an option<br>23.4: Communication with mass flow controller can be disabled by setting full range to zero | August 30, 2007   | FLA      |
|                       |                                                                                                                                                                                                                                                                                                                                                                   |                   |          |